

BATTERY STORAGE AND NATURAL GAS GENERATOR MARKET POWER

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Abstract

This paper estimates the impact of utility-scale battery storage on the market power of incumbent fossil fuel generators. Using unit-level data from the Texas electricity market, where over 8 GW of storage has entered since 2021, I exploit the staggered entry of storage across local markets to identify its competitive effect. Local storage entry reduces natural gas peaker markups by 14 to 19 percent during evening ramp hours, operating through a flattening of residual demand rather than through changes in operating efficiency. This disciplining effect is concentrated among small-capacity generators, while large incumbents with dominant local shares are largely unaffected and see their inframarginal rents rise. As a result, storage entry raises aggregate producer rent by \$200 million annually rather than reducing it. The net social benefit of storage instead comes almost entirely from displaced emissions, which I estimate at \$2.4 billion annually.

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1 Introduction

Battery storage is viewed as a linchpin of the energy transition because it complements intermittent renewable generation: by shifting energy across time, storage smooths the variability of wind and solar output, paving the way for renewables to supply a larger share of generation (Karaduman, 2023; Andres-Cerezo and Fabra, 2025). The welfare gains from this complementarity are typically framed as reductions in environmental externalities when storage-backed renewables displace fossil-fuel generation. An underappreciated benefit of storage, however, is that it competes directly with the conventional generators it will eventually replace. Fossil-fuel plants, particularly the marginal units dispatched during high-demand hours, earn markups that reflect local market power (Borenstein, Bushnell and Wolak, 2002; Woerman, 2023). Because batteries discharge precisely when residual demand is least elastic, their entry increases the elasticity of demand facing incumbent generators and disciplines these markups. This competitive effect is a welfare margin distinct from the reduction in external costs, and it has not been quantified. Whether this competitive channel matters in aggregate is ultimately an empirical question.

In this paper, I quantify how battery storage entry affects the market power of incumbent natural gas generators. I study the Texas electricity market (ERCOT), an energy-only market where frequent transmission congestion enables local generators to extract scarcity rents. ERCOT provides an ideal setting for this analysis, having integrated over 8 GW of battery capacity since 2021, providing a scale of market penetration substantially larger than those evaluated in prior studies.¹ To identify the competitive effects of this entry, I construct a unit-level panel and define local competitive markets using high-resolution electrical grid topologies (Arkolakis and Walsh, 2024; Mercadal, 2022; Woerman, 2023). This network-based framework allows me to construct precise measures of local market size and test how incumbent firm scale mediates the competitive impact of storage.

To formalize these mechanisms, I develop a two-period optimal bidding model where a strategic natural gas generator competes against a price-taking storage fleet. The model highlights two distinct channels. Through the *market power channel*, storage reshapes the generator’s residual demand—flattening it during peak discharge periods and steepening it during off-peak charge periods. This yields a direct empirical prediction: markups should fall when batteries discharge and rise when they charge, though the magnitude depends on the generator’s market share and network connectivity. Through the *dynamic efficiency channel*, increased cycling degrades a plant’s physical operating efficiency. Because storage simultaneously reshapes pricing power and potentially alters physical costs, these two channels confound overall firm profitability. Identifying the pure competitive effect therefore requires separating them. I achieve this by constructing a measure of realized markups that purges these mechanical efficiency shocks, leaving only the firm’s strategic pricing response.

As storage enters ERCOT, incumbent natural gas generators are exposed to different levels

¹This capacity is roughly ten to one hundred times higher than those analyzed in Butters, Dorsey and Gowrisankaran (2025), Karaduman (2023), and Kirkpatrick (2025).

of new competition across counties. My research design compares the behavior and markups of generators facing substantial local battery entry to those that remain unexposed, both before and after storage become commercially operating. Exploiting this spatial and temporal variation to identify the causal effect of storage on market power, I find four main results. First, storage entry significantly erodes the market power of incumbent natural gas generators, but only during peak discharge hours. Local storage entry reduces peaker generator markups by \$6 to \$9.5/MWh (a 14% to 19% reduction) during the evening peak period, and by roughly \$2.3 to \$2.9/MWh during the morning peak. Conversely, when scaling the treatment by continuous cumulative storage capacity, I find a small positive markup effect during nighttime charging periods for smaller peakers in large markets, confirming the theoretical prediction of outward-shifting residual demand when batteries charge. This treatment effect is concentrated almost entirely among small capacity generators, across both peaker and non-peaker technologies, with large-capacity generators largely unaffected.

Second, the main mechanism reducing markups is a flattening of the residual demand curve. Event study estimates on generator heat rates show no statistically significant operational shift, which rules out the dynamic efficiency channel. At the current scale of storage entry, I do not find evidence that suggests that storage has an unintended consequence of reducing the scale efficiency of natural gas units. Decomposing the realized markup by using pre-storage heat rate confirms that the reduction operates entirely through the market power channel. Using the submitted bid to test this, I find that peakers respond to entry by steepening their bid curves, attempting to reprice the marginal megawatts that do clear even as their overall realized markups fall.

Third, I document significant heterogeneity based on storage ownership and operational characteristics. Intuitively, storage disciplines gas markups less when its owner has a reason to discharge cautiously: either to protect the revenue of its other assets, or to preserve the arbitrage spread when it holds a large share of storage capacity. The data support the first channel and reject the second. The disciplining effect on natural gas markups is stronger when storage is operated independently rather than co-located with a solar PV generator, and when it is managed by independent firms rather than multi-asset owners. These multi-asset owners are typically wind and solar developers rather than fossil fuel generators, and their other assets need not sit in the storage unit's local market. The relevant channel is therefore not a direct cannibalization of local gas markups, but the owner's incentive to avoid depressing the broader price environment in which its renewable portfolio is also compensated. Entry by the top five largest storage firms delivers a larger markup reduction than entry by smaller firms, which I read as a scale effect rather than storage-side market power: at current penetration, the largest owners behave as price-takers on the discharge side and shift the competitive fringe more than smaller, dispersed entrants can.

Fourth, I quantify the welfare implications of these competitive and operational shifts. Scaling the estimates across the ERCOT market reveals a critical distributional nuance: while storage successfully disciplines the markups of small natural gas units, it simultaneously increases the inframarginal rents of large-capacity non-peakers, resulting in a net \$200 million annual increase in producer rent market-wide. The true social value of storage therefore lies outside of market

power mitigation: I estimate a total annual social benefit of \$2.2 billion, driven almost entirely by \$2.4 billion in annual environmental benefits from displaced fossil-fuel emissions. This benefit exceeds the roughly \$1 billion annualized cost of the currently operating fleet, so storage entry is net welfare-positive at current penetration, but the case rests on its unpriced environmental value rather than on curbing market power: even though it substantially disciplines the subset of generators identified above, the redistribution of rent toward large incumbents means aggregate producer rent rises rather than falls.

This paper contributes to three large strands of literature. First, this work contributes to the broad industrial organization literature on the effect of entry on incumbent firm behavior. Classic empirical studies demonstrate that incumbents respond dynamically to the threat or realization of entry across various industries, from retail supercenters (Arcidiacono et al., 2020; Busso and Galiani, 2019) and retail gasoline (Taylor and Muehlegger, 2026) to airlines (Goolsbee and Syverson, 2008) and pharmaceuticals (Ellison and Ellison, 2011). Closest to my setting, Arcidiacono et al. (2020) and Taylor and Muehlegger (2026) use high-frequency microdata to trace how incumbents adjust to entry in retail supermarkets and gasoline. I extend this framework to wholesale energy markets, where I observe the incumbent’s response at the level of the individual bid rather than inferring it from coarse prices, and where the entrant is not a symmetric rival but an intertemporal arbitrageur that reshapes residual demand as it charges and discharges. Furthermore, by examining the heterogeneity of the treatment effect across different storage configurations, I demonstrate how the organizational boundaries of the entering technology, such as whether the battery is co-located with a solar host or owned by a multi-asset portfolio firm, dictate the magnitude of the incumbent’s competitive response.

Second, this paper contributes to the literature on competition and market power in wholesale electricity markets. A robust body of work documents how generators exercise market power under transmission constraints and localized congestion (Borenstein, Bushnell and Wolak, 2002; Hortaçsu and Puller, 2008; Ito and Reguant, 2016; Woerman, 2023), and recent studies show that transmission expansion (Ryan, 2021; Doshi, 2024), financial arbitrageurs (Mercadal, 2022), and renewable price exposure (Fabra and Imelda, 2023) can reduce it. Closer to this paper, two recent studies examine how large-scale solar entry affects conventional generators: Jha and Leslie (2025) show it softens evening competition through start-up costs, while Bushnell and Novan (2021) show it shifts the marginal technology toward less efficient units. I study an analogous large-scale entry, storage, and show that it disciplines markups through the same channel this literature identifies as central to transmission-driven discipline: a flattening of the incumbent’s residual demand curve.

Finally, this paper speaks to the growing literature on the economics of utility-scale storage in wholesale electricity markets. This work has largely quantified storage’s aggregate, market-level impact, documenting how it substitutes for or complements renewable capacity (Karaduman, 2023; Andres-Cerezo and Fabra, 2025), compresses intra-day price spreads (Lamp and Samano, 2022), shifts long-run investment equilibria (Butters, Dorsey and Gowrisankaran, 2025), and relieves local congestion (Kirkpatrick, 2025). Much less is known about how entry restructures the strategic

incentives of specific incumbents. I depart from this focus by using disaggregated, unit-level operation data, to my knowledge the first such analysis in this literature. These data let me trace the channel from an individual storage unit’s entry to the bidding behavior and realized markup of specific, proximate generators, and to characterize heterogeneity by generator capacity, market structure, and storage ownership. While prior structural models characterize the theoretical price impact of storage (Andrés-Cerezo and Fabra, 2023), I provide causal, reduced-form evidence that observed storage dispatch erodes realized markups, and I use a welfare framework that separates distributional transfers from net social gains.

The remainder of this paper is organized as follows. Section 2 provides an overview of the Texas electricity market and develops a theoretical model to derive predictions of how storage entry affects the market power of natural gas generators. Section 3 describes the data sources, panel construction, and descriptive statistics. Section 4 presents the empirical strategy. Section 5 presents the empirical results, covering the average treatment effect, heterogeneity, and the mechanism tests that isolate the market power channel. Section 6 discusses the welfare analysis and policy implications, and Section 7 concludes.

2 Background and Theoretical Motivation

In this section, I provide an overview of the Texas electricity market, highlighting the unique features of ERCOT that make it an ideal setting to study the impact of storage entry on natural gas generator market power. I then develop a theoretical model to derive predictions of how storage entry affects these competitive dynamics.

2.1 Storage and The Texas Electricity Market

The Electric Reliability Council of Texas (ERCOT) manages the flow of electric power to more than 26 million customers in Texas, accounting for about 90% of the state’s electric load. ERCOT operates a wholesale electricity market and schedules the dispatch of over 1500 generators in real time. ERCOT has limited interconnection with other markets, which means that the market is relatively insulated from external shocks. ERCOT’s generation mix and market design make it an ideal setting to study the competitive interactions between storage and natural gas generators. First, ERCOT has a large share of natural gas generation, which accounts for about 44.3% of the generation capacity, followed by wind (25.2%) and solar (13.2%) (ERCOT, 2025).² Second, ERCOT has experienced massive growth in battery storage capacity in recent years, which is beginning to alter competition dynamics. Finally, as an energy-only market with no capacity payments, ERCOT creates strong incentives for generators to respond strategically to real-time market conditions.

²Figure F.24 shows the spatial distribution of each unit based on their fuel type in ERCOT. Table E.1 summarizes the generator units in ERCOT territory as of the end of 2024, while Figure F.25 details the installed capacity over my sample period: Natural gas units dominate the installed capacity (38.9% or 65.23 GW), followed by wind and solar PV units, respectively. Natural gas and coal capacity remain roughly the same over my sample, while solar PV and wind capacity grow significantly by about 25 GW and 20 GW.

Market Design. Unlike PJM, MISO, and ISO-NE, ERCOT has no centralized capacity market. Instead, generators are paid only for the energy they produce and the ancillary services they provide.³ ERCOT operates a nodal market where the price of electricity, the locational marginal price (LMP), is determined at each node on the grid reflecting the cost of supplying an additional unit of electricity to that node, taking into account generation costs, transmission constraints, and losses.

ERCOT runs two sequential markets: a day-ahead market and a real-time market. In the day-ahead market, a generator or load can offer or bid energy as well as ancillary services. ERCOT then clears the quantity and price for these services, establishing the responsibility of the resource for each hour of the next day.

On the day of electricity delivery, the real-time market runs every 5 minutes to allow generators to adjust for any changes in the realization of demand and renewable generation. Generators and loads submit offer or bid curves considering their energy forward positions and ancillary service responsibilities carried over from the day-ahead market. ERCOT then performs a Security Constrained Economic Dispatch (SCED), which solves for the least-cost dispatch quantity for each 5-minute interval. This process results in the real-time LMP for each node.⁴

Congestion and Market Power Dynamics. Transmission congestion is a frequent occurrence in ERCOT, fragmenting the broader grid into smaller local markets with divergent prices (LaRiviere and Lyu, 2022; Fell, Kaffine and Novan, 2021; Woerman, 2023; Doshi, 2024). This isolation shields local generators from external competition, allowing even small units to become pivotal and exercise market power. Firms primarily exploit this through economic withholding, where they artificially raise offer prices well above marginal cost for higher output quantities. By shifting their offer curves upward, generators force the market to clear at higher price intervals, making the entry of localized battery storage a critical competitive threat to these strategies.

This incentive to withhold is amplified by ERCOT’s energy-only market design. Without centralized capacity payments, the financial consequences of withholding are borne entirely in the real-time energy price, providing a strong incentive to optimize short-run bidding strategies. Because this design creates substantial revenue uncertainty during non-scarcity hours, natural gas generators depend heavily on securing outsized scarcity rents during tight, congested hours to cover their fixed operating costs. This is true not only for peakers, whose entire revenue model rests on scarcity pricing, but also for non-peaker generators whose position in the local competitive set exposes them to similar pricing incentives during ramp hours.

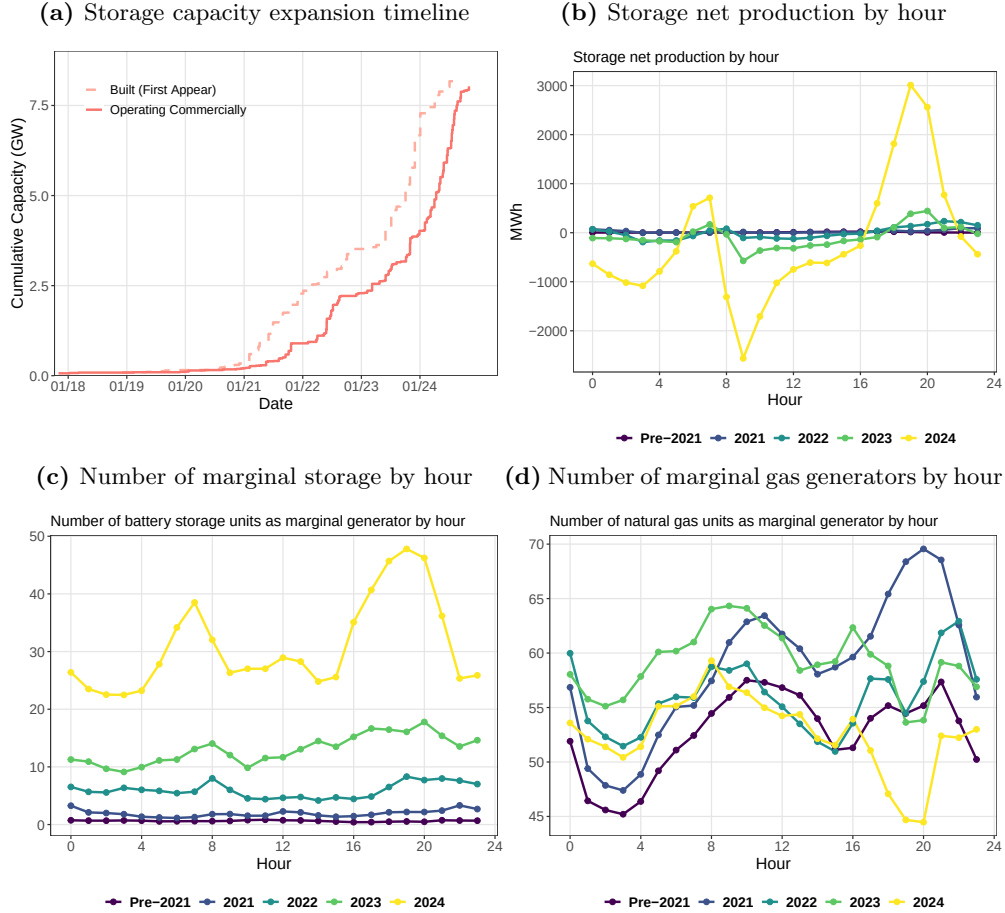
Furthermore, ERCOT grants generators immense latitude to execute these strategies due to

³Ancillary services in ERCOT include regulation services (regulation up, REGUP, or regulation down, REGDN) and reserve services (responsive reserve, RRS; non-spin reserve, NSPIN; or ERCOT contingency reserve service, ECRS). A generator can provide REGUP service by importing energy to the grid, while load provides REGDN by exporting energy from the grid. Generators can also be paid by reserving capacity ready to be dispatched when needed.

⁴LMP is assigned at the settlement point, which is essentially a node in the grid. Each generator is mapped to a settlement point and receives the LMP of that settlement point.

its highly flexible bidding rules and loose price caps. While most centralized electricity markets tightly regulate offers to mitigate market power and impose rigid price caps around \$1,000/MWh or less(Joskow, 2022), ERCOT has historically featured caps shifting between \$3,000/MWh and \$9,000/MWh (and up to \$5,000/MWh at the end of my study period).⁵ This regulatory design allows generators to raise prices sharply when local grid conditions tighten.

Figure 1: Storage expansion and its competition trends with gas generators in ERCOT



Notes: Panel A shows the cumulative storage capacity built and commercially operating over time. Panel B shows the aggregate storage net production pattern by hour. Net production is defined as aggregate storage discharge less aggregate storage charge. Panel C shows the number of storage units that receive the highest offered price in each hour. Panel D shows the number of natural gas generators that receive the highest offered price in each hour.

Storage Participation and Competition. Storage units in ERCOT operate as fully dispatchable generation, acting as generators when exporting energy and as loads when importing. They participate extensively in both the day-ahead and real-time markets, receiving dispatch instructions and compensation for energy and ancillary services.⁶

⁵Under FERC Order 831, non-ERCOT markets allow offer caps to increase to \$2,000/MWh to set the market clearing price, but strictly on the condition that these higher offers are justified by verified short-run marginal costs.

⁶When charging, storage can be instructed to provide REGDN, and when discharging, it can also be instructed to provide REGUP, RRS, NSPIN, or ECRS. For example, during summer 2023, revenue from ECRS dominated

Figure 1 illustrates this rapid expansion, with ERCOT storage capacity growing to 7.6 GW since 2021, second only to California.⁷ Because battery storage responds rapidly to real-time market signals, it is structurally optimized to target the exact congested, high-revenue hours where incumbent generators historically exercised market power. This is evident in Panel B, which shows storage discharging heavily during the evening ramp period (hours 16–20). Panels C and D further document this emerging pattern of direct substitution. During peak discharge hours, the number of storage units receiving the highest offered price has significantly increased, while the corresponding number of marginal natural gas generators has decreased. This clear shift in competition pattern motivates the central question of this paper: how does storage entry affect the market power of natural gas generators?

2.2 Theoretical Framework: Storage Entry and Generator Market Power

To address this question, I write a simple two-period model to illustrate how storage entry alters the competitive behavior of natural gas generators (gas generators, hereafter). The model identifies two primary mechanisms: a market power channel, where storage entry flattens the incumbent’s residual demand curve, and a dynamic efficiency channel, where storage-induced operational changes lower the generator’s heat rate and marginal costs. Importantly, the model treats the gas generator as a representative unit whose position in the local competitive set determines how strongly it is disciplined by storage entry. The technology of the generator (peaker versus non-peaker) affects the dynamic efficiency channel through the shape of the heat-rate curve, but the market power channel operates through the generator’s share of local capacity and the size of the local market, both of which vary within and across technology types.

Setup. There are two periods $t \in \{L, H\}$ with inelastic load $D_L < D_H$, clearing at a single locational price p_t (the LMP) within a market. Three groups supply energy.

A *competitive fringe* of inframarginal, price-taking units (nuclear, coal, must-run gas) has upward-sloping aggregate supply $S_t^c(p) = \alpha_t + \beta p$ with $\beta > 0$ (Fabra and Imelda, 2023; Ito and Reguant, 2016). A *strategic natural gas firm* f chooses output q_{ft} taking its residual demand as given. Its marginal cost

$$(1) \quad c_{ft} = p_t^g hr_{ft}$$

is the fuel price p_t^g times its heat rate hr_{ft} . For the sake of exposition, I fix heat rate and exclude environmental costs in this section. A *battery storage* with power capacity $\bar{P}$ (in MW, the rate at

total revenue, contributing more than 75%, while real-time energy settlement contributed just under 15%, and RRS followed closely. Revenue from operating reserve demand curve (ORDC) and regulation services contributed only a small share of the total. See Miller and Nakajima (2024).

⁷In Q2 2025, ERCOT surpassed the total installed capacity in California. Both markets have since experienced significant growth in storage capacity. The fundamental difference in ERCOT is that the rapid expansion is driven by market forces and interconnection speed, rather than state-level procurement mandates or subsidies as in California (Mathrani, 2025; Ahmed et al., 2026).

which it can charge or discharge), energy capacity $\bar{K}$ (in MWh, the maximum amount of energy it can store), and round-trip efficiency $\eta \leq 1$ injects s_t into the market in period t : it buys in the low-demand period ($s_L \leq 0$, charging) and sells in the high-demand period ($s_H \geq 0$, discharging).⁸ The market clears in each period,

$$(2) \quad S_t^c(p_t) + q_{ft} + s_t(p_t) = D_t.$$

The strategic firm chooses q_{ft} to maximize its profit $\pi_{ft} = (p_t - c_{ft}) q_{ft}$, taking the residual demand as given.

The economics of storage is pure arbitrage subject to capacity limits. A single price-taking battery's marginal cost of delivering peak energy is the *opportunity* cost, $m_s = p_L/\eta$.⁹ Individual batteries are inelastic at their limits, but a fleet with heterogeneous arbitrage thresholds delivers a smooth, upward-sloping aggregate supply. This slope is larger when more storage sits near its threshold and zero when the relevant capacity is saturated. Appendix A.1 states the optimal storage problem and derives the net injections s_t , the opportunity cost m_s , and σ_t . The aggregate storage supply can be written as follows:

$$(3) \quad s_t(p_t) = \bar{s}_t + \sigma_t p_t, \quad \sigma_t = \frac{\partial s_t}{\partial p_t} \geq 0.$$

Next, I derive the strategic firm's optimal markup.

Generator's optimal markup. A generator with market power does not bid its marginal cost; it withholds, trading off a higher price against lost sales. The strategic firm's optimal markup is

$$(4) \quad \mu_{ft} \equiv p_t - c_{ft} = \frac{q_{ft}}{E_t}, \quad E_t \equiv \left| \frac{\partial q_{ft}}{\partial p_t} \right| = \beta + \sigma_t,$$

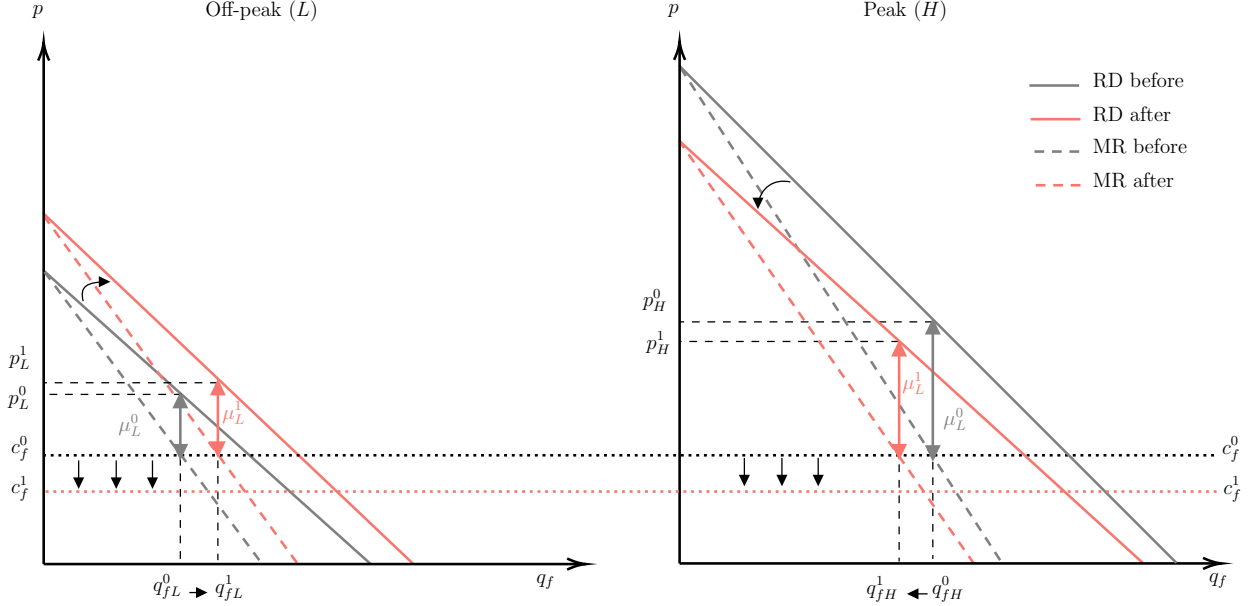
where q_{ft} is the residual quantity it serves and E_t is the (absolute) slope of its residual demand. The optimal markup is proportional to how much of the residual demand it serves and inversely proportional to how easily the rest of the market substitutes away from it.¹⁰ Equation (4) governs how storage affects natural gas generator markups. Storage can move the markup directly by moving the *level* q_{ft} in the numerator or the *slope* E_t in the denominator. Another possible channel is through the marginal cost c_{ft} if storage entry affects the generator's operational efficiency. Figure 2 renders these two channels graphically as a residual monopolist problem. I derive the comparative statics of the markup with respect to storage entry for each channel in turn.

⁸Due to round-trip efficiency, the energy it stores is $-s_L$ and it delivers $\eta(-s_L)$ at the peak. The two injections are equal and opposite only when storage is lossless, $\eta = 1$.

⁹I extend the analysis to allow for strategic storage in Appendix A.5.

¹⁰Appendix A.2 states the generator's problem and derive the optimal markup. This is the standard result in the literature on market power in electricity markets (Jha and Leslie, 2025; Woerman, 2023; Ryan, 2021; Doshi, 2024; Fabra and Imelda, 2023).

Figure 2: Illustration of the market power and dynamic efficiency channels



Notes: This figure illustrates the market power and dynamic efficiency channels pre- (gray) and post-storage entry (red). Left and right panels display the natural gas generator's residual demand during charge (off-peak) and discharge (peak) periods. Setting $MR = c$ yields markup μ . The market power channel (holding c fixed) shifts residual demand outward off-peak (raising μ) and inward on-peak (lowering μ). The dynamic efficiency channel captures shifts in c : reduced ramping intensity lowers the average heat rate (shifting c down), but operating further from optimal efficiency raises it (shifting c up). The net effect on μ depends on the relative magnitudes of these two channels.

Market power channel. Let the pre-storage residual-demand intercept be $A_t^0 \equiv D_t - \alpha_t$ (storage absent) and slope β . Holding marginal cost fixed to isolate the market power channel, the change in markup from storage entry is

$$(5) \quad \Delta\mu_H = \frac{A_H^0 - \bar{s}_H}{2(\beta + \sigma_H)} - \frac{A_H^0}{2\beta} < 0 \quad (\text{discharge period}),$$

$$(6) \quad \Delta\mu_L = \frac{A_L^0 - \bar{s}_L}{2(\beta + \sigma_L)} - \frac{A_L^0}{2\beta} > 0 \quad (\text{charge period}),$$

In the discharge period, storage is a rival supplier. It serves part of the load that gas would otherwise have served, so it shifts the gas firm's residual demand inward (lower q_{fH}) and, because a flexibly-dispatched battery is itself price-responsive, makes that residual demand flatter (higher E_H). Both effects shrink the markup. In the charge period, storage is instead a buyer: it raises load, shifting residual demand outward (higher q_{fL}) and raising the markup. The off-peak markup rises only if the *proportional* outward shift in residual demand ($-\bar{s}_L/A_L^0$) exceeds the *proportional* increase in its elasticity (σ_L/β). When storage charges at a binding limit, $\sigma_L = 0$ and the condition holds for any $\bar{s}_L < 0$; the increase weakens as storage becomes a more flexible, price-responsive buyer off-peak.

Prediction 1 summarizes the theoretical predictions of the market power channel.

Prediction 1: (Market power channel) Holding marginal cost fixed, local storage entry reduces the markup of natural gas generators in the discharge period unambiguously, and increases the markup in the charge period provided the outward shift in residual demand dominates the increase in its elasticity.

Two features of the framework shape how this prediction applies across generators. First, the first-order condition in Equation (4) does not depend on generator technology; technology enters the model only through the dynamic efficiency channel of Section 2.2. The market power channel therefore applies to any natural gas generator that serves positive residual demand in a market where storage entry alters A_t (intercept) or E_t (slope). Second, storage enters the generator’s residual demand only if the two clear together. Three cases follow from the relative location of storage and the gas firm: local storage produces the full effect; remote storage on a slack line still disciplines the generator through the integrated fringe; and remote storage behind a binding constraint is priced separately and leaves the markup unchanged.¹¹

Congestion also determines the generator’s competitive set. Storage flattens the inverse semi-elasticity from $1/\beta$ to $1/(\beta + \sigma_t)$, a proportional reduction of $\sigma_t/(\beta + \sigma_t)$ which is the storage’s share of the competitive set.¹² Two features of the local market moderate this discipline. The first is *local market size*, following Woerman (2023): residual demand is less elastic in smaller markets (lower β), which raises pre-entry markups. Storage is then a larger share of the competitive set and erodes a larger markup, so the effect is stronger in smaller markets. The second is the generator’s *own share of local capacity*. A generator holding a large share remains pivotal after storage entry, because the alternative supply available at the relevant quantity is limited. Storage of a given size represents a smaller share of the competitive set relative to what the dominant generator supplies, and does less to erode its market power.

The two moderators are complementary. Both raise the pre-entry markup that storage can erode, and both determine how much of that markup storage compresses. The effect is largest for smaller generators in smaller markets, where pre-entry markups are high and storage is a substantial share of the local competitive set, and vanishes for larger generators in larger markets, where the generator remains pivotal and storage is a small share of the competitive set.¹³

Prediction 2: (Market size and generator capacity scale the market power channel) The market power effect is jointly moderated by two features of the local competitive set: local market size, which sets the pre-entry markup available to erode, and the generator’s own share of local capacity, which determines whether it remains pivotal after storage entry. The effect is largest for smaller generators in smaller markets, where both moderators reinforce the effect, and vanishes for larger generators in larger markets. The two moderators are theoretically distinguishable but empirically complementary.

¹¹ Appendix A.4 derives these three cases from a two-node extension of the model.

¹² Formally, the proportional reduction is $(\frac{1}{\beta} - \frac{1}{\beta + \sigma_t}) / \frac{1}{\beta} = \sigma_t / (\beta + \sigma_t)$.

¹³ Local market concentration, measured by the Herfindahl-Hirschman Index (HHI), is highly correlated with market size in this setting because congestion drives variation in both. I focus on market size as the moderator entering the theoretical prediction; empirical results are qualitatively similar under HHI splits.

Dynamic efficiency channel. Storage entry can affect the generator’s marginal cost by changing the distribution of its operating points. Following [Demirer and Karaduman \(forthcoming\)](#), I model heat rate as a function of the generator’s utilization $\bar{q}_{ft}$ (capacity factor) and cycling intensity ρ_{ft} (variation in capacity factor):

$$(7) \quad hr_{ft} = hr_g(\bar{q}_{ft}, \rho_{ft}), \quad \frac{\partial hr_g}{\partial \bar{q}} < 0, \quad \frac{\partial hr_g}{\partial \rho} > 0,$$

where g indexes technology. Heat rate is decreasing in utilization: a unit is most efficient near full load and least efficient at minimum stable generation; and increasing in cycling intensity from ramps and starts. The magnitudes of both derivatives vary across technologies, as combined-cycle units and simple-cycle peakers are optimized for different operating profiles.

The generation-weighted average changes in heat rate decomposes into a composition term and a cycling term:

$$(8) \quad \Delta \bar{hr}_{ft} = \underbrace{\sum_{\bar{q}} (\Delta w_{\bar{q}}) hr_g(\bar{q}, \cdot)}_{\text{composition}} + \underbrace{\sum_{\bar{q}} w_{\bar{q}} \frac{\partial hr_g}{\partial \rho} \Delta \rho}_{\text{cycling}},$$

where $w_{\bar{q}}$ is the share of output at utilization level $\bar{q}$. Storage entry alters both terms. The composition term reflects the shift in the distribution of operating points: storage can displace inefficient part-load operation (lowering the mean heat rate) or push generators into new inefficient regions (raising it). The cycling term reflects changes in ramping intensity, which typically rise with storage entry because generators must chase a more volatile residual load profile. The two terms need not move together and can operate in opposite directions.¹⁴ Prediction 3 summarizes the dynamic efficiency channel.

Prediction 3: (Dynamic efficiency channel) The net change in a unit’s generation-weighted heat rate, $\Delta \bar{hr}_{ft}$, depends on the balance of the composition and cycling terms in Equation (8). The relative magnitude of the two terms varies across technologies through the shape of the heat-rate curve, specifically, the steepness of $|\partial hr_g / \partial \bar{q}|$ and the cycling sensitivity $\partial hr_g / \partial \rho$. The sign and magnitude of $\Delta \bar{hr}_{ft}$ are therefore theoretically ambiguous and must be determined empirically for each technology.

Realized markup and its decomposition. My empirical analysis focuses on the realized markup, $\mu_{ft} = p_t - c_{ft}$, which is the difference between the LMP and the generator’s marginal

¹⁴A related composition mechanism has been documented at the market level: [Bushnell and Novan \(2021\)](#) show that solar ramping shifts aggregate dispatch in California such that the marginal generating technology across the system moves from efficient combined-cycle to less efficient simple-cycle units, raising shoulder-hour prices through this technology-mix effect. My composition term in Equation (8) operates at the unit level instead, capturing whether an individual generator’s own distribution of operating points shifts, holding fixed which units are dispatched. Whether storage entry produces a comparable within-unit composition shift, or whether the offsetting cycling channel dominates, is an empirical question.

cost. The change in the realized markup decomposes into the two channels:

$$(9) \quad \Delta\mu_{ft} = \underbrace{\Delta\left(\frac{q_{ft}}{E_t}\right)}_{\text{competition (Channel 1)}} + \underbrace{(\Delta c_{ft})}_{\text{relative efficiency (Channel 2)}}.$$

The efficiency term moves $c_{ft} = p_t^g \bar{hr}_{ft}$ through changes in heat rate, so a markup built from the realized heat rate reflects the combined effect of both channels. My main empirical specification uses this realized-heat-rate markup, since it corresponds to the actual price-cost margin earned by the generator and is the object of interest for measuring changes in market rent. To isolate the pure competitive effect for the mechanism analysis, I additionally construct a fixed-heat-rate markup that holds each generator's heat rate at its pre-storage level, which shuts down Channel 2 by construction. The comparison between the two versions in Section 5.4.2 identifies the relative contribution of the market power and efficiency channels.

Extension: Strategic storage and ownership structure. The baseline model treats storage as a competitive arbitrageur that discharges whenever the spread justifies it. In practice, storage owners' discharge decisions depend on two additional considerations that reduce the storage supply elasticity σ_t and, through it, the strength of market power channel. Following [Andrés-Cerezo and Fabra \(2023\)](#), I incorporate both into the storage owner's problem in Appendix A.6.

The first is *cooptimization*. When storage is dispatched jointly with another asset, either physically (colocated with generation) or financially (part of a multi-asset owner's portfolio), the owner internalizes the price impact of discharge on the other asset's revenue. The storage owner's problem carries an extra term equal to the price impact of discharge times its co-owned peak output; the larger that co-owned output, the less the owner discharges competitively.

The second is *storage-side market power*. A firm with a large share of storage capacity faces a downward-sloping residual demand for its own discharge and has the incentive to withhold or time discharge to preserve the spread. This is the storage-side analog of the withholding behavior modeled for natural gas generators in Section 2.2.

Both forces lower the equilibrium responsiveness σ_t and attenuate the disciplining effect on gas markups. In practice they often coincide, because larger storage owners tend to hold more diversified portfolios.

Prediction 4: (Storage type heterogeneity) The disciplining effect on natural gas markups is stronger when storage is dispatched more competitively. Two directional implications follow. First, storage subject to cooptimization with another asset produces a weaker effect than storage dispatched independently. Second, storage held by owners with a large share of storage capacity produces a weaker effect than storage held by smaller-share owners.

Section 5.3.2 tests these implications using four ownership contrasts: standalone versus colocated (physical cooptimization), independent versus multi-asset owner (financial cooptimization), and non-top-5 versus top-5 by storage capacity (storage-side market power).

Summary of predictions and empirical analysis Table 1 maps the four theoretical predictions onto the empirical analyses that test them. Prediction 1 on the sign of the markup response is tested using the main results in Section 5.1. Prediction 2 on the two moderators of the response is tested using the heterogeneity results by generator capacity and market size in Section 5.2, and refined using the continuous treatment analog in Section 5.3.1. Prediction 3 on heat rate is tested using the mechanism analysis in Section 5.4. Prediction 4 on ownership-driven heterogeneity is tested using the heterogeneity results by storage characteristics in Section 5.3.2.

Table 1: Mapping of theoretical predictions to empirical analysis

Prediction	Statement	Tested in
1	Discharge markup falls; charge markup weakly rises	Section 5.1 and 5.3
2	Effect scales with market size and inversely with own capacity share	Section 5.2 and 5.3
3	Heat rate change is ambiguous and varies by technology	Section 5.4
4	Cooptimization and storage-side market power attenuate the effect	Section 5.3

Notes. This table summarizes how the four theoretical predictions from Section 2.2 map onto the empirical analyses in Section 5.

3 Data and Descriptive Evidence

The objective is to compile a dataset that allows me to analyze the impact of storage entry on natural gas generator market power. In this section, I describe the data sources and the construction of the panel dataset. I also provide descriptive statistics for storage units and its operations patterns then turn to natural gas generators units and describe their characteristics, operations, and market power.

3.1 Data Sources and Sample Construction

I combine multiple datasets to construct a panel at the generator level covering January 1, 2018 to December 1, 2024. The main components are the generator and storage unit characteristics, the offers or bids they submit to the market, the associated market outcomes, and the generator’s marginal cost. The novel component of this dataset is that I observe each storage unit’s power capacity and its storage duration, the offers and bids they submit, and the electricity charged or discharged at the unit level.

Generator and battery storage data. I collect the list of unique generators from ERCOT’s 60-day Security Constrained Economic Dispatch (SCED) Report and the Monthly Outlook for Resource Adequacy (MORA) Report.¹⁵ For all generators, I define effective capacity as the maximum

¹⁵Two or more generators may be associated with a single storage or generation unit. A generator’s name might also change before and after testing. For combined-cycle units, I use the maximum production from the generator combination with the highest possible output as the effective capacity.

production observed in the SCED Report, and I use the first date a generator produces electricity outside of testing as its initial operating date.

Because ERCOT reports power capacity but not energy capacity for battery storage, I complement this data using EIA Form 860M, the MODO Energy Battery Buildout Report, and other public sources to determine each unit’s storage duration. Furthermore, because storage acts as both a generator and a load, it operates under different names when submitting supply offers versus load bids; I manually match these names for each storage unit. Finally, I use EIA Form 860 to determine the load zone and county for each generator, and the Decision Making Entity from the SCED MORA Report to identify the parent company.¹⁶

Electricity market data. I collect data on generator offer curves (and load bid curves for storage units), dispatch instructions, and operating status at 15-minute intervals from the SCED Report. In each interval, ERCOT collects these offers, evaluates resulting transmission constraints, and solves the dispatch problem through SCED. ERCOT then issues a dispatch instruction for each generator, dictating its required production (or load consumption) and operating status. The SCED process also produces a locational marginal price (LMP) for each settlement point every five minutes, which I map to each generator and aggregate to 15-minute intervals. ERCOT also reports the generators with the highest accepted offer for each interval, allowing me to identify the marginal generators.

Marginal cost. I follow [Hausman \(2025\)](#) and [Doshi \(2024\)](#) to construct the marginal cost of natural gas generators as the sum of variable fuel cost plus emission permit costs:

$$c_{it} = \underbrace{hr_{it} \cdot p_{it}^{Fuel}}_{\text{Fuel Cost}} + \underbrace{ER_{it}^{CO_2} p_t^{CO_2} + ER_{it}^{NO_x} p_t^{NO_x}}_{\text{Emission Cost}}.$$

I use hourly generation data from SCED and collect information on heat content and emissions to compute heat rates and emission rates at the plant-by-type-by-week level from the Environmental Protection Agency’s (EPA) Continuous Emission Monitoring System (CEMS) from 2018 to 2024.¹⁷ To compute fuel cost, I collect daily natural gas price data from S&P Capital IQ. Natural gas units source their fuel from different hubs.¹⁸ To compute emission cost, I collect permit prices for SO₂ and NO_x from S&P Capital IQ and apply the emission cost to units subject to the Acid Rain SO₂ Program and CSAPR NO_x Program.

¹⁶I manually crosscheck this ownership data with EIA Form 860 and employ AI tools to determine the parent company for each generator. Generators owned by multiple companies are coded under a single parent company.

¹⁷Heat rate and emission rate vary depending on the operating condition of a generator, as discussed by [Demirer and Karaduman \(forthcoming\)](#). I compute the heat rate and emission rate for each plant-by-type for each week: I divide the weekly total heat content and emissions for that week by the weekly total net generation for each plant-type combination. For units with no observation in CEMS, I assign the average heat rate for the plant with the same Type (Combined Cycle, Simple Cycle, or Gas Steam Turbine). I winsorize the extreme values to be within the reported heat rate range for the same generator type from EPA eGrid, to account for unreasonably high and low values from CEMS.

¹⁸I use the natural gas price location information for each unit from SNL Energy. For units missing this information in the S&P Capital IQ database, I assign Waha Hub for units in the West load zone, NGPL TX-OK for units in the North load zone, TETCO STX for units in the South load zone, and Katy for units in the Houston load zone.

Markup. In the main analysis, I use realized markup as the main outcome variable to measure market power, defined as:

$$\mu_{it} = \mathbf{1}[q_{it} > 0] (p_{it} - c_{it}).$$

This markup is zero when the generator is not producing electricity, and equal to the difference between its LMP p_{it} and marginal cost c_{it} when it is producing. This construction captures both the extensive margin (whether the generator is producing) and the intensive margin (how much markup it charges when producing).

To separate the market power channel from the dynamic efficiency channel, I also construct a fixed-heat-rate version of the markup that holds each generator’s heat rate at its pre-storage level. This isolates the pure competitive effect from any efficiency-driven change in marginal cost.

Local market definition. Following [Mercadal \(2022\)](#) and [Woerman \(2023\)](#), I define local markets dynamically at each 15-minute interval based on LMP divergence and network connectivity. Generators are first clustered if their LMPs are within \$1/MWh or 1% of the larger price. I then use U.S. electrical grid data from [Arkolakis and Walsh \(2024\)](#) to verify transmission connectivity within these clusters.¹⁹ Generators sharing both similar LMPs and grid connections constitute a local market, reflecting unconstrained electrical flow on the margin.

Figure [B.3](#) shows examples of market definition for two intervals, one with no congestion and one with congestion. The number of local markets varies from 1 to 137 across intervals, with a median of 8.²⁰ I use this local market definition together with parent company information and generator capacity to construct the local market statistics that underpin the heterogeneity analysis: market size, market concentration, and the number of firms in each local market.

Sample construction. I construct an unbalanced panel at the generator-interval level. The sample includes all natural gas generators in ERCOT from January 1, 2018 to December 1, 2024. For the main analysis, I aggregate the 15-minute interval data to the generator-by-hour-by-week-of-sample level and use the average markup within each hour-by-week as the main outcome variable. I use the 15-minute sample to show robustness of the main results. When analyzing the dynamic efficiency mechanism, I use the subset of generators observed in CEMS to compute the heat rate, which is around 80% of the total natural gas generators in ERCOT.

3.2 Descriptive Statistics of Battery Storage Units

This section provides descriptive statistics on the battery storage units in ERCOT. I document storage characteristics, ownership structure, and siting patterns. I then describe storage operation

¹⁹[Arkolakis and Walsh \(2024\)](#) collect satellite data from OpenStreetMap and Google Earth to identify high-voltage transmission lines, substations, and their characteristics.

²⁰Figure [B.2](#) presents the distribution within my sample and across hours by year of sample. The number of local markets has increased over the years as more generators are added to the ERCOT grid.

patterns observed in the data.

Table 2: Summary statistics of storage units

	All Storage (1)	Storage in County w/ NG Plant (2)	First Storage in County w/ NG Plant (3)	Co-located w/ Solar (4)	Co-located w/ Wind (5)
Panel A: Storage Characteristics					
Avg. Capacity (MW)	49.3 (51.3)	51.0 (53.3)	61.4 (61.7)	61.7 (38.9)	53.0 (53.1)
Avg. Duration (hr)	1.37 (0.60)	1.36 (0.64)	1.34 (0.47)	1.46 (0.72)	1.60 (0.85)
Year of First Appearance	2023.0 (1.4)	2023.1 (1.5)	2022.6 (1.6)	2022.9 (1.2)	2022.6 (1.0)
Year of First Operation	2023.1 (1.5)	2023.0 (1.4)	2022.9 (1.6)	2023.4 (1.5)	2023.8 (0.5)
Cumulative Capacity Built (GWh)	13.0	6.3	2.7	2.4	0.5
Cumulative Capacity Operational (GWh)	12.3	6.0	2.7	2.0	0.5
Panel B: Ownership Structure					
By Parent Company					
Share w/ NG Portfolio	0.05	0.07	0.07	0.04	0.14
Share w/ Wind Portfolio	0.43	0.35	0.43	0.67	1.00
Share w/ Solar Portfolio	0.50	0.47	0.57	1.00	1.00
Share Independent	0.50	0.53	0.43	0.00	0.00
By Co-location (ERCOT Designation)					
Share Co-located w/ Natural Gas	0.02	0.05	0.03	0.00	0.00
Share Co-located w/ Solar	0.15	0.13	0.17	1.00	0.00
Share Co-located w/ Wind	0.04	0.00	0.00	0.00	1.00
Panel C: Sample Composition					
# of Battery Units (Operational)	165	75	30	24	7
# of Battery Units (Built)	211	106	33	26	7
# of Counties	70	30	30	16	4
# in Houston Zone	19	19	4	2	0
# in North Zone	30	16	10	14	0
# in South Zone	53	30	12	6	0
# in West Zone	63	10	4	2	7

Notes. This table includes summary statistics of storage units in ERCOT as of December 1, 2024. Mean and counts are reported, with standard deviations in parentheses. Column (1) reports all storage units. Column (2) reports units in the same county as a natural gas generator (used as the treatment dose in the continuous difference-in-differences analysis). Column (3) shows the first storage units entering those counties (used as the binary treatment indicator in the main difference-in-differences analysis). Columns (4) and (5) report storage colocated with solar PV and wind generators, respectively.

Table 2 presents summary statistics of storage units in my sample. Column (1) describes characteristics of all storage units. At the end of the sample period, a total of 9 GW storage capacity (211 units) is built in ERCOT, with 8.1 GW operating commercially (165 units). The average capacity is around 49.3 MW with an average duration of 1.34 hours, indicating that storage in ERCOT typically consists of one- to two-hour duration units, with very few long-duration units. Storage units are relatively evenly distributed across the state, with the highest concentration in the West load zone, followed by the South load zone. Firms that own storage also own wind and solar PV plants, but they are less likely to own natural gas units. As detailed in Table E.1, the

largest storage firm owns 22.8% of storage capacity, while the second- to fifth-largest own between 5.5% and 14.8%.

In Column (2), I present the same statistics for storage units that enter the same county as a natural gas generator. There are a total of 4.4 GW storage capacity built (106 units) with 3.8 GW operating commercially (75 units) in the same county as a natural gas generator. The average capacity and duration mirror the full sample, but because natural gas generators cluster in populated areas, these units are more likely to be located in the Houston and South load zones, and less likely to be in the West. I use this cumulative storage capacity as the treatment dose in the continuous difference-in-differences model. Column (3) describes the first storage units that enter the same county as a natural gas generator, which I use to define the treatment indicator in the main difference-in-differences model.

Columns (4) and (5) present summary statistics for storage colocated with other generators at the same grid node. Storage units are frequently colocated with solar PV plants, primarily in the North load zone, and rarely with wind plants, which are located entirely in the West load zone. Storage and natural gas units rarely colocate at the same node.

Trends in storage operation pattern. I next describe how storage in ERCOT operates by exploring the relationship between storage capacity factor and various market conditions. I define storage capacity factor as the ratio of actual energy charged or discharged to the maximum possible energy charged or discharged for each interval; a positive value indicates charging, while a negative value indicates discharging.

Figure B.4 shows the storage operation pattern by market conditions. The pattern is consistent with the optimal storage behavior of arbitraging price differences. Storage is more likely to charge when LMPs are low, when demand is low, and when renewable generation is abundant. Storage units also charge when the market is more competitive (lower HHI) and less congested (larger market size), while they discharge under more concentrated market conditions and in smaller markets. As documented by [Woerman \(2023\)](#), firms tend to exercise market power in smaller markets. The fact that storage is more likely to discharge in smaller and more concentrated markets suggests that storage entry might affect the competitive dynamics between storage and natural gas generators in these markets, supporting the market-size moderator in Prediction 2.

3.3 Descriptive Statistics on Natural Gas Generators

This section provides descriptive statistics on natural gas generators in ERCOT, detailing their operational patterns prior to significant storage entry and examining the localized market conditions they face. I also document that pre-entry markup varies systematically with generator capacity and local market size.

Table 3 summarizes these operational patterns. My sample consists of 558 natural gas generators (347 non-peakers and 211 peakers), with an average capacity of 366 MW and an average marginal cost of \$28/MWh. Geographically, these generators cluster heavily in the Houston, South,

Table 3: Summary statistics of natural gas generators

	All Generators			Treated Generators			Never-Treated Generators		
	All	12AM –12PM	12PM –12AM	All	12AM –12PM	12PM –12AM	All	12AM –12PM	12PM –12AM
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Outcome Variables (2018–2021)									
LMP (\$/MWh)	28.7 (93.3)	21.7 (49.9)	35.7 (121.7)	29.0 (94.7)	21.9 (51.1)	36.1 (123.4)	27.5 (87.1)	20.9 (44.4)	34.2 (114.5)
Capacity Factor	0.11 (0.18)	0.08 (0.16)	0.13 (0.20)	0.10 (0.17)	0.08 (0.15)	0.12 (0.19)	0.13 (0.20)	0.11 (0.17)	0.16 (0.22)
Probability Marginal	0.07 (0.26)	0.08 (0.27)	0.06 (0.25)	0.08 (0.27)	0.09 (0.28)	0.07 (0.26)	0.05 (0.22)	0.06 (0.23)	0.04 (0.20)
Markup (\$/MWh)	2.13 (74.91)	-1.60 (32.31)	5.86 (100.75)	1.64 (74.34)	-2.02 (31.95)	5.29 (100.04)	4.16 (77.13)	0.12 (33.71)	8.20 (103.59)
Panel B: Market Structure (2018–2021)									
HHI	1011 (1286)	955 (1156)	1068 (1401)	1002 (1326)	942 (1191)	1063 (1447)	1044 (1122)	1003 (1017)	1086 (1217)
Market Size (GW)	86.6 (31.1)	88.1 (29.9)	85.1 (32.2)	88.2 (29.4)	89.7 (28.0)	86.6 (30.7)	80.7 (36.1)	82.0 (35.5)	79.5 (36.7)
# Firms per Market	80.0 (28.0)	81.5 (27.1)	78.4 (28.8)	81.2 (26.9)	82.9 (25.8)	79.6 (27.9)	75.3 (31.2)	76.6 (30.8)	74.1 (31.6)
Panel C: Generator Characteristics									
Heat Rate (MMBtu/MWh)		10.7 (3.0)			10.9 (3.0)			10.2 (3.1)	
CO ₂ Rate (tons/MWh)		0.6 (0.2)			0.6 (0.2)			0.5 (0.2)	
NO _x Rate (lbs/MWh)		1.151 (1.967)			1.122 (1.746)			1.247 (2.564)	
SO ₂ Rate (lbs/MWh)		0.010 (0.044)			0.011 (0.050)			0.007 (0.006)	
Marginal Cost (\$/MWh)		31.74 (11.35)			32.36 (10.43)			29.75 (13.80)	
Avg. Capacity (MW)		366 (446)			350 (453)			419 (420)	
Panel D: Sample Composition									
# of Unique Generators		558			427			131	
# Peaker (SC)		211			177			34	
# Non-Peaker (CC + ST)		347			250			97	
# of Counties		50			32			23	
# in Houston Zone		220			196			24	
# in North Zone		136			68			68	
# in South Zone		152			135			17	
# in West Zone		50			28			22	

Notes. This table presents summary statistics of natural gas generators in ERCOT. Each entry reports the mean and counts of natural gas plants, with standard deviations in parentheses. Panels A–C report statistics computed over 2018–2021, the period before significant storage build-out in ERCOT. Panel D reports sample composition over the full estimation period (2018–2024). Columns (1)–(3) report statistics for all natural gas plants; Columns (4)–(6) for plants sharing the same county with storage; Columns (7)–(9) for plants located not in the same county with any storage. Panel A describes operation patterns within a day. Panel B describes market conditions. Panel C describes plant characteristics and marginal costs. Panel D describes sample composition.

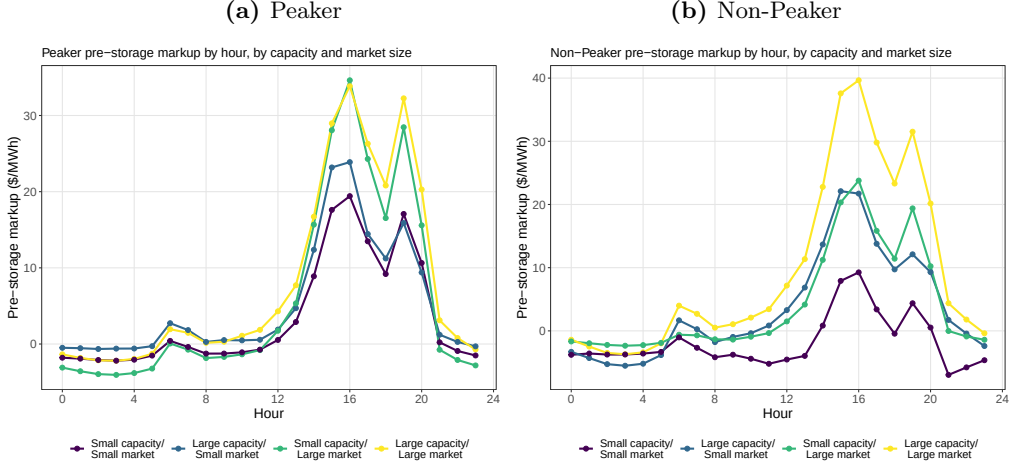
and North load zones. Columns (1)–(3) detail daily market outcomes: later hours (12 PM–12 AM) consistently feature higher LMPs, capacity factors, and markups compared to early hours. Panel B links these elevated afternoon and evening markups directly to tighter market conditions, specifically, smaller local market sizes and higher concentration (HHI) that enhance generator market power.

Columns (4)–(6) present identical statistics for generators that share the same county with storage units, while Columns (7)–(9) present statistics for generators located in different counties from any storage units. Because the operational patterns and market conditions of these groups closely mirror the overall sample, the treatment group in my difference-in-differences design is highly representative of ERCOT natural gas generators. The LMP spread between early and late periods does not differ significantly between the treatment and control groups, alleviating concerns regarding endogenous storage siting. Detailed balance tables comparing treated and control groups are provided for peakers (Appendix Table E.2) and non-peakers (Appendix Table E.3).

Trends in natural gas generator markups. Figure B.5 presents the diurnal markup pattern for peakers and non-peakers over my sample period. Both technology types exhibit their highest markups during the late afternoon and evening ramp, aligning with periods of peak demand and high prevalence of transmission congestion. Markups rise steeply between hour 16 and hour 20, reflecting an increase in market power during peak hours. The year-on-year overlay shows that markups during the evening ramp have generally increased over the study period. A naive interpretation of these aggregate trends might suggest that the concurrent storage build-out drives up markups. This visual steepening is likely dominated by the rapid expansion of solar capacity, which systematically suppresses midday margins and concentrates peak market power into the early evening (Bushnell and Novan, 2021; Jha and Leslie, 2025). These aggregate trends thus conflate the effect of storage entry with concurrent changes in renewable penetration and demand growth.

Beyond the aggregate diurnal pattern, market power in the pre-storage period also varies across generators within local markets. Figure 3 plots the average markup by hour over 2021 and earlier, grouped by whether the generator’s capacity is above the technology-specific median and whether its local market size is above the median. Two patterns emerge from the group averages. Generators in larger markets tend to earn higher markups during the evening ramp than generators in smaller markets. Within a given market-size group, capacity is associated with smaller differences in markup, particularly for peakers. These grouped patterns are broadly consistent with the markup rule in Equation (4): generators with more residual demand to serve and facing a less elastic competitive fringe sustain higher markups. Both the aggregate trends and the cross-sectional variation motivate the empirical strategy in Section 4.

Figure 3: Markup by hour in the pre-storage period, by capacity and market size



Notes: These figures show the average markup by hour during 2021 and earlier, before significant storage build-out in ERCOT. Generators are grouped by whether their capacity is above the technology-specific median and whether their local market size is above the median. Markups are capacity-weighted averages within each group.

4 Empirical Strategy

This section details the empirical strategy to estimate the effect of storage entry on natural gas generator markups. Following [Baker et al. \(2026\)](#), I structure this discussion around the target parameter, treatment definition and estimation, and threats to identification.

4.1 Two-Stage Difference-in-Differences Design

Target parameter. Because operational characteristics and intra-day competitive dynamics vary, I estimate the treatment effect separately for peakers and non-peakers across each hour of the day. The target parameter is the average treatment effect on the markups of generator type $g \in \{\text{Peaker}, \text{Non-Peaker}\}$ at hour $h \in \{0, \dots, 23\}$:

$$(10) \quad \tau_{gh} = E[Y_{it}(1) - Y_{it}(0) | \text{Hour}_t = h, \text{Tech}_i = g],$$

where $Y_{it}(1)$ and $Y_{it}(0)$ are the potential markups for generator i at time t with and without storage entry.

Market definition and treatment timing I define the local market at the county level. A gas generator is treated on and after the Commercial Operation Date (COD): the first date a storage unit within the same county is awarded a positive quantity in the real-time market. While the county boundary is an approximation, it captures highly localized competitive effects, and storage units may frequently interact with generators outside their immediate county (see [Figure F.26](#)). [Figure F.27](#) illustrates the distribution of the time gap between a unit's initial appearance in the SCED data (during testing and commissioning) and its actual COD, which has a median of 204

days.

Estimation strategy. To reduce high-frequency noise and computational burden, I aggregate the 15-minute interval data to the generator by week-of-sample by hour level.²¹ I estimate the following specification using a two-stage difference-in-differences approach (Gardner et al., 2024):

$$(11) \quad Y_{it}^{gh} = \alpha_i^{gh} + \mu_t^{gh} + \beta^{gh} D_{it} + \mathbf{X}_{it} \theta^{gh} + \epsilon_{it}^{gh},$$

where Y_{it} is the average realized markup or average generation, α_i are generator fixed effects, and μ_t are week-of-sample fixed effects. The treatment indicator $D_{it} = \mathbb{1}[Storage_{m(i),t} > 0]$ equals one after the first storage COD in county $m(i)$. Regressions are capacity-weighted with standard errors clustered at the county level.

The vector $\mathbf{X}_{it}$ includes time-varying controls to isolate the competitive effect of storage from concurrent market shifts. Controls include cubic polynomials for system-wide demand and ambient temperature, alongside linear terms for local load, local uncurtailed solar and wind generation, and Henry Hub natural gas prices. Month fixed effects absorb residual seasonality.²² To prevent anticipation effects (e.g., threat-of-entry behavior outlined by Goolsbee and Syverson, 2008 and Ellison and Ellison, 2011) from biasing β^{gh} , I include an indicator for the interim period between a storage unit’s first SCED appearance and its COD.

Estimation proceeds in two stages: I first estimate the fixed effects and control coefficients using untreated observations ($D_{it} = 0$), and subsequently regress the first-stage residuals on the treatment indicator to obtain β^{gh} . This accommodates potential biases from staggered treatment timing and heterogeneous effects. I exclude the periods of Winter Storms Uri and Elliott from the main sample.²³

The key coefficient of interest, β^{gh} , captures the average treatment effect on treated (ATT) natural gas generators for each generator type g at hour h . Based on Prediction 1, I expect the sign of β^{gh} to be negative during the storage discharge period, which based on observed storage operation patterns include the morning ramp (6 AM–7 AM) and the evening ramp (6 PM–9 PM). The sign during the charging period is weakly positive.

Identification. Identification of the ATT relies on the conditional parallel trends assumption:

$$(12) \quad \mathbb{E}[\epsilon_{it}^{gh} | D_{it}, \mathbf{X}_{it}, \alpha_i^{gh}, \mu_t^{gh}] = 0.$$

²¹Aggregation is common in empirical settings with high frequency data (Demirer and Karaduman, forthcoming; Cicala, 2022; Ghanem and Smith, 2021; Gonzales, Ito and Reguant, 2023). The level of aggregation chosen reflects the expected variation in the target parameter. I run the regression using 15-minute interval data as a robustness check and find similar results.

²²Since I run this regression separately for each hour, the fixed effects and control coefficients are allowed to vary by technology type and hour.

²³Including these extreme weather events and controlling for them via event-specific generator fixed effects yields qualitatively similar results.

To provide a falsification test for this assumption, I estimate an the following event-study analog of Equation (11):

$$(13) \quad Y_{it}^{gh} = \alpha_i^{gh} + \mu_t^{gh} + \sum_{s \in \{-3, 3\}} \delta_s^{gh} D_{i(t'+s)} + \mathbf{X}_{it} \theta^{gh} + \epsilon_{it}^{gh},$$

where $D_{i(t'+s)}$ is an indicator variable relative to a year prior to the first storage COD in the same county.²⁴ The lead coefficients test for pre-trends, while the lags capture dynamic treatment effects.²⁵ Because actual COD occurs well after initial SCED appearance, and SCED data is published with a 60-day lag, anticipatory adjustments by gas generators prior to the testing phase are highly unlikely.

The stable unit treatment value assumption (SUTVA) requires that the treatment effect of storage entry does not affect generators in other counties. This assumption might be violated if there are spillover effects to generators in other counties. To address this concern, I employ a difference-in-differences specification that allows for spillover effects following Butts (2023a) and Butts (2023b) in Section 5.1.2.

Extensions. The binary difference-in-differences design identifies the average treatment effect of storage entry on natural gas generator markups, pooling across generators exposed to different amounts of storage capacity. Two extensions build on this baseline. First, I estimate a saturated version of Equation (11) that interacts the treatment indicator with generator capacity and local market size, to test whether the effect scales with the two features of the local competitive set identified in Prediction 2. Second, I replace the binary treatment indicator with cumulative storage capacity in the local market, which scales the treatment intensity and allows me to identify heterogeneous effects across storage types. Both extensions are presented in Section 5 together with their results.

5 Results

I organize the empirical results around the theoretical predictions developed in Section 2.2. Section 5.1 presents the average effect of storage entry on natural gas generator markups, testing Prediction 1. Section 5.2 decomposes the heterogeneous effects by generator capacity and local market size, testing Prediction 2 and showing that the effect is concentrated among smaller generators across both technology types. Section 5.3 uses the continuous treatment specification to sharpen the magnitudes and to test Prediction 4 on the ownership dimensions of storage-side heterogeneity. Section 5.4 turns to the mechanism, testing Prediction 3 on heat rate and separating

²⁴ $D_{i(t'-1)}$ is the omitted indicator variable. Specifically, $D_{i(t'-2)}$ is an indicator for 13 to 24 months before storage entry, $D_{i(t')}$ is an indicator for 0 to 12 months post storage entry, $D_{i(t'+1)}$ is an indicator for 13 to 24 months post storage entry, and so on.

²⁵I also perform an event study using leads and lags of the entry indicator relative to the month prior to COD. Even though the estimates are more noisy due to seasonality and periods of very high markup values, the estimates are quantitatively similar.

the market power channel from the dynamic efficiency channel.

5.1 Average Effect on Natural Gas Generator Markups

Prediction 1 states that local storage entry reduces the markup of natural gas generators during the storage discharge period, and weakly raises it during the charge period. I test the discharge-period component by estimating Equation (11) separately for peakers and non-peakers on each hour of the day. Under this prediction, β^{gh} should be negative during the discharge hours identified in Figure 1b, which include the evening ramp (hours 17–20) and the morning ramp (hours 6–7).

5.1.1 Peaker and Non-Peaker Results

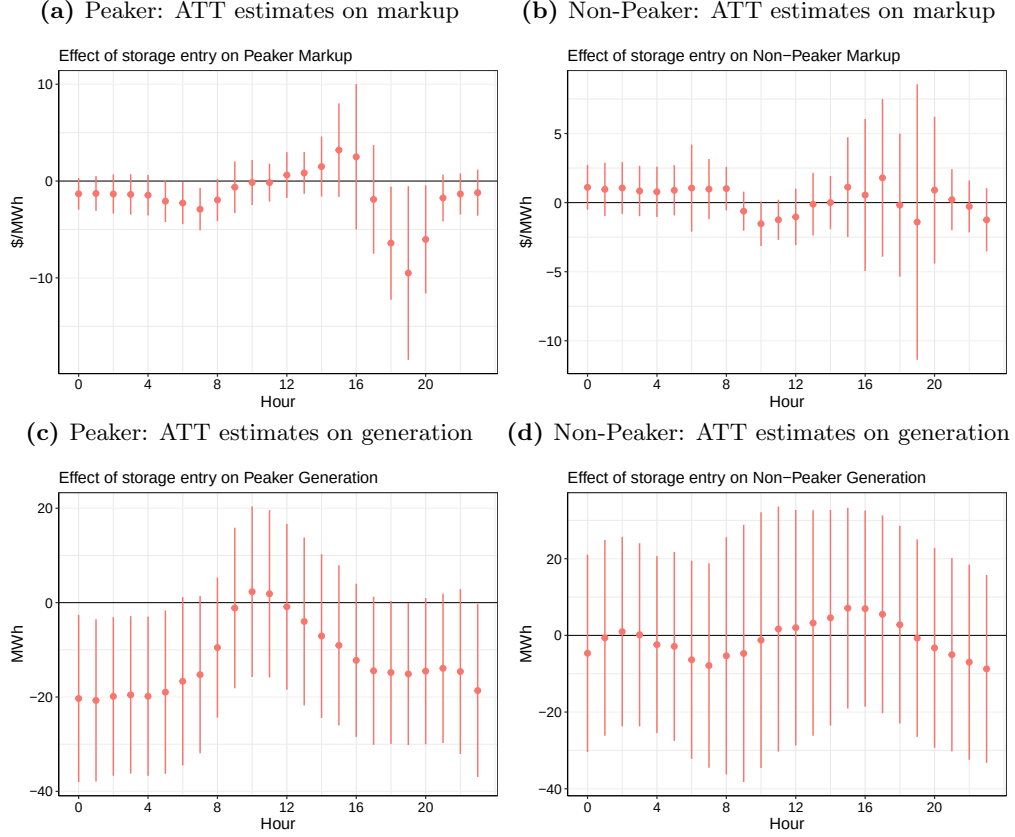
Table 4: ATT estimates during ramp hours

Hour	Peaker		Non-Peaker	
	Markup (\$/MWh) (1)	Generation (MWh) (2)	Markup (\$/MWh) (3)	Generation (MWh) (4)
Panel A: ATT estimates				
6 AM	−2.3** (1.1)	−16.7* (9.1)	1.1 (1.6)	−6.4 (13.2)
7 AM	−2.9*** (1.1)	−15.3* (8.5)	1.0 (1.1)	−7.9 (13.6)
6 PM	−6.4** (3.0)	−14.8* (7.7)	−0.2 (2.6)	2.8 (13.2)
7 PM	−9.5** (4.6)	−15.1** (7.7)	−1.4 (5.1)	−0.7 (13.1)
8 PM	−6.0** (2.9)	−14.5* (7.9)	0.9 (2.7)	−3.3 (13.3)
Panel B: Pre-storage baseline mean				
6 AM	2.8	24.3	4.6	324.1
7 AM	3.0	25.0	4.5	334.8
6 PM	33.4	46.1	45.5	387.1
7 PM	67.7	45.9	86.4	388.3
8 PM	39.6	44.2	48.9	383.3

Notes. This table reports point estimates and standard errors (in parentheses) of β^{gh} from estimating Equation (11) for peakers in columns (1)–(2) and non-peakers in columns (3)–(4), during the morning ramp (6–7 AM) and evening ramp (6–8 PM). Panel A reports ATT estimates with standard errors clustered at the county level in parentheses. Panel B reports capacity-weighted pre-entry means of the outcome by generator type and hour, computed over the 12 months immediately preceding each generator’s treatment date. The full set of hourly estimates are shown in Figure 4. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4 summarizes the ATT estimates for peakers and non-peakers during the ramp hours, and Figure 4 shows the full hourly profile. For peakers, storage entry erodes markups by roughly \$2.3 to \$2.9/MWh during the morning ramp (6–7 AM) and by \$6 to \$9.5/MWh during the evening ramp (6–8 PM), as reported in columns (1)–(2) of Table 4. Against the pre-entry baselines in Panel B of the table, these represent more than an 80% reduction in the morning ramp, from a small baseline of \$2.8 to \$3/MWh, and a more modest 14 to 19% reduction in the evening ramp, from a

Figure 4: Main results for Peaker and Non-Peaker generators



Notes. Figures 4a and 4b present the coefficient estimates of β^{gh} from estimating Equation (11) for peakers ($g = \text{Peaker}$) and non-peakers ($g = \text{Non-Peaker}$), with markup as the outcome. Figures 4c and 4d show the analogous estimates with generation as the outcome. The line shows the 95% confidence interval clustered at the county level. The peaker sample includes 177 treated generators and 34 never-treated generators; the non-peaker sample includes 250 treated generators and 97 never-treated generators. Balance tables are reported in Appendix Tables E.2 and E.3.

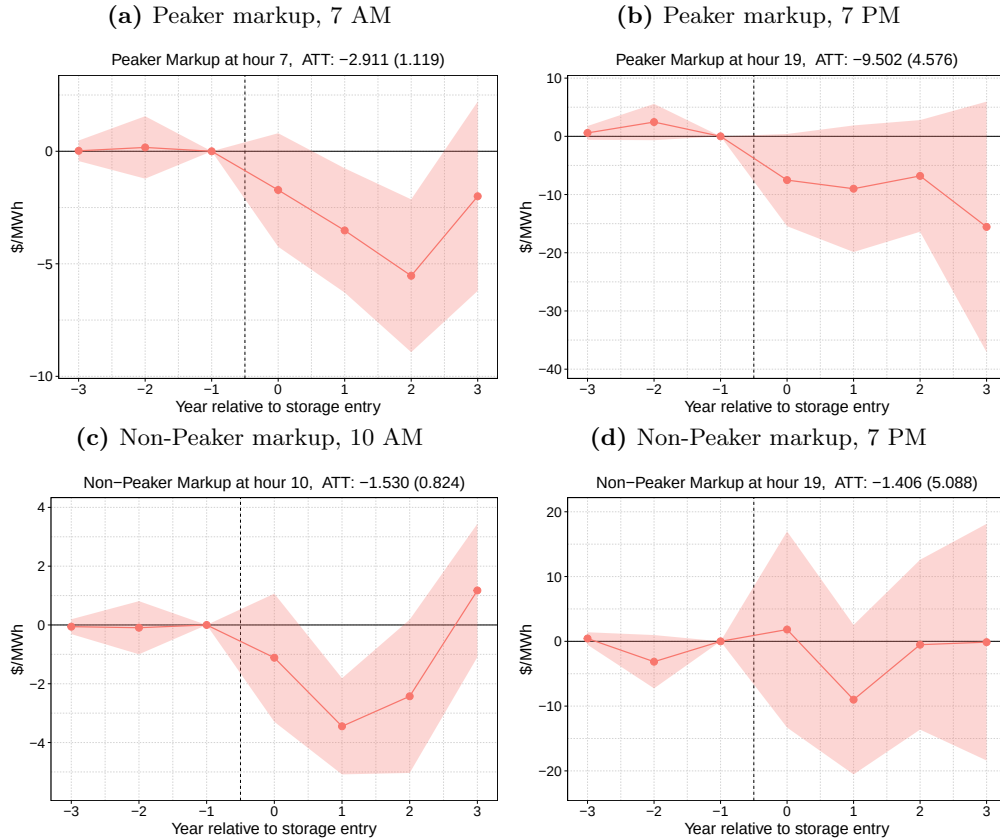
larger baseline of \$33 to \$67/MWh. Figure 4a shows that these effects are concentrated in exactly the ramp hours; estimates in other hours are indistinguishable from zero. These are the hours when peakers exercise market power, as Figure 3 documents.

The non-peaker results, in columns (3)–(4) of Table 4 and Figure 4b, are less immediately striking: the point estimates are negative during the evening ramp but statistically indistinguishable from zero across most hours, with only the 10 AM estimate marginally significant at a reduction of about \$1.5/MWh. Taken at face value, these estimates would suggest that storage entry has no meaningful effect on non-peaker markups. I show in Section 5.2 that this pooled null reflects off-setting effects across capacity size rather than an absence of effect. Smaller non-peakers experience markup reductions similar in magnitude to smaller peakers, while larger non-peakers are unaffected, and the two average out in the pooled specification. This pattern is anticipated by Prediction 2: the storage effect scales with generator capacity and local market size, so the pooled non-peaker sample averages across heterogeneous configurations and masks the within-group heterogeneity.

Figures 4c and 4d show the effect on generation. Storage displaces around 20 MWh of peaker output overnight (11 PM–5 AM), but its effect during the morning and evening ramp hours is not statistically significant. The non-peaker generation response is close to zero across all hours in the pooled specification. Together with the markup results, this pins down the mechanism in the peaker case: in the ramp hours the markup falls with modest decrease in output, so the effect runs primarily through the slope of residual demand E_t rather than the residual quantity q_{ft} .²⁶

The event study estimates for peaker markup at 7 AM and 7 PM are presented in Figure 5. The lead coefficients are not statistically significant, which supports the validity of the parallel trends assumption. The lag coefficients show a qualitatively similar effect beyond the first year of storage entry, suggesting that the competitive effect on peaker generators has lasting impacts on the market dynamics. Considering multiple entries of storage units, the event study estimates embed a cumulative effect of storage entry within a county.

Figure 5: Event study estimates: Peaker and Non-Peaker markup



Notes. These figures present the coefficient estimates of $\delta_s^{g,h}$ from estimating the event study in Equation (13) for peakers ($g = \text{Peaker}$, Figures 5a–5b) and non-peakers ($g = \text{Non-Peaker}$, Figures 5c–5d). The line shows the 95% confidence interval clustered at the county level. Event study estimates for other hours and monthly-bin versions are presented in Appendix Figures F.29–F.34.

²⁶To confirm that this is purely an intensive margin effect, I estimate the same specification using an indicator of whether the generator is running in Appendix D.1 and find that the estimated effect is not statistically significant.

Summary of main results The average effect results support the discharge-period component of Prediction 1 for peakers: markups fall in the exact hours storage discharges, while peaker output is only modestly displaced. The mechanism therefore operates primarily through the residual demand slope E_t rather than the residual quantity q_{ft} . The pooled non-peaker average is not significant, but as I show in the next section, this masks a substantial effect for smaller non-peakers. The charge-period component of Prediction 1 receives no support in the pooled binary specification for either technology. I return to whether the charge-period effect emerges under continuous treatment in Section 5.3.

The insignificant markup estimates in some hours do not represent precise nulls as the confidence intervals includes economically meaningful effects. One explanation is that heterogeneity in the treatment effect is masked in the pooled estimates, which I examine directly through the heterogeneity analysis in the next section. A second explanation is that the binary treatment indicator obscures substantial variation in the capacity of the first storage edit, injecting noise into the estimates. This motivates the continuous treatment specification, which I turn to in Section 5.3.

5.1.2 Robustness

I perform a battery of robustness checks to examine the sensitivity of the main results. The results are robust to using 15-minute level data, controlling for spillover effects, and using alternative staggered difference-in-differences estimators. Details are reported in the Appendix.

Robustness to allowing for spillover effect In the main specification, I define the local market as the county, which is an approximation. To test the robustness of my results, I follow Butts (2023a) and Butts (2023b) and use the frequency that the storage units interact with natural gas generators in the same local market pre-entry as a notion of distance. This allows generators outside the county to receive spillover effects of storage if they are rarely separated by a binding transmission constraint. Details are provided in Appendix C.2. Overall, the results suggest that while there may be some spillover effects, the main competitive effect of storage entry is concentrated on generators in the same county.

Robustness to using 15-minute level data I test the robustness of my results to using 15-minute level data. The benefit of higher-frequency data is that it allows a more detailed examination of the treatment effects over time. I re-estimate Equation (11) without control variables and compare to my main specification. The results, reported in Appendix C.1, are consistent with my main results indicating that my aggregation approach does not suffer from attenuation bias.

Robustness to alternative staggered difference-in-differences estimators I test the robustness of my results to alternative estimators designed for staggered difference-in-differences settings (Callaway and Sant’Anna, 2021). These estimators account for potential heterogeneity

in treatment effects across cohorts and over time. The results, reported in Appendix C.3, are consistent with my main results.

5.2 Heterogeneity by Generator Capacity and Local Market Size

Prediction 2 states that the market power effect scales with two moderators: local market size and the generator’s share of local capacity. Under this prediction, the effect should be largest for smaller generators in smaller markets and vanish for larger generators in larger markets. Because these moderators depend on the generator’s position in the local competitive set, I expect the same pattern should hold for both peakers and non-peakers.

I test these predictions by estimating a variation of Equation (11) that interacts the treatment indicator with two binary indicators. Let Z_i^C be an indicator equal to one if generator i ’s capacity is above the median capacity within its Type g , and zero otherwise. Let Z_i^M be an indicator equal to one if generator i is located in a county whose local market size is above the median, and zero otherwise.²⁷ Both are fixed generator-level characteristics. Specifically, I estimate:

$$(14) \quad Y_{it}^{gh} = \alpha_i^{gh} + \mu_t^{gh} + \beta^{gh} D_{it} + \gamma_C^{gh} D_{it} \times Z_i^C + \gamma_M^{gh} D_{it} \times Z_i^M \\ + \gamma_{CM}^{gh} D_{it} \times Z_i^C \times Z_i^M + \mathbf{X}_{it} \theta^{gh} + \epsilon_{it}^{gh},$$

where D_{it} is the treatment indicator. The level main effects of Z_i^C and Z_i^M are absorbed by the generator fixed effects α_i^{gh} . Saturating the treatment in this way lets a single regression identify the effect in each of the four capacity-by-market-size group, estimated off a common set of controls and never-treated comparison units, while the interaction γ_{CM}^{gh} provides a direct test of whether the large-capacity, large-market cell departs from what the two single splits predict.

Prediction 2 maps onto the signs of these combinations for the markup outcome. The baseline small-capacity, small-market effect is expected to be negative and largest in magnitude: these generators face the strongest competitive pressure from storage. A larger local market dampens this effect ($\gamma_M^{gh} > 0$), because storage of a given size represents a smaller share of a larger competitive set. A larger own capacity share also dampens the effect ($\gamma_C^{gh} > 0$): a generator holding a large share of local capacity remains closer to pivotal after storage entry. The two attenuating channels reinforce one another in the large-capacity, large-market cell, whose total effect should be closest to zero. The sign of γ_{CM}^{gh} is *a priori* ambiguous: positive if the capacity and market-size channels are super-additive, negative if they are partial substitutes.

5.2.1 Heterogeneity Results

Figure 6 presents the heterogeneity results focusing on the evening-ramp hours for both peakers and non-peakers. In this section, I discuss the binary treatment estimates in Figures 6a and 6b, and return to the continuous treatment results in Figures 6c and 6d in Section 5.3.

²⁷Median peaker capacity is 90 MW, while median non-peaker capacity is 350 MW. I use county average market size for 2021 and earlier, the period before major storage entry in Texas.

Markup Figure 6a shows the binary-treatment markup effect. For peakers, the reduction is concentrated in the two small-capacity group: small-capacity peakers in small markets and small-capacity peakers in large markets both experience statistically significant markup reductions of similar magnitude, while large-capacity peakers in either market size are essentially unaffected. Non-peakers show the same pattern: the markup reduction is concentrated in the small-capacity group and absent for large-capacity generators. This finding confirmed that the pooled non-peaker null in Figure 4b masks a substantial effect for small non-peakers of similar magnitude to small peakers, consistent with the technology-agnostic markup rule in Equation (4). Generator capacity is the dominant moderator in this specification: the reduction in market rent operates through the smaller, competitive-margin generators of both technologies rather than through the larger units. The complete hourly estimates for markup shown in Panel A of Appendix Figures D.16–D.17 documents that the effect is concentrated in the evening ramp hours where storage discharges.

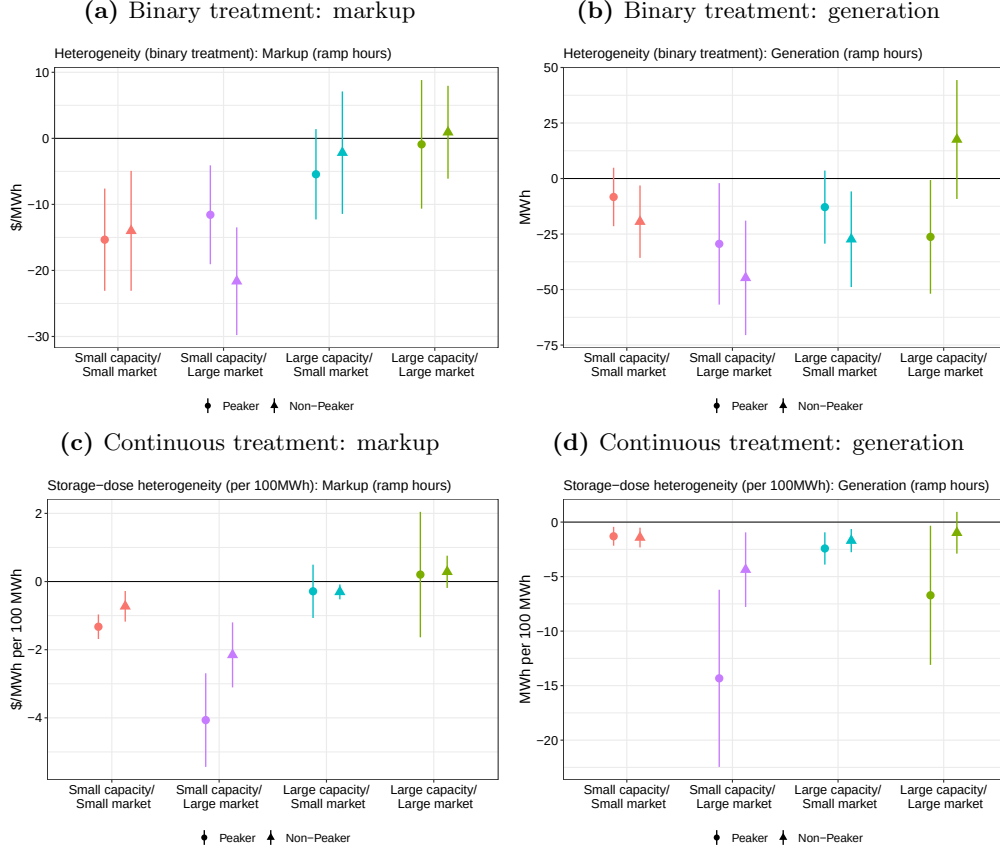
Generation Figure 6b presents the effect on generation. Peaker generation is displaced by storage entry across all groups, but the point estimates is only statistically significant for both small-capacity peakers in larger markets. Interestingly, non-peakers generation is displaced by roughly a similar magnitude for three of the four cells, except for large-capacity non-peakers in large markets which are unaffected. Large-capacity non-peakers in large markets are the only group that is not displaced by storage, consistent with the prediction, as generator in this group still holds a dominant share of capacity after entry of relatively smaller capacity storage. The complete hourly estimates for generation in Panel B of Appendix Figures D.16–D.17 show that this displacement extends from the evening ramp through the end of the morning ramp for the affected groups, consistent with the storage discharge window.

Summary of heterogeneity results The heterogeneity results support Prediction 2. The markup reduction is concentrated in small-capacity generators of both technologies, confirming that the market power channel is governed by capacity and by whether the generator is on the margin. The generation displacement is broader: storage substitutes for peaker output across all groups, and for non-peaker output in every cell except the large-capacity, large-market group. Two implications follow. First, welfare gains through reduced market power are concentrated on the smaller generators of both technologies. Second, the environmental benefit from displaced fossil generation is broadly distributed across the peaker fleet and across most of the non-peaker fleet, sparing only the largest units in the largest markets.

5.3 Continuous Treatment Results

The binary treatment indicator obscures substantial variation in the size of storage entry, treating a 10 MW facility identically to a 100 MW facility. This has two consequences. First, it injects noise into the pooled effect estimates, contributing to the wide confidence intervals in Section 5.1. Second, it cannot separate the effect by storage type, because a single indicator cannot represent

Figure 6: Heterogeneity results by capacity and market size, ramp hours



Notes. These figures show the four heterogeneous effects, pooled across the evening ramp hours (6–8 PM), constructed from the linear combinations: small capacity, small market: β^{gh} ; large capacity, small market: $\beta^{gh} + \gamma_C^{gh}$; small capacity, large market: $\beta^{gh} + \gamma_M^{gh}$; large capacity, large market: $\beta^{gh} + \gamma_C^{gh} + \gamma_M^{gh} + \gamma_{CM}^{gh}$, from estimating Equation (14) (Figures 6a and 6b) and the equivalent variant of Equation (15) (Figures 6c and 6d) for $g = \text{Peaker, Non-Peaker}$. Standard errors are obtained by the delta method and clustered at the county level. The line shows the 95% confidence interval. Hourly estimates are presented in Appendix Figures D.16–D.17 for binary treatment and in Appendix Figures D.18–D.19 for continuous treatment.

multiple simultaneously entering storage types with different capacities. Both concerns motivate a continuous treatment specification that scales the effect by cumulative storage capacity.

I substitute the binary indicator in Equation (11) with $\text{Storage}_{m(i),t}$, the cumulative capacity of storage in the county at time t , expressed in 100 MW units:

$$(15) \quad Y_{it}^{gh} = \alpha_i^{gh} + \mu_t^{gh} + \tilde{\beta}^{gh} \text{Storage}_{m(i),t} + \mathbf{X}_{it} \theta^{gh} + \epsilon_{it}^{gh}.$$

The saturated heterogeneity specification in Equation (14) carries over analogously, with $\text{Storage}_{m(i),t}$ replacing D_{it} . Identification relies on the same conditional parallel trends assumption, with the addition that the level of storage capacity in a county is as good as random conditional on observed covariates and fixed effects. Estimation proceeds in two stages as before, with the first stage done on subsample where $\text{Storage}_{m(i),t} = 0$.

Interpretation of the continuous treatment estimand Following [Gardner et al. \(2024\)](#), and building on the result of [Yitzhaki \(1996\)](#), the two-stage difference-in-differences estimator with a continuous treatment identifies a positively weighted average of the derivatives of the causal response function. Formally, letting $Y_{it}(d)$ denote the potential outcome for generator i at time t under cumulative county-level storage capacity d (in 100 MW units), the estimator recovers

$$(16) \quad \tilde{\beta} = \mathbb{E} \left[\sum_d w(d) \cdot \frac{\partial \mathbb{E}[Y_{it}(d) - Y_{it}(0) \mid D_{it} = d]}{\partial d} \right],$$

where $w(d) \geq 0$ are positive weights that reflect the empirical distribution of treatment intensity among treated generators. My specification therefore recovers a locally linear approximation to the derivative of the causal response function, averaged across observed doses. The estimated coefficient captures the marginal effect of an additional 100 MW of storage on markup, averaged across the observed distribution of cumulative capacity in treated counties.

Two features of this estimand warrant emphasis. First, $\tilde{\beta}$ is a marginal effect rather than a total effect: it captures the slope of the causal response function averaged over observed doses, not the impact of moving from zero to 100 MW of storage. Second, cumulative storage capacity in my setting is time-varying by construction, as additional storage facilities enter each county over the sample period. The recent continuous difference-in-differences literature, including [Callaway, Goodman-Bacon and Sant’Anna \(2024a,b\)](#), primarily develops identification results under the assumption that the dose is a time-invariant unit characteristic, and does not currently accommodate time-varying doses of the form used here. I therefore rely on the above interpretation to characterize $\tilde{\beta}$ as a weighted average of marginal effects.

Because cumulative storage capacity is mechanically zero for all treated generators prior to their first neighboring storage entry, the pre-treatment coefficients in the dose-interacted event study are not separately identified from the reference period. I therefore rely on the binary event study in [Section 5.1](#) as the primary parallel-trends diagnostic.

5.3.1 Continuous Heterogeneity by Capacity and Market Size

The continuous treatment specification sharpens the heterogeneity estimates from [Section 5.2](#) by scaling the effect by the size of storage entry, which allows a direct comparison against the pre-storage markup base. I refer back to [Figures 6c and 6d](#) in [Figure 6](#), which present the continuous heterogeneity results pooled across the evening ramp hours for both peakers and non-peakers.

Markup [Figure 6c](#) shows the continuous-treatment markup effect. The pattern from the binary specification carries through: the reduction is concentrated in the two small-capacity group for both peakers and non-peakers, while large-capacity generators of both technologies are essentially unaffected. Scaling by the storage dose reveals a difference across the small-capacity group that the binary specification did not resolve. For smaller peakers, the evening-ramp reduction is substantially larger in large markets than in small markets, at roughly \$4/MWh per 100 MW of storage versus

about \$1.4/MWh per 100 MW (the largest effect at 7PM of \$10/MWh versus \$3/MWh per 100 MW storage). For smaller non-peakers, the same ordering appears: about \$2/MWh per 100 MW of storage in large markets against \$0.7/MWh per 100 MW in small markets (the largest effect at 7PM of \$5/MWh versus \$2.5/MWh per 100 MW storage).

The larger evening-ramp reduction in larger markets may appear to run against Prediction 2, which anticipates a stronger effect in smaller markets. Part of this pattern reflects differences in the pre-storage markup rather than differences in the competitive pressure that storage exerts. As Figure 3 shows, smaller peakers in larger markets earn a substantially higher pre-storage markup during the evening ramp, on the order of \$28/MWh, than their counterparts in smaller markets, at roughly \$17/MWh. Scaling the reductions by these baselines narrows the gap without closing it: the reduction is about 40% of the pre-storage markup for smaller peakers in larger markets, against under 20% for those in smaller markets. Storage therefore compresses a larger share of the pre-storage markup in larger markets, consistent with the tighter local supply conditions during the ramp that Prediction 2 emphasizes.

The complete hourly estimates for markup are shown in Appendix Figures D.18a and D.19a, which reveal a second finding not visible in the binary specification: a positive markup effect during the nighttime and post-morning-ramp charging periods for smaller peakers in larger markets, of about \$1/MWh per 100 MW of storage, and during the charging period preceding the evening ramp for smaller peakers in smaller markets. This is the first evidence in support of the charge-period component of Prediction 1, which the binary specification could not detect.

Generation Figure 6d presents the effect on generation. The binary pattern carries through: peaker generation is displaced across all groups, while non-peaker generation is displaced in every group except the large-capacity, large-market group. Scaling by the storage dose sharpens the magnitudes. For smaller peakers in larger markets, generation falls by 10 to 20 MWh per 100 MW of storage; for smaller peakers in smaller markets the effect is close to zero and precisely estimated, while larger peakers in smaller markets show a modest negative effect. For non-peakers, generation falls by roughly 3 MWh per 100 MW of storage in the three affected groups. Larger non-peakers in larger markets remain unaffected, consistent with their dominant share of local capacity after the entry of relatively smaller storage capacity.

The generation displacement without a markup response for larger non-peakers in smaller markets is consistent with a pure quantity-displacement effect: storage takes their volume without materially affecting their pricing power. The complete hourly estimates for generation in Appendix Figures D.18b and D.19b show that this displacement extends from the evening ramp through the end of the morning ramp for the affected groups, consistent with the storage discharge window.

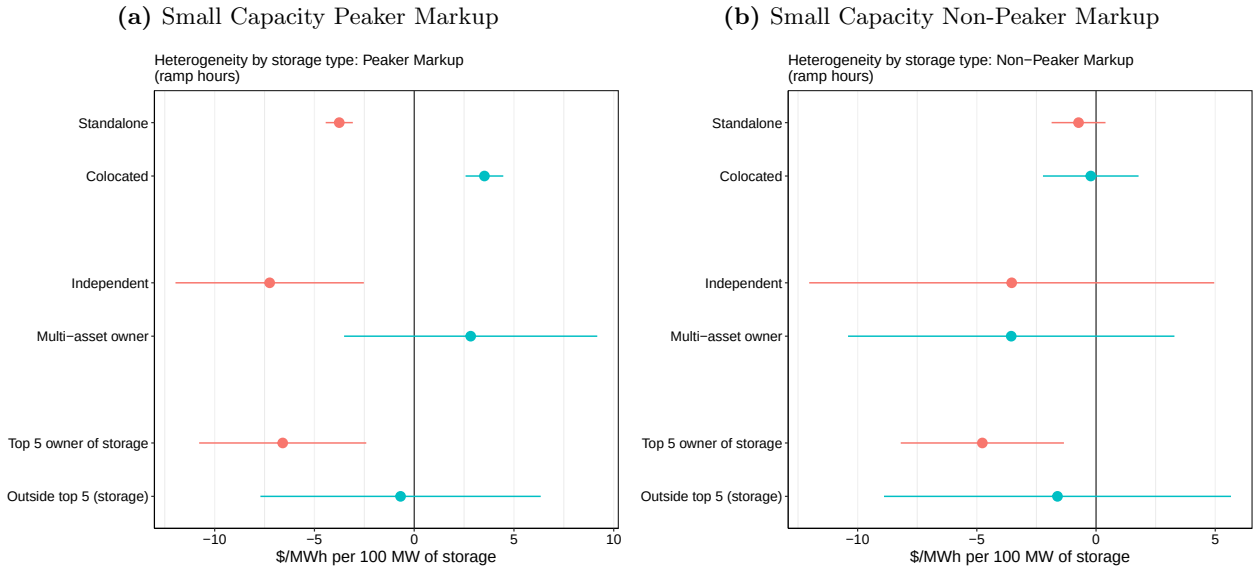
Summary of continuous heterogeneity The continuous specification confirms and sharpens the qualitative pattern from the binary heterogeneity results, while adding two new findings. First, the charge-period effect anticipated by Prediction 1 emerges under continuous treatment for smaller

peakers. Second, scaling by the pre-storage markup base clarifies that smaller peakers in larger markets experience the largest proportional markup compression during the evening ramp.

5.3.2 Heterogeneity by Storage Characteristics

Prediction 4 states that the disciplining effect on natural gas markups is stronger when storage is dispatched more competitively, through two forces: cooptimization with other assets and storage-side market power. I test these implications using three ownership contrasts: standalone versus colocated storage (physical cooptimization), independent versus multi-asset owner (financial cooptimization), and top-5 versus outside top-5 by storage capacity (storage-side market power). I focus on smaller generators and pool across the evening ramp hours (6–8 PM). Results are shown in Figure 7 with Peaker in Panel A and Non-Peaker in Panel B.

Figure 7: Heterogeneity by storage characteristics



Notes. These figures show the coefficient estimates from estimating a variant of Equation (15) with two continuous treatment variables per contrast, each representing cumulative capacity in 100 MW units for the corresponding storage type. Estimates are pooled across the evening ramp hours (6–8 PM) and restricted to small-capacity generators. The line shows the 95% confidence interval clustered at the county level.

Standalone versus colocated storage Standalone storage delivers a larger markup reduction than colocated storage, consistent with the physical cooptimization channel in Prediction 4. Colocated storage is dispatched jointly with a solar host, so its discharge tracks the solar profile rather than the peaker markup. Standalone storage has no such tie and discharges when prices are high, which is exactly when the marginal peaker exercises its markup. For small non-peakers, the two are not statistically distinguishable. Since small non-peakers are largely off the margin during the evening ramp, the mechanism has limited room to show up in this window.

The positive point estimate for colocated storage on peaker markups admits two explanations,

both operating through the net output of the colocated facility. First, solar output collapses during the ramp, so if the battery is used to firm the solar host, the net facility contributes less energy than a standalone battery of the same MW rating would. Second, colocated storage may charge from its solar host in the late afternoon, extending its charging window into the ramp; charging raises residual demand and mechanically raises peaker markups.²⁸ Both point to the same conclusion: colocated storage reshapes the solar profile rather than competing against natural gas directly.

Independent versus multi-asset owner The same logic operates at the ownership level, forming the financial cooptimization channel. A multi-asset owner’s storage is typically paired with wind or solar assets rather than natural gas, and those assets need not sit in the same local market as the storage unit. Aggressive discharge during the ramp still carries a cost for this owner: it depresses the broader price environment in which its renewable generation is also compensated, so the owner has some incentive to moderate discharge relative to a storage unit with no such exposure. An independent storage owner faces no such tradeoff and discharges to maximize arbitrage revenue alone. Consistent with this, independent storage delivers a larger markup reduction on peakers than multi-asset-owner storage. For small non-peakers, the two are again not distinguishable, again consistent with small non-peakers being off the margin during these hours.

Top-5 storage firm versus outside top-5 If top-5 storage firms withhold discharge to exercise their own market power, they should reduce peaker markups less than smaller firms. The data show the opposite: top-5 storage firms deliver a larger markup reduction. Two readings are possible. Under a scale interpretation, top-5 firms move enough MW that their entry meaningfully shifts the competitive fringe, while outside-top-5 storage is too dispersed to accumulate. Under a market power interpretation, top-5 firms would withhold discharge and attenuate the effect. The data are consistent with the scale reading and against the market power reading: at current penetration levels in ERCOT, the largest storage owners appear to be price-takers on the discharge side. For small non-peakers, the contrast is not distinguishable during the evening ramp.

Summary of storage heterogeneity For peakers during the evening ramp, all three contrasts point the same way: storage disciplines markups more when the owner does not internalize the price impact of discharge on other assets and does not withhold at scale. For small non-peakers during the same window, none of the three contrasts is statistically distinguishable, even though the aggregate storage effect on small non-peaker markups is confirmed in Section 5.3.1. This asymmetry suggests that the storage-owner behavior channel identified in Prediction 4 operates most sharply on peakers, whose markup is highly sensitive to the exact shape of residual demand during the ramp because they are running only when residual demand is high enough to clear their offer. Small non-peakers respond to aggregate storage entry but appear less sensitive to differences across storage-owner types. The storage-owner behavior channel therefore identifies primarily on peakers,

²⁸The charging channel is testable through the state-of-charge profile of colocated batteries and is addressed in Section 5.3.

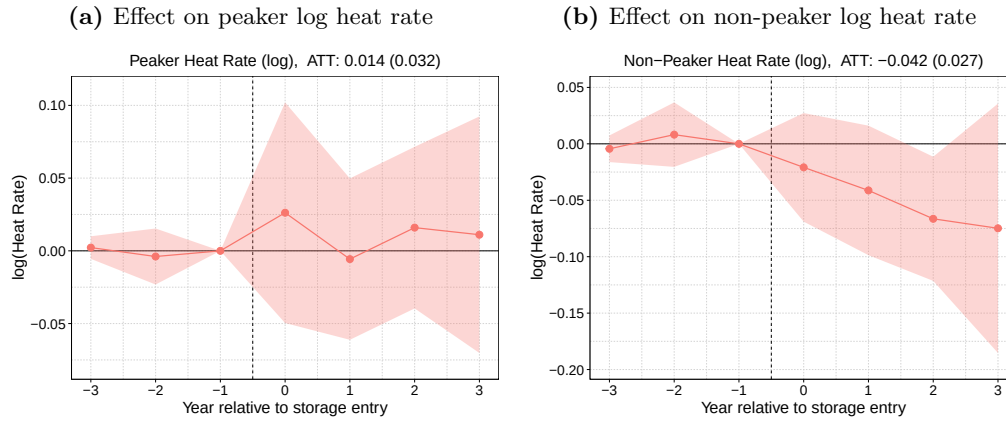
while the markup response documented in Section 5.3.1 confirms that the underlying mechanism reaches non-peakers as well.

5.4 Mechanism

The results in Sections 5.1–5.3 establish that storage entry compresses markups for smaller natural gas generators, and that in the ramp hours the peaker markup response operates through the residual demand slope rather than through quantity displacement. In this section I probe the mechanism further, testing Prediction 3 on the dynamic efficiency channel and directly decomposing the realized markup response into the market power and efficiency components using Equation (9).

5.4.1 Dynamic Efficiency Results

Figure 8: Effect of storage entry on generator heat rate



Note. These figures show the estimates of the effect of storage entry on log weekly average heat rate, estimated using Equation (13). Since the data is at the generator by week of sample level, the controls include average weekly maximum temperature, maximum load, maximum local load, maximum local uncurtailed wind and solar generation, and the same set of fixed effects. The sample includes only generators available in the CEMS dataset (128 of 153 treated peakers, 198 of 298 treated non-peakers). The line shows the 95% confidence interval clustered at the county level.

Prediction 3 states that the net change in a generator’s heat rate is theoretically ambiguous, with the sign varying by technology through the shape of the heat-rate curve. I test this by estimating Equation (13) with log heat rate as the outcome.

Figure 8 presents event study estimates on log weekly average heat rate. The ATT is not statistically significant for either peakers or non-peakers, and the significant lag coefficients for non-peakers beyond three years do not indicate a systematic shift. At current storage penetration levels in ERCOT, storage entry does not materially change the operating point of either technology along its heat-rate curve, so the heat rate response is not a meaningful channel through which storage affects realized markups.

5.4.2 Decomposing the Market Power and Efficiency Channels

The realized markup $\mu_{ft} = p_t - c_{ft}$ can be decomposed into a market power channel and a dynamic efficiency channel, following Equation (9). Since $c_{ft} = p_t^g \bar{hr}_{ft}$, changes in heat rate move the realized markup through the efficiency channel. As documented by the heat rate results, the realized markup response documented in Sections 5.1–5.3 operates primarily through the market power channel rather than through dynamic efficiency, thus I expect no difference between the realized and fixed heat rate markup results as shown in Table 5.

Table 5: Isolating the competitive channel from the efficiency channel

Hour	Peaker		Non-Peaker	
	Markup (\$/MWh)	Markup with fixed heat rate (\$/MWh)	Markup (\$/MWh)	Markup with fixed heat rate (\$/MWh)
	(1)	(2)	(3)	(4)
6 AM	−2.3** (1.1)	−2.8** (1.1)	1.1 (1.6)	1.2 (1.8)
7 AM	−2.9*** (1.1)	−3.1** (1.2)	1.0 (1.1)	1.0 (1.3)
6 PM	−6.4** (3.0)	−6.6 (4.5)	−0.2 (2.6)	−0.9 (2.6)
7 PM	−9.5** (4.6)	−9.4 (8.2)	−1.4 (5.1)	−1.9 (4.8)
8 PM	−6.0** (2.9)	−8.3* (5.0)	0.9 (2.7)	0.3 (2.8)

Notes. This table reports point estimates and standard errors (in parentheses) of β^{gh} from estimating Equation (11) for peakers in columns (1)–(2) and non-peakers in columns (3)–(4), during the morning ramp (6–7 AM) and evening ramp (6–8 PM). Column (1) and Column (3) reports markup results using realized heat rate, while Column (2) and Column (4) reports markup results using pre-storage heat rate. The full set of hourly estimates are shown in Appendix Figure D.20. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.4.3 Change in Bidding Behavior

I use the bid data from the SCED process to test whether storage entry affects the bidding behavior of natural gas generators. This provides a direct check on the market power channel: if generators respond strategically to storage entry, the estimates should reflect a significant change in the price level at which they offer their capacity.

I estimate the same specification as in Equation (11) using submitted markup and the slope of the bid curve as outcomes. Detailed results are presented in Appendix D.5. I compute the submitted markup at 35% up to 85% of the generator’s capacity and find that storage entry does not affect the submitted markup for either peakers or non-peakers.

Using bid prices at different capacity levels, I then compute the slope of the bid curve and estimate the effect of storage entry on this slope. A steeper bid curve means the generator is bidding higher prices on marginal MW as it approaches its capacity. I find suggestive evidence that peakers increase their bid slope by about \$25/MWh per 25% increment of capacity at both lower and higher utilization levels, while non-peakers do not change their bid slope. This pattern

is consistent with a repricing response: with storage displacing peakers from the margin during the discharge window, peakers appear to bid more aggressively on the marginal MW that do clear, even as their realized markup falls in those hours.

Summary of mechanism The mechanism results identify the market power channel as the source of the realized markup response to storage entry. Storage entry does not shift generators along their heat-rate curves, and the realized and fixed heat rate markup estimates in Table 5 are close to identical, ruling out the efficiency channel. The bidding results add behavioral evidence: peakers do not lower their submitted markups but steepen their bid curves, consistent with peakers repricing the marginal MW that clear while realized markups fall through compression of the residual demand slope E_t . Non-peakers show no bidding response.

6 Discussion

In this section, I discuss the implications of my findings for the electricity market and for policy makers. I first quantify the welfare impacts of storage entry by estimating the changes in excess rent and emissions, and then discuss the overall welfare impacts. Finally, I discuss the policy implications of my findings.

6.1 Welfare Analysis of Storage Entry

In this section, I quantify the welfare impacts of storage entry per 100 MW of storage capacity through two channels: the environmental benefit from displaced generation and the change in market-power rent. I report both per affected generator, separately for small and large peakers and non-peakers, so that the composition of effects across generator types is transparent.²⁹

I measure the change in excess rent as the change in producer surplus of natural gas generators. For each group, I combine the hourly effects on generation and markup with baseline hourly generation and markup to form the change in hourly rent, which decomposes into a price effect ($q_0\Delta\mu$), a quantity effect ($\mu_0\Delta q$), and an interaction term. I sum across hours and annualize.³⁰ I then quantify the environmental benefit through the effect of storage entry on total weekly generation, multiplied by the average emission rate of untreated generators in each cell for NO_x , SO_2 , and CO_2 , and valued at the social cost of each pollutant.³¹

²⁹I recover each group’s effect by interacting the continuous storage dose, measured in cumulative MW entering the generator’s county, with a fixed capacity indicator, and recovering the small- and large-capacity effects by the delta method. Splitting by capacity is necessary because, as documented in Section 5.2, the response to storage entry is concentrated in smaller generators, so a pooled estimate would understate the effect on the generators that drive the welfare change.

³⁰I compute standard errors by drawing the hourly markup and generation effects from their estimated sampling distributions and propagating them through the rent decomposition with 10,000 Monte Carlo draws.

³¹I value emissions at \$190 per ton for CO_2 (EPA, 2023) and \$38.70 and \$13.90 per pound for SO_2 and NO_x from the AP4 model (Tschofen, Azevedo and Muller, 2019), and annualize the weekly effect by 52. Using the untreated-period average holds the emission rate fixed at baseline, so the benefit reflects only the removal of generation at its baseline dirtiness. Section 5.4 shows storage does not move generators along their heat-rate curves, so the emission-rate channel is negligible.

Table 6: Environmental and Market-Power Benefits of Storage Entry by Generator Type and Capacity

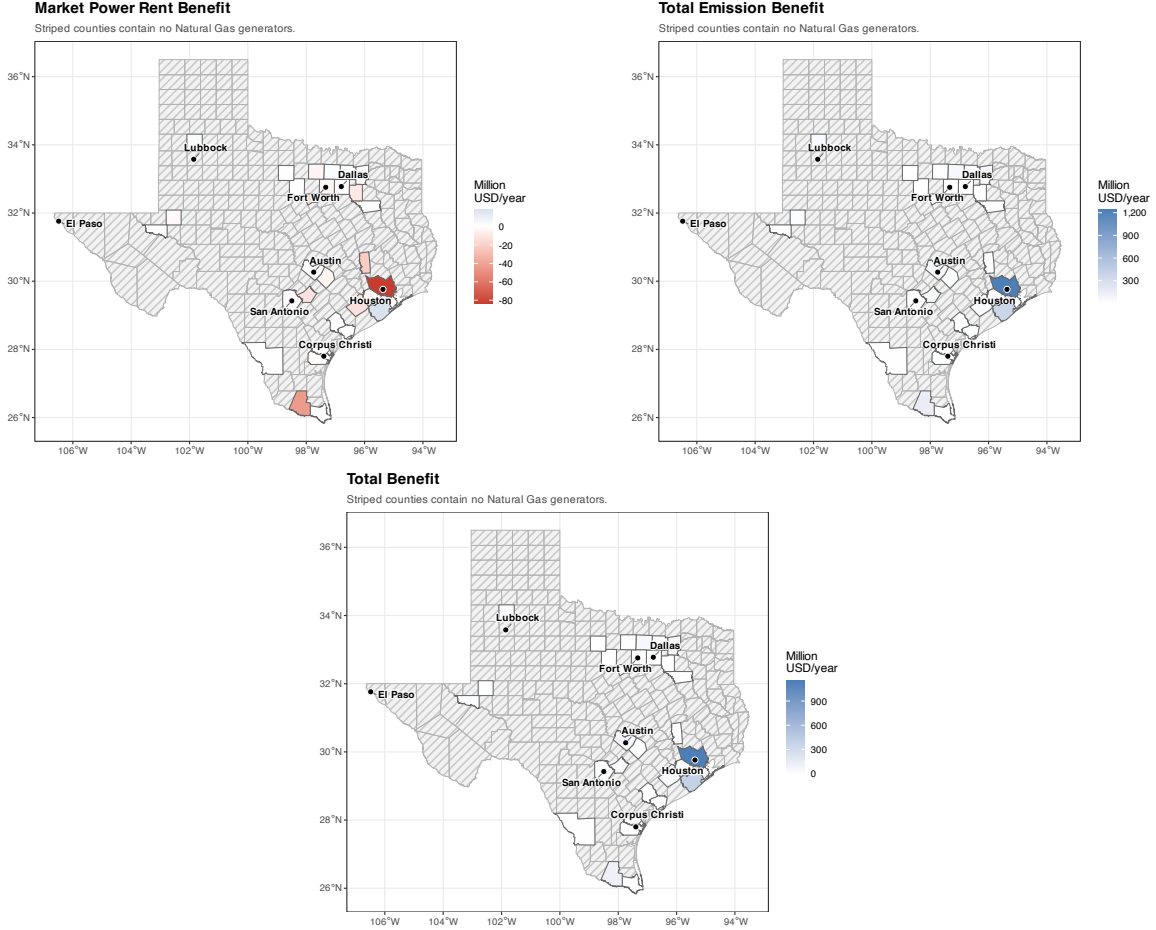
Panel A: Empirical estimates (per 100 MW storage entrant)				
	Peaker		Non-Peaker	
	Small (≤ 90 MW)	Large (> 90 MW)	Small (≤ 350 MW)	Large (> 350 MW)
Δ weekly generation (MWh)	-241.13 (162.08)	-367.01 (149.74)	-551.90 (155.35)	-163.91 (176.05)
Δ ramp markup, 6–8 PM (\$/MWh)	-1.51 (0.32)	0.04 (0.54)	-2.22 (0.67)	1.89 (0.48)
Panel B: Assumptions and parameters				
	Value		Source	
CO ₂ damage (\$/ton)	\$190		EPA	
SO ₂ damage (\$/lb)	\$38.70		AP4	
NO _x damage (\$/lb)	\$13.90		AP4	
Panel C: Annual benefit per affected generator (per 100 MW, million USD/year)				
	Peaker		Non-Peaker	
	Small (≤ 90 MW)	Large (> 90 MW)	Small (≤ 350 MW)	Large (> 350 MW)
Environmental benefit	1.66 [−0.18, 3.51]	2.58 [0.84, 4.34]	4.46 [2.39, 6.56]	0.94 [−0.71, 2.60]
Market-power rent reduction	0.04 [−0.01, 0.09]	0.17 [0.11, 0.24]	0.14 [0.11, 0.17]	−1.23 [−1.47, −0.98]

Notes. Annual benefit of a representative 100 MW storage entrant, per affected generator, by technology and capacity. Panel A reports the effect of storage entry on weekly generation and on the evening-ramp (6–8 PM) markup, per 100 MW, with county-clustered standard errors in parentheses. Panel B reports the emission damage values. Panel C reports the annual benefit: (1) the environmental benefit from displaced NO_x, SO₂, and CO₂, valued at the untreated-period emission rate and the social cost of each pollutant; and (2) the market-power rent reduction, largely a transfer to consumers, where a negative value indicates storage raises that group’s rent. Brackets report 95% intervals from 10,000 Monte Carlo draws.

Table 6 summarizes the annual benefit of storage entry per 100 MW, per affected generator. The environmental benefit is the dominant channel, positive for all four groups and largest for small non-peakers and large peakers, which display the greatest generation displacement. The market-power rent reduction is smaller and differs in sign across groups: positive for peakers and small non-peakers, consistent with the disciplining of marginal-unit markups in Section 5.1, but negative for large non-peakers, indicating that storage raises the inframarginal rent of the largest baseload-like units. This follows from their large baseline generation: a small increase in the price they receive, amplified by their large output, elevates their inframarginal rent by more than the modest quantity displacement reduces it.

To put these welfare estimates into context, I compute the total annual benefit of storage entry in ERCOT by multiplying the per-generator estimates by the number of affected generators in each county. Figure 9 shows the distribution of the changes in market-power rent and emission benefits. Only a few counties gain from reduced market-power rent; in most, rent rises, because aggregate generation still comes predominantly from large non-peaker generators. On net, market-power rent increases by \$200 million. The environmental benefit is positive in all counties, and largest where generation displacement is greatest. The total annual benefit of storage entry in ERCOT is

Figure 9: Distribution of Annual Environmental and Market-Power Benefits by County



Notes: This figure show counties siting natural gas generators in Texas, the color of the county show the benefit from changes in excess rent, the benefit from changes in emissions, and the total benefits, computed using end of sample cumulative storage capacity for each county.

\$2.2 billion, dominated by the environmental benefit of \$2.4 billion.

To weigh this annual benefit against the cost of the fleet, I annualize the \$9 billion capital cost of the 4.4 GW of storage currently operating over its service life.³² Applying a capital recovery factor for a 15-year asset life, following the NREL Annual Technology Baseline assumption for utility-scale lithium-ion storage, at a 7% real discount rate gives an annualized cost of about \$1.0 billion per year.³³ The change in market-power rent is a transfer between generators and consumers and nets out of social welfare. The environmental benefit is the net social gain, and because the annualized cost of about \$1.0 billion is well below the \$2.4 billion environmental benefit, the net welfare impact

³²The investment cost comprises a power cost of 1,440 \$/kW and an energy cost of 461.08 \$/kWh (EIA, 2024). I compute the total investment cost by multiplying these unit costs by the actual total power capacity and the sample-average storage duration of 1.36 hours.

³³I annualize using the capital recovery factor $CRF = r(1 + r)^N / [(1 + r)^N - 1]$, with asset life $N = 15$ and real discount rate $r = 0.07$, so that $CRF \approx 0.110$.

of storage entry is positive at current penetration levels once its emission externality is valued.³⁴

Discussion of limitations My analysis has three main limitations. First, it considers only the indirect effect of storage on natural gas generators through the market-power, efficiency, and total emission channels; storage also helps integrate renewables, improve reliability, and relieve congestion, so my analysis likely understates the total welfare impact. Second, it considers only natural gas generators, not other fossil generators such as coal. Finally, storage revenue in my sample is dominated by ancillary-service revenue, which may not persist as more storage enters and that market saturates; with more revenue then coming from arbitrage, the welfare impact could be larger than my estimates.

6.2 Policy Implication

The findings in this paper yield three primary policy implications for wholesale electricity market design and the regulation of utility-scale energy storage.

First, storage’s social value rests almost entirely on its environmental benefit, not on market power discipline. Of the \$2.2 billion in aggregate annual benefit from ERCOT storage entry, \$2.4 billion comes from displaced fossil-fuel emissions, while net producer rent rises by \$200 million rather than falling. Although the environmental benefit exceeds the annualized cost of the fleet, this net social gain is not captured by storage owners: private arbitrage and ancillary-service revenues fall well short of the annualized investment cost, and competitive discipline cannot close the gap. Private markets will therefore under-provide storage unless policy internalizes the environmental externality directly, through carbon pricing or targeted capacity subsidies.

Second, the results inform the regulation of localized market power. Storage disciplines the smaller peaker and small non-peaker units that set prices during congested ramp hours, through the same residual-demand-flattening mechanism central to transmission-driven discipline. This effect is concentrated on the margin, not market-wide: large-capacity non-peakers, protected by their dominant local share, see inframarginal rents rise, so aggregate producer rent increases rather than falls. Storage therefore cannot be relied on as a market-power remedy at the market level; grid expansion and structural reforms remain necessary to address rents earned by dominant incumbents.

Third, the ownership heterogeneity points to a trade-off between hybrid and standalone storage deployment. Standalone storage disciplines incumbent markups more effectively than storage co-located with solar, likely because hybrid storage is often dispatched to firm renewable output rather than to pursue pure price arbitrage during scarcity hours. Hybrid resources lower interconnection costs and aid renewable integration, but policymakers should weigh this against standalone storage’s stronger competitive discipline.

³⁴Including the average annual storage revenue of \$1.06 million per 100 MW, which reflects private arbitrage and ancillary-service value, only widens this margin. I compute this figure by summing the total revenue of all storage units in the sample and dividing by total storage capacity, then scaling to 100 MW.

7 Conclusion

Utility-scale battery storage is transforming wholesale electricity markets by shifting energy across time and space. While existing literature emphasizes its role in integrating renewables and reducing price spreads, this paper provides the first empirical evidence on how storage entry alters the strategic bidding behavior and market power of incumbent natural gas generators, using a novel unit-level ERCOT panel and a staggered difference-in-differences design.

I find that local storage entry systematically disciplines the market power of small-capacity natural gas generators, operating entirely through a flattening of the residual demand slope rather than through changes in heat rate. This effect is highly heterogeneous: it is concentrated among smaller generators in congested markets, while large-capacity generators with dominant local shares remain largely unaffected. Its magnitude also depends on the storage asset’s operational boundaries, with standalone, independently operated storage exerting the strongest competitive pressure relative to solar-hybrid facilities.

Finally, the welfare analysis shows that storage’s social value is substantial but driven predominantly by displaced emissions rather than by reduced market-power rents, which rise rather than fall in aggregate. As the energy transition progresses, storage will continue to discipline marginal generators’ pricing through this residual-demand mechanism, even as it raises the inframarginal rents of dominant incumbents. Optimal deployment therefore requires policies that target its environmental value directly, rather than treating competitive discipline as the primary rationale for its buildout.

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Supplementary Appendix

A Detailed Theoretical Derivations

This appendix provides the formal derivations underlying the theoretical framework in Section 2.2. Section A.1 derives the storage owner's optimization problem and characterizes the aggregate storage supply. Section A.2 derives the natural gas generator's optimal markup. Section A.3 derives the comparative statics of the markup with respect to storage entry that underpin Prediction 1. Section A.4 extends the analysis to the two-node case to characterize how transmission congestion mediates the market power channel. Section A.5 generalizes storage from a price taker to a strategic actor, and Section A.6 derives the ownership wedges underlying Prediction 4.

A.1 Storage Problem

A price-taking battery starts empty. Its net injection is $s_L \leq 0$ in the low-demand period (it buys $-s_L$) and $s_H = -\eta s_L \geq 0$ in the high-demand period (it delivers $\eta(-s_L)$). With each period a block of τ hours, power capacity $\bar{P}$, and energy capacity $\bar{K}$, storage maximizes arbitrage profit subject to physical constraints:

$$(A.1) \quad \max_{s_L \leq 0} (\eta p_H - p_L)(-s_L) \quad \text{s.t.} \quad \underbrace{-s_L \leq \bar{P}\tau}_{\text{charge power constraint}}, \quad \underbrace{s_H \leq \bar{P}\tau}_{\text{discharge power constraint}}, \quad \underbrace{-s_L \leq \bar{K}}_{\text{energy constraint}}.$$

Optimal storage operation. Because the objective is linear in s_L , the optimum is a corner solution. Writing the stored energy as $-s_L$ and the corner capacity as $\bar{k} \equiv \min\{\bar{P}\tau, \bar{K}\}$,

$$(A.2) \quad -s_L^* = \begin{cases} \bar{k} & \text{if } \eta p_H > p_L, \\ 0 & \text{if } \eta p_H < p_L, \end{cases} \quad s_H^* = \eta(-s_L^*),$$

where the discharge-power constraint $s_H \leq \bar{P}\tau/\eta$ is slack since $\eta \leq 1$. Storage arbitrages at full capacity whenever the round-trip spread is positive and is idle otherwise. Writing the *duration* $D \equiv \bar{K}/\bar{P}$ (hours of discharge at rated power), the binding limit is governed by duration:

$$(A.3) \quad \bar{k} = \begin{cases} \bar{K} & \text{if } D < \tau \quad (\text{short-duration: energy-constrained}), \\ \bar{P}\tau & \text{if } D > \tau \quad (\text{long-duration: power-constrained}). \end{cases}$$

Utility-scale batteries are typically 1 to 4 hours duration. In ERCOT the average duration is below 2 hours, so against a multi-hour peak they are energy-constrained.

Storage marginal cost. To deliver one MWh at the peak the battery must buy $1/\eta$ MWh off-peak, so its marginal cost is the intertemporal opportunity cost

$$(A.4) \quad m_s = \frac{p_L}{\eta},$$

and it discharges whenever $p_H \geq m_s = p_L/\eta$. A single price-taking battery's offer is perfectly elastic at the floor m_s up to quantity $\eta\bar{k}$, then inelastic at its limit ($\partial s_t/\partial p_t = 0$). Figure F.23 confirms this pattern in the actual storage bid data: Panel (a) shows offer curves when storage is charging (negative net injection) and Panel (b) shows offer curves when storage is discharging (positive net injection). In both cases, the awarded quantity lies close to the vertical segment of the supply curve, consistent with the corner solution.

Aggregate storage supply. Now consider a battery fleet with heterogeneous arbitrage thresholds: a higher peak price pulls more units across their threshold, so aggregate net injection splits into a price-independent level and a price-responsive margin,

$$(A.5) \quad s_t(p_t) = \bar{s}_t + \sigma_t p_t, \quad \sigma_t = \frac{\partial s_t}{\partial p_t} \geq 0.$$

The first term $\bar{s}_t$ is the inframarginal capacity that clears whatever the marginal price, and σ_t is the slope—positive when storage sits near its threshold and zero once capacity saturates. The energy capacity limits storage supply during the discharge period: it can supply at most $\eta\bar{k}$ MWh. Aggregating across the fleet in this way reconciles the corner-solution representation at the unit level with the smooth, upward-sloping supply used in the main text.

A.2 Natural Gas Generator Problem

I now solve the natural gas generator's optimization problem under the setup in Section 2.2. Substituting the fringe supply and storage's net injection into the clearing condition (2), the residual demand facing the strategic firm is

$$(A.6) \quad q_{ft} = D_t - s_t(p_t) - S_t^c(p_t) = A_t - E_t p_t, \quad E_t \equiv \beta + \sigma_t,$$

where the intercept collects the autonomous (price-independent) terms,

$$(A.7) \quad A_t = A_t^0 - \bar{s}_t, \quad A_t^0 \equiv D_t - \alpha_t,$$

and $\bar{s}_t$ is storage's price-independent net injection. In the high-demand period $\bar{s}_H \geq 0$ lowers the intercept (storage adds supply), and in the low-demand period $\bar{s}_L \leq 0$ raises it (storage adds demand, since $-\bar{s}_L \geq 0$). The slope E_t absorbs both the fringe slope β and storage's price responsiveness σ_t from (A.5). Storage therefore shifts both the intercept A_t (down in the high-demand period, up in the low-demand period) and the slope E_t (upward in both periods).

The firm chooses q_{ft} to maximize $\pi_{ft} = p_t(q_{ft})q_{ft} - C_{ft}(q_{ft})$ with constant marginal cost c_{ft} . Writing the inverse residual demand as $p_t = (A_t - q_{ft})/E_t$, the first-order condition is

$$(A.8) \quad \begin{aligned} \frac{\partial \pi_{ft}}{\partial q_{ft}} &= p_t + q_{ft} \frac{\partial p_t}{\partial q_{ft}} - c_{ft} = p_t - \frac{q_{ft}}{E_t} - c_{ft} = 0 \\ \implies p_t - c_{ft} &= \frac{q_{ft}}{E_t}, \end{aligned}$$

which is the optimal price-cost markup rule in Equation (4), equivalent to the Lerner-index formulation common in the electricity market literature (Jha and Leslie, 2025; Woerman, 2023; Ryan, 2021; Doshi, 2024; Fabra and Imelda, 2023). The second-order condition is $\partial^2 \pi_{ft} / \partial q_{ft}^2 = 2 \partial p_t / \partial q_{ft} = -2/E_t < 0$, so the optimum is a maximum and, with linear demand and constant marginal cost, unique.

Solving the first-order condition jointly with residual demand (A.6) gives the closed-form equilibrium

$$(A.9) \quad p_t^* = \frac{1}{2} \left(\frac{A_t}{E_t} + c_{ft} \right), \quad q_{ft}^* = \frac{1}{2} (A_t - E_t c_{ft}), \quad \mu_t^* = \frac{A_t}{2E_t} - \frac{c_{ft}}{2},$$

which has interior solutions whenever $A_t > E_t c_{ft}$. Three comparative statics follow immediately and map to the three forces discussed in Section 2.2:

$$(A.10) \quad \begin{aligned} \frac{\partial \mu_t^*}{\partial A_t} &= \frac{1}{2E_t} > 0 \text{ (residual demand level),} \\ \frac{\partial \mu_t^*}{\partial E_t} &= -\frac{A_t}{2E_t^2} < 0 \text{ (residual demand slope),} \\ \frac{\partial \mu_t^*}{\partial c_{ft}} &= -\frac{1}{2} < 0 \text{ (efficiency).} \end{aligned}$$

A larger residual demand raises the markup; a more elastic residual demand lowers it; a lower marginal cost raises it.

A.3 Comparative Statics of the Markup under Storage Entry

This section derives the results in Prediction 1. Let the pre-entry residual-demand intercept be A_t^0 (storage absent) and slope β . Holding marginal cost fixed to isolate the market power channel, entry changes the markup from $\mu^0 = A_t^0/(2\beta) - c/2$ to μ^* in Equation (A.9).

High-demand period. Storage lowers the intercept to $A_H = A_H^0 - \bar{s}_H$ and raises the slope to $E_H = \beta + \sigma_H$. Then

$$\Delta \mu_H = \frac{A_H^0 - \bar{s}_H}{2(\beta + \sigma_H)} - \frac{A_H^0}{2\beta} < 0,$$

because the numerator falls and the denominator rises. The markup falls *unambiguously*, so long as gas still serves positive residual demand ($A_H^0 > \bar{s}_H$).

Low-demand period. Storage raises the intercept to $A_L = A_L^0 - \bar{s}_L$ (with $-\bar{s}_L = |\bar{s}_L| > 0$, since charging adds demand) and raises the slope to $E_L = \beta + \sigma_L$. Then

$$\begin{aligned}\Delta\mu_L &= \frac{A_L^0 - \bar{s}_L}{2(\beta + \sigma_L)} - \frac{A_L^0}{2\beta} > 0 \\ \iff \beta(-\bar{s}_L) &> A_L^0 \sigma_L \\ \iff \frac{-\bar{s}_L}{A_L^0} &> \frac{\sigma_L}{\beta}.\end{aligned}$$

The markup rises if the *proportional* outward shift in residual demand ($-\bar{s}_L/A_L^0$) exceeds the *proportional* increase in its elasticity (σ_L/β). When storage charges at a binding limit, $\sigma_L = 0$ and the condition holds for any $\bar{s}_L < 0$; the increase weakens as storage becomes a more flexible, price-responsive buyer off-peak.

A.4 Congestion and the Local Market

The comparative statics above are for a single node with unconstrained transmission. With congestion, the effect of storage entry depends on where storage sits relative to the gas firm and its load. Let two regions W and E be joined by a line of capacity $\bar{K}$ carrying flow τ from W to E . Region E holds the strategic gas firm f , a competitive fringe $S_E^c(p_E) = \alpha_E + \beta_E p_E$, and load D_E ; region W holds a fringe $S_W^c(p_W) = \alpha_W + \beta_W p_W$, load D_W , and the battery storage, whose injection contributes a level $\bar{s}_t$ and a price-responsive part with slope σ_t . Nodal balance and the line constraint are

$$(A.11) \quad (E) \quad q_{ft} + \alpha_E + \beta_E p_E + \tau = D_E,$$

$$(A.12) \quad (W) \quad \alpha_W + (\beta_W + \sigma_t)p_W + \bar{s}_t = D_W + \tau,$$

$$(A.13) \quad \tau \leq \bar{K}, \quad \lambda \geq 0, \quad \lambda(\bar{K} - \tau) = 0, \quad p_E - p_W = \lambda,$$

so the nodal price gap equals the congestion shadow price. The gas firm's residual demand follows from Equation (A.11), $q_{ft} = D_E - \alpha_E - \beta_E p_E - \tau$, and what storage does to it depends entirely on how τ responds to s .

Slack line ($\lambda = 0$). The two regions clear at one price, $p_W = p_E \equiv p$. Adding Equations (A.11) and (A.12) eliminates τ :

$$(A.14) \quad q_{ft} = (D_E + D_W) - (\alpha_E + \alpha_W + \bar{s}_t) - (\beta_E + \beta_W + \sigma_t)p,$$

so f faces a residual-demand slope of $\beta_E + \beta_W + \sigma_t$: storage's responsiveness σ_t enters in full, because τ adjusts freely to hold $p_W = p_E$ and carries the storage MWh into E . The markup falls

exactly as in the single-node case, now with the combined fringe: $\partial\mu_f/\partial\sigma_t < 0$.

Binding line ($\lambda > 0$). The import into E is pinned at $\tau = \bar{K}$, so Equation (A.11) gives $q_{ft} = D_E - \alpha_E - \beta_E p_E - \bar{K}$ and f faces slope β_E alone—neither β_W nor σ_t appears. Storage now clears entirely within W (Equation (A.12) pins p_W), depressing p_W and widening $\lambda = p_E - p_W$, but leaving q_{ft} , the slope, and the markup untouched: $\partial\mu_f/\partial\sigma_t = \partial\mu_f/\partial s_t = 0$.

Local storage (region E). If storage sits in E it enters Equation (A.11) directly, $\beta_E \rightarrow \beta_E + \sigma_t$, so f faces slope $\beta_E + \sigma_t$ regardless of the line. The generator experiences the full effect independent of λ .

These are the three cases discussed in Section 2.2. In terms of storage's share of f 's relevant competitive set, that share is $\sigma_t/(\beta_E + \sigma_t)$ for local storage; $\sigma_t/(\beta_E + \beta_W + \sigma_t)$ for remote storage on a slack line, which is smaller because the integrated fringe is larger; and 0 for remote storage behind a binding constraint. One implication is worth noting. Storage in W lowers p_W and so raises the desired export τ , meaning storage entry can itself tip the line into the binding regime. Congestion is thus partly endogenous to where storage is sited. I abstract from this endogeneity by implicitly assuming that no constraint is marginally binding, following Woerman (2023), and leave it as a direction for future work.

A.5 Strategic Storage

Sections A.1 and A.3 treat storage as a price taker: it sets quantity against given prices, and its responsiveness σ_t enters the gas firm's residual-demand slope. Here, I let storage be strategic and solve the equilibrium jointly with the gas firm, nesting the price-taking case as a limit. I follow Andres-Cerezo and Fabra (2025) and Andrés-Cerezo and Fabra (2023), in which strategic storage internalizes the price impacts of its charging and discharging decisions. I focus on the discharge period, where strategic storage matters most; the charge period is symmetric, with storage acting as a strategic buyer.

Netting out the fringe, the discharge-period inverse demand faced by the two strategic quantity-setters, the gas firm (q_f) and storage (s), is

$$(A.15) \quad p = \frac{a - q_f - s}{\beta}, \quad a \equiv D_H - \alpha_H,$$

so $\partial p/\partial q_f = \partial p/\partial s = -1/\beta$. The gas firm has marginal cost c_f ; storage has marginal cost $m_s = p_L/\eta$ of delivering a discharge MWh from Equation (A.4).

I introduce a *conduct parameter* $\theta \in [0, \infty)$ that scales how much storage internalizes its own price impact, so storage perceives marginal revenue $p + \theta s \partial p/\partial s = p - \theta s/\beta$. The two first-order

conditions are

$$(A.16) \quad \underbrace{p - c_f = \frac{q_f}{\beta}}_{\text{gas firm}}, \quad \underbrace{p - m_s = \theta \frac{s}{\beta}}_{\text{storage, conduct } \theta}.$$

At $\theta = 0$, storage is a price taker ($p = m_s$: it supplies until price equals its marginal cost, the maximal-discipline case); at $\theta = 1$ it is a Cournot competitor of the gas firm; intermediate θ interpolates. Solving Equations (A.15)–(A.16),

$$(A.17) \quad s^*(\theta) = \frac{a + \beta c_f - 2\beta m_s}{1 + 2\theta}, \quad q_f^*(\theta) = \frac{1}{2}(a - s^* - \beta c_f), \quad \mu_f^*(\theta) = \frac{q_f^*}{\beta}.$$

The comparative static in conduct is the key result:

$$(A.18) \quad \frac{\partial s^*}{\partial \theta} < 0, \quad \frac{\partial \mu_f^*}{\partial \theta} = \frac{a + \beta c_f - 2\beta m_s}{\beta(1 + 2\theta)^2} > 0 \quad (\text{whenever } s^* > 0).$$

Strategic storage weakens discipline. More strategic storage (higher θ) discharges less and leaves a higher gas markup. The two endpoints bracket the effect: at $\theta = 0$, competitive storage delivers the maximal markup reduction, while as $\theta \rightarrow \infty$, storage fully withholds ($s^* \rightarrow 0$) and the gas markup returns to its no-storage monopoly level $\mu_f^* \rightarrow (a - \beta c_f)/(2\beta)$. Storage that is present but strategic can therefore provide arbitrarily little competitive discipline.

This puts “strategic storage” on a defined equilibrium footing: the ownership wedges derived in Appendix A.6 are exactly what raise θ (storage-side market power from a large own position) or raise m_s (a gas-heavy owner’s opportunity cost of discharging includes the cannibalized rents on its own peak generation). Both push μ_f^* up, so top-5 owners and multi-asset owners discipline the gas firm less than independent storage-only owners. The price-taking analysis in the main text is the $\theta = 0$ slice; there, the smooth, upward-sloping storage supply—and the slope σ_t that enters $E_t = \beta + \sigma_t$ —comes from aggregating the heterogeneous bang-bang arbitrageurs of Appendix A.1, not from any single unit’s cost curve. This reconciles the two representations.

A.6 Ownership: Cooptimization and Storage-Side Market Power

Let the storage owner also hold generation selling G_H at the peak and G_L off-peak, and let storage be large enough to move prices, with $\partial p_t / \partial s_t = -1/E_t$ from Equation (A.6). The owner chooses its discharge $s_H \geq 0$ (buying $s_L = -s_H/\eta$ off-peak) to maximize portfolio profit

$$(A.19) \quad \Pi = p_H(s_H + G_H) + p_L(s_L + G_L) - (\text{gen. cost}),$$

with first-order condition (after multiplying through by η)

$$(A.20) \quad \underbrace{\eta p_H - p_L}_{\text{arbitrage spread}} + \underbrace{\left(\frac{s_L}{E_L} - \eta \frac{s_H}{E_H} \right)}_{\text{storage-side market power} < 0} + \underbrace{\left(\frac{G_L}{E_L} - \eta \frac{G_H}{E_H} \right)}_{\text{cooptimization with generation}} = 0.$$

A price-taking arbitrageur ($G = 0$, ignoring its own price impact) keeps only the arbitrage spread: it is positive, so the battery discharges to capacity (the corner solution of Appendix A.1). The other two terms capture the two forces underlying Prediction 4.

The second term is the *storage-side market power* wedge. With $s_L < 0 < s_H$, it is negative: a large storage position makes the owner internalize that its own discharge depresses the peak price and its own charging raises the off-peak price. The owner therefore withholds, and equilibrium discharge s_H^* and its responsiveness σ_H in Equation (A.5) both fall, attenuating Channel 1. This underpins Implication (ii) of Prediction 4, which the top-5-by-storage-capacity contrast tests empirically.

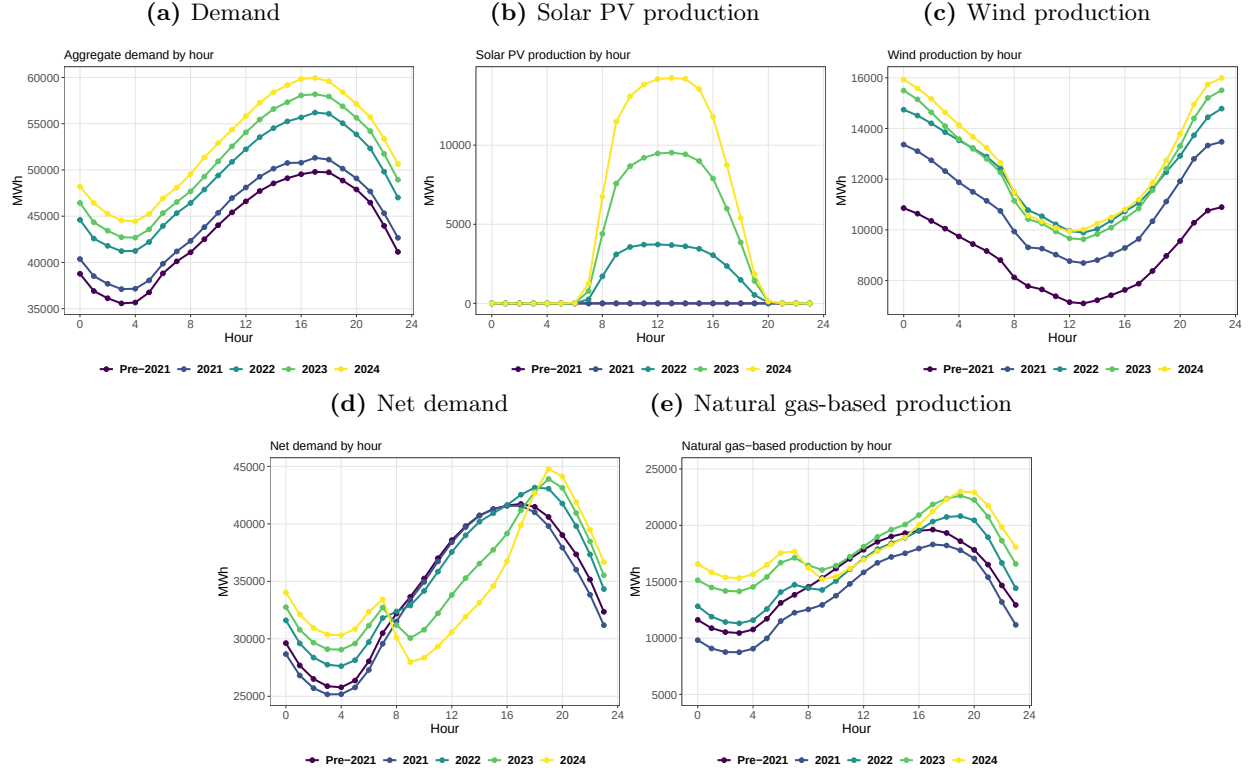
The third term is the *cooptimization* wedge. Its peak component $-\eta G_H/E_H < 0$ shows that an owner with large peak generation G_H loses rents on that generation when storage discharges. The owner therefore withholds relative to a pure arbitrageur. This underpins Implication (i) of Prediction 4, which encompasses three empirical contrasts. A firm holding a large fossil-generation portfolio (top-5 by total capacity) internalizes cannibalization directly through this term. A multi-asset firm holding wind or solar internalizes cannibalization symmetrically: discharging depresses the peak price at which its other generation clears. Finally, a solar-located battery has a partly mechanical dispatch tied to its own solar profile, which reduces its price responsiveness σ through a related but not identical channel: the dispatch is physically constrained rather than strategically chosen, but the equilibrium consequence—lower σ —is the same. All three empirical contrasts therefore test the same underlying cooptimization force operating through different institutional mechanisms.

Either wedge lowers the equilibrium discharge s_H^* and its responsiveness σ_H , attenuating Channel 1. In the strategic-storage representation of Appendix A.5, these wedges appear as increases in the conduct parameter θ (storage-side market power) or in the effective marginal cost m_s (cooptimization), both of which raise μ_f^* .

B Additional Descriptive Statistics

B.1 Trends in Demand and Supply Patterns

Figure B.1: Hourly demand and supply patterns in ERCOT

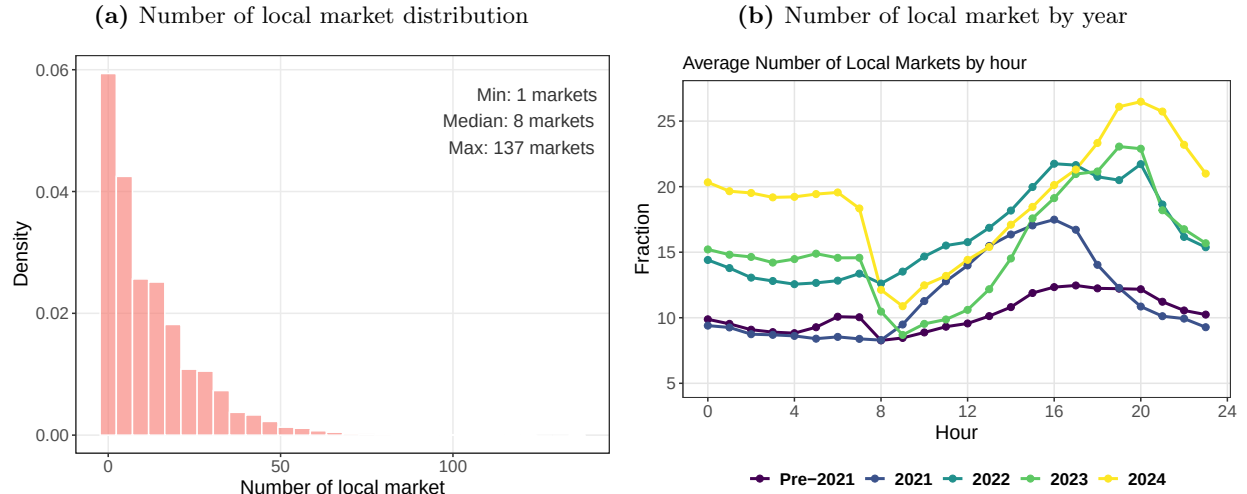


Notes: These figures show the average renewable generation, load, net load, and natural gas-based generation for each hour of the day. The net load is defined as the total load minus the total renewable generation.

Figure B.1 documents the hourly feature of the Texas electricity market over my sample period. First, in Panel B and C, the daily renewable production pattern have changed significantly over the years. Before 2021, the average renewable generation is relatively flat throughout the day, with its highest production at night due wind domination in the renewable generation. As more solar PV capacity is installed, we see a significant increase in renewable generation during the daytime hours, peaking around 1-2 pm. Figure B.1a shows that the hourly demand pattern has also evolved over the years, generally increasing across all hour of day. The hourly pattern shows a typical daily cycle with peaks in the morning and evening hours. These two things combined has implications for the net demand pattern in Figure B.1d, which has evolved from a relatively flat pattern to a more pronounced “duck curve” pattern, with a significant dip during the daytime hours when solar PV production is high and a steep ramp in the morning and evening hours when demand is high and solar PV production is low.

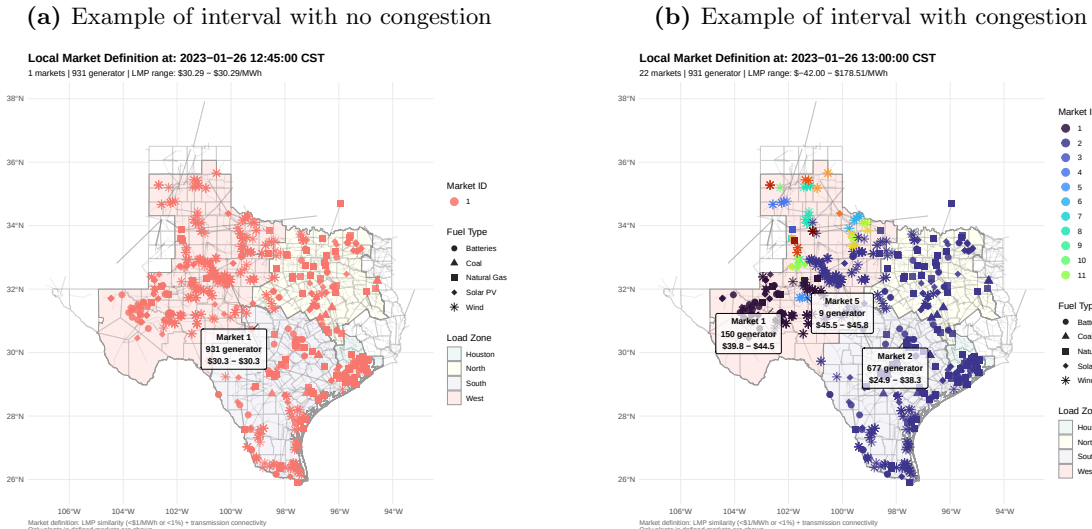
B.2 Local Market Descriptive Statistics

Figure B.2: Local market statistics



Notes: Panel (a) and (b) summarize the distribution of the number of local markets and the trends in hourly variation of the number of local markets by year.

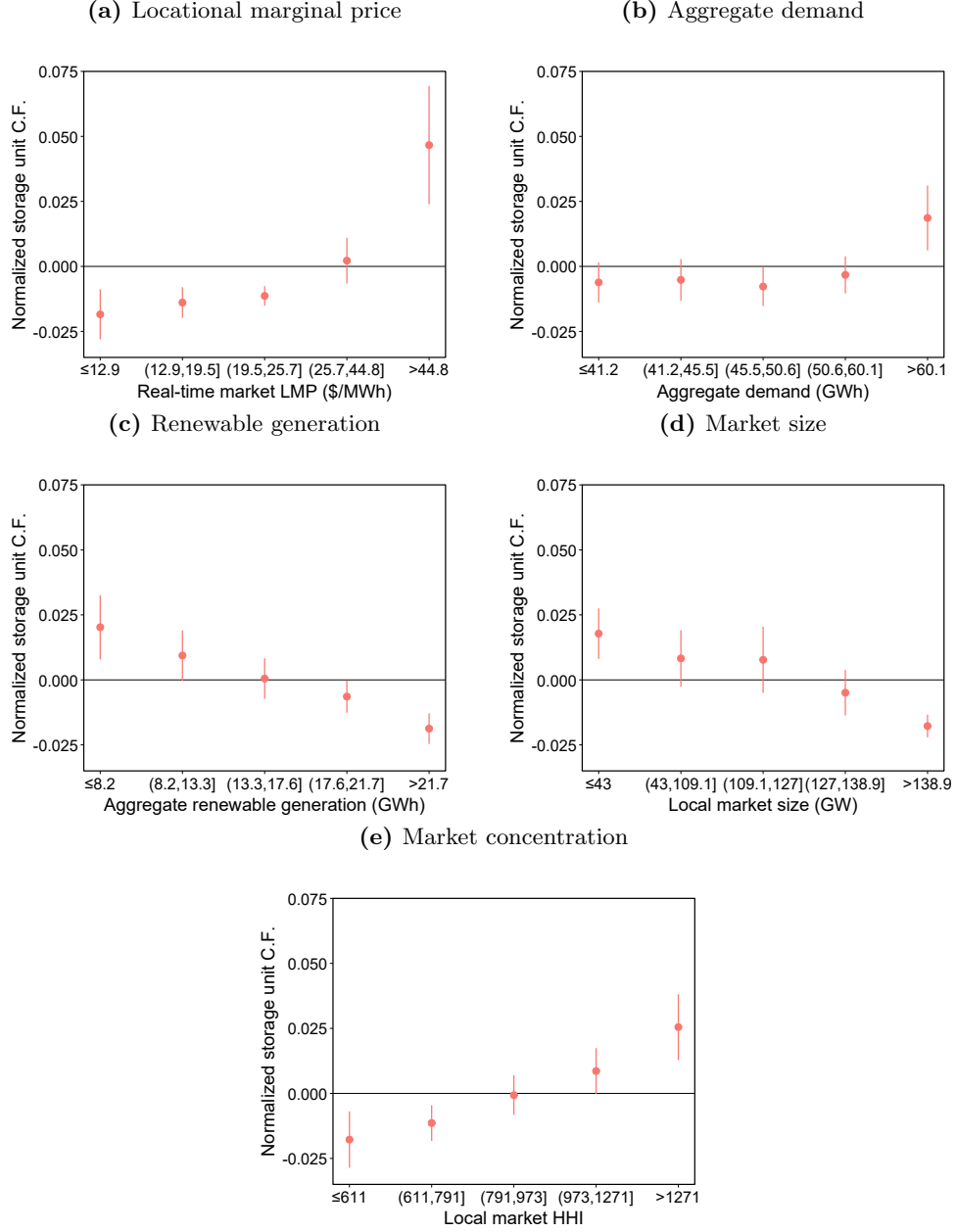
Figure B.3: Example of local market definition



Notes: Panel (a) and (b) show an example of market definition for intervals 15 minutes apart. Each dot is a generator, with colors representing distinct local markets. In panel (c), there is no congestion and all generators are in the same local market, while in panel (d), congestion splits the generators into 22 local markets. Across the sample, the number of local markets varies from 1 to 137, with a median of 8. Figure B.2 presents the distribution of local markets statistics within my sample.

B.3 Storage Operation Patterns by Market Conditions

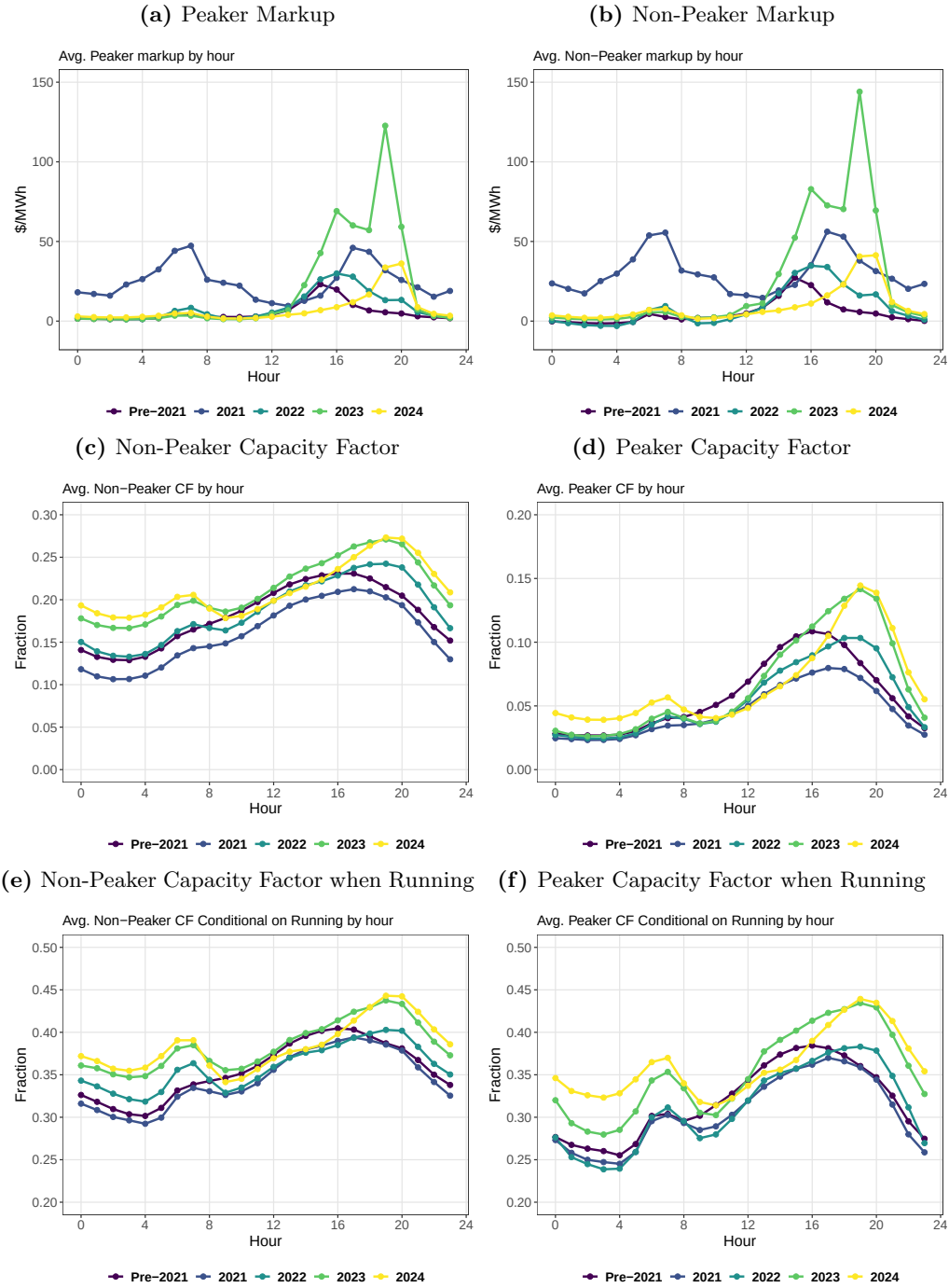
Figure B.4: Battery storage capacity factor by market conditions



Notes: Each panel shows the estimated conditional mean and 95% confidence interval of storage capacity factor for each quantile of the explanatory variables. Market size is measured by total capacity installed, while market concentration is measured by the Herfindahl-Hirschman Index (HHI) computed from installed capacity share within each local market. I estimate the following model in the spirit of [Lamp and Samano \(2022\)](#) with storage unit capacity factor as the dependent variable: $\text{StorageCF}_{it} = \sum_q \beta_q \mathbb{1}(X_{it} \in q\text{-th quantile of } \mathbf{X}) + \varepsilon_{it}$, where X_{it} is the explanatory variable of interest for generator i at time t . LMP, market size, and market concentration vary by generator and interval, while aggregate demand and renewable generation vary by interval. Standard errors are two-way clustered at the generator and day level.

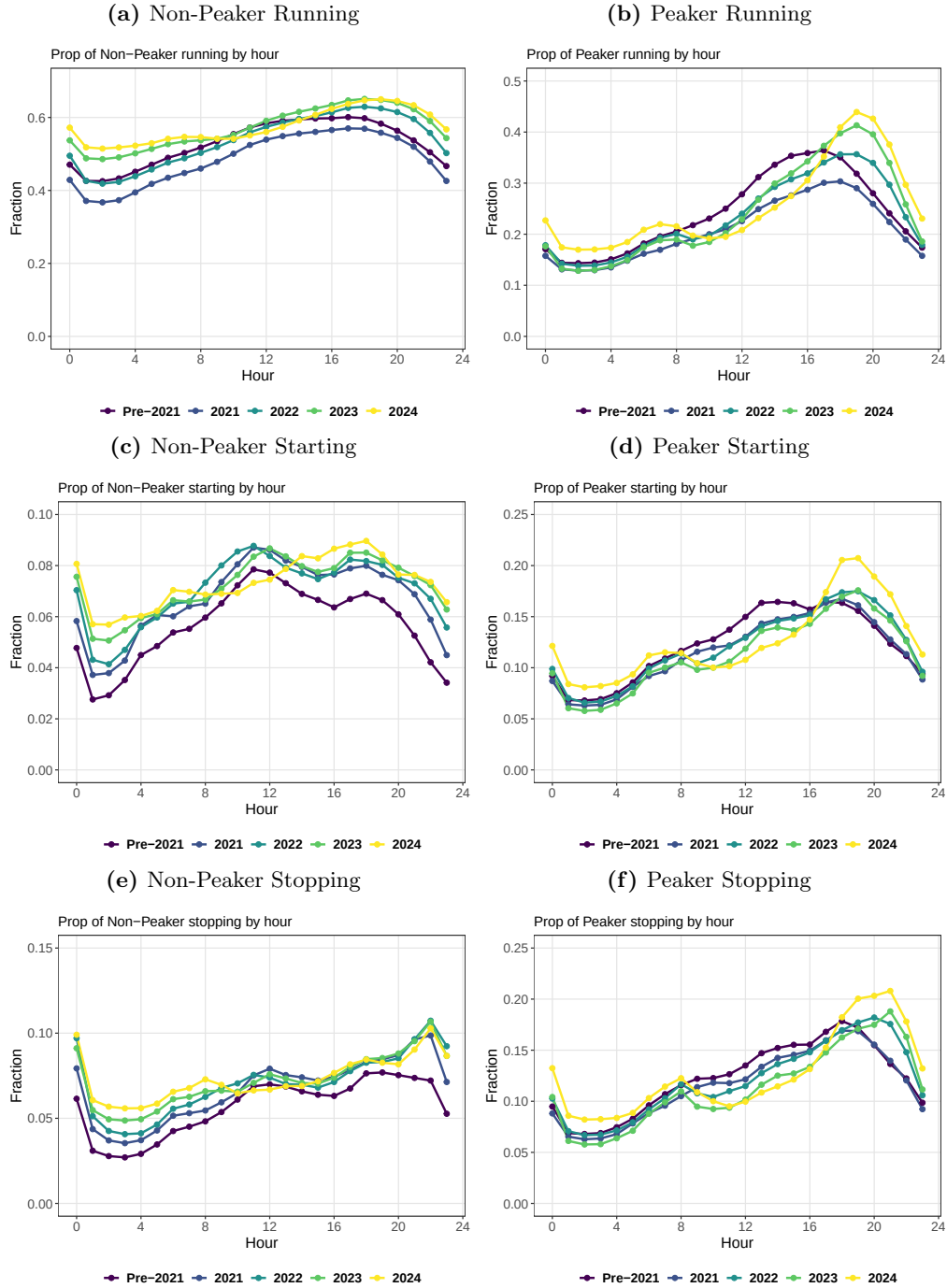
B.4 Natural Gas Generators Descriptive Statistics

Figure B.5: Natural gas unit markup and capacity factor by hour of day



Notes: Each panel shows the average of each variable for each hour of the day, plotted separately for the pre-2021 period and each year from 2021 through 2024.

Figure B.6: Natural gas unit operation pattern (running, starting, stopping) by hour of day

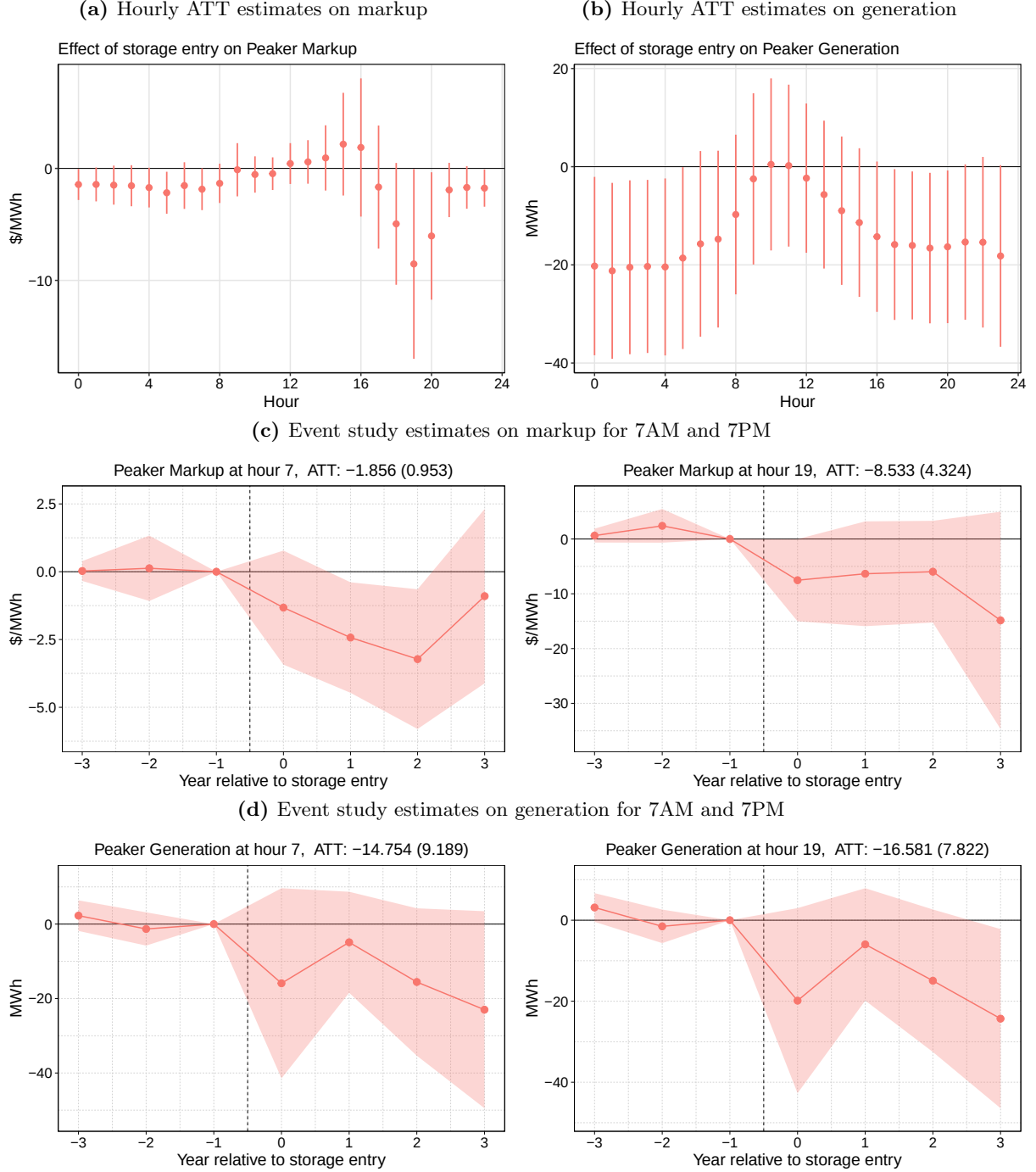


Notes: Each panel shows the average for each indicator variable for each hour of the day, plotted separately for the pre-2021 period and each year from 2021 through 2024.

C Detailed Robustness Check Results

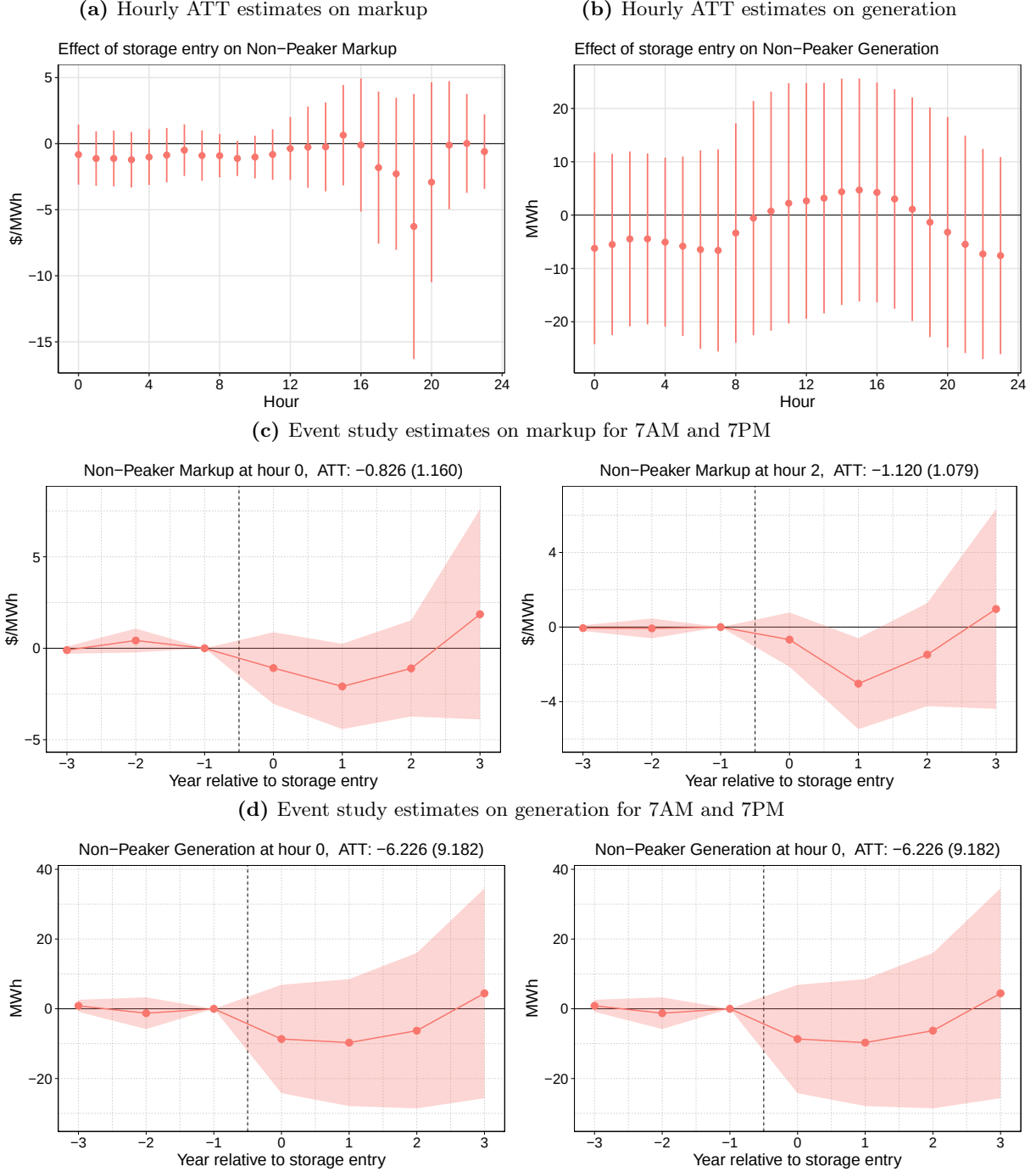
C.1 Main Results with 15-minutes Interval Data

Figure C.7: Main results for Peaker generators using 15-minute level data



Note. Panel A and B presents the coefficient estimates of β^{gh} for $g = \text{Peaker}$ from estimating equation (11). Panel C and D presents figures show the coefficient estimates of δ_s^{gh} for $g = \text{Peaker}$ for $h = 7, 19$ from estimating an event study in equation (13). The line shows the 95% confidence interval clustered at the daily level. The peaker sample include 177 treated generator and 34 never treated generator. Table E.2 presents the balance table for peaker generators.

Figure C.8: Main results for Non-Peaker generators using 15-minute level data



Note. Panel A and B presents the coefficient estimates of β^{gh} for $g = \text{Peaker}$ from estimating equation (11). Panel C and D presents the coefficient estimates of δ_s^{gh} for $g = \text{Peaker}$ for $h = 7, 19$ from estimating an event study in equation (13). The line shows the 95% confidence interval clustered at the daily level. The non-peaker sample include 250 treated generator and 97 never treated generator. Table E.3 presents the balance table for non-peaker generators.

C.2 Main Results allowing for Spillover

In the main specification, I define the local market as county, which might be an approximation. To test the robustness of my results to this approximation, I follow the approach of Butts (2023a) and Butts (2023b) and use the frequency that the storage units interact with natural gas generators in the same local market pre-entry, i.e. treatment intensity, as a notion of distance. This allows generators that are not located in the same county to receive spillover effects of storage if they rarely separated by a binding transmission constraint.

I re-estimate equation (11) by adding an interaction term for treatment intensity bin with indicator for control groups. Specifically, I estimate the following specification separately for each hour h and technology type g :

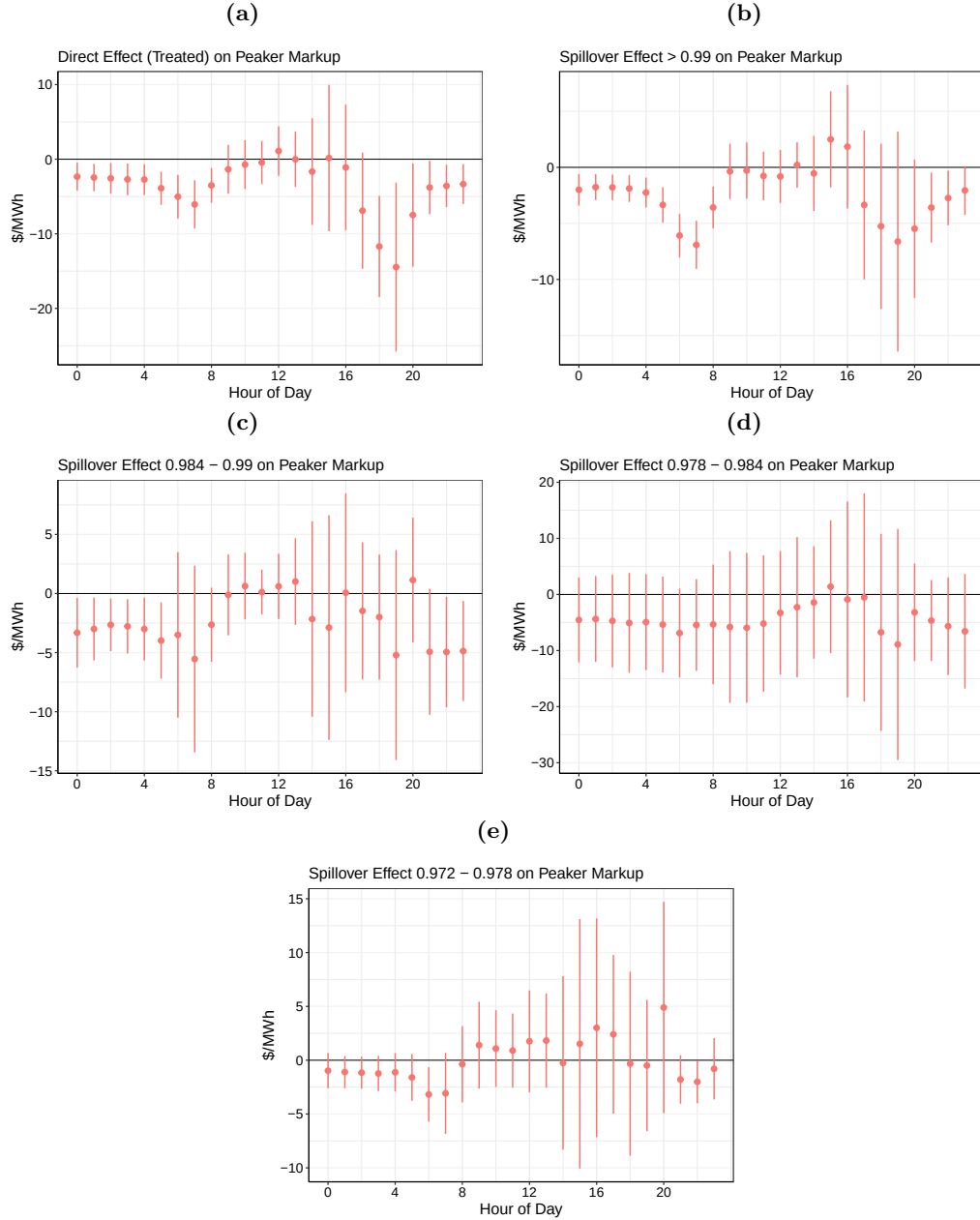
$$(C.21) \quad Y_{it}^{gh} = \alpha_i^{gh} + \mu_t^{gh} + \beta^{gh} D_{it} + \sum_j \gamma_j^{gh} \mathbb{1}[\text{Intensity}_{it} \in \mathcal{F}_j] \times (1 - D_{it}) + \mathbf{X}_{it} \theta^{gh} + \epsilon_{it}^{gh},$$

where Intensity_{it} is the fraction of interval that generator i shares the same local market with a storage unit before the storage's COD. The coefficient γ_j^{gh} captures the spillover effect of storage to natural gas generators that are not in the same county, but the Intensity_{it} falls into intensity bin $\mathcal{F}_j$. If γ_j^{gh} is not significant, this suggests that spillover effect is not a concern and that the main results are robust to the definition of local market. If γ_j^{gh} is significant, this suggests that spillover effect is present and that the main results might be biased if we do not account for it. In this case, the estimates of β^{gh} would capture the effect of storage entry on generators that are located in the same county. The implementation of the two-stage estimation stay the same except that the first stage estimation now use only generators that is assumed to not affected by the spillover at all (generators for which the treatment intensity falls into the excluded bin).

I focus on hour 19 when the treatment effect for peaker is most pronounced. Figure C.10 presents the results of this robustness exercise. In Panel A, I start by including only generator with intensity greater than 99% in the spillover group. In Panel B through D, I then continually add bins with an increment of 0.6% intensity, and find that the spillover decay beyond 98.4% intensity. Specifically, the estimated spillover for generators in different county with intensity between 97.8% and 98.4% which serves as a placebo is statistically zero. For generators with intensity greater than 98.4%, the spillover effect is significant and positive, which suggests that storage entry has a positive effect on the markups of these generators, but the magnitude of this effect is not statistically different from the negative effect on generators in the same county. This suggests that while there might be some spillover effect of storage entry on nearby generators, the main competitive effect of storage entry is still concentrated on generators in the same county.

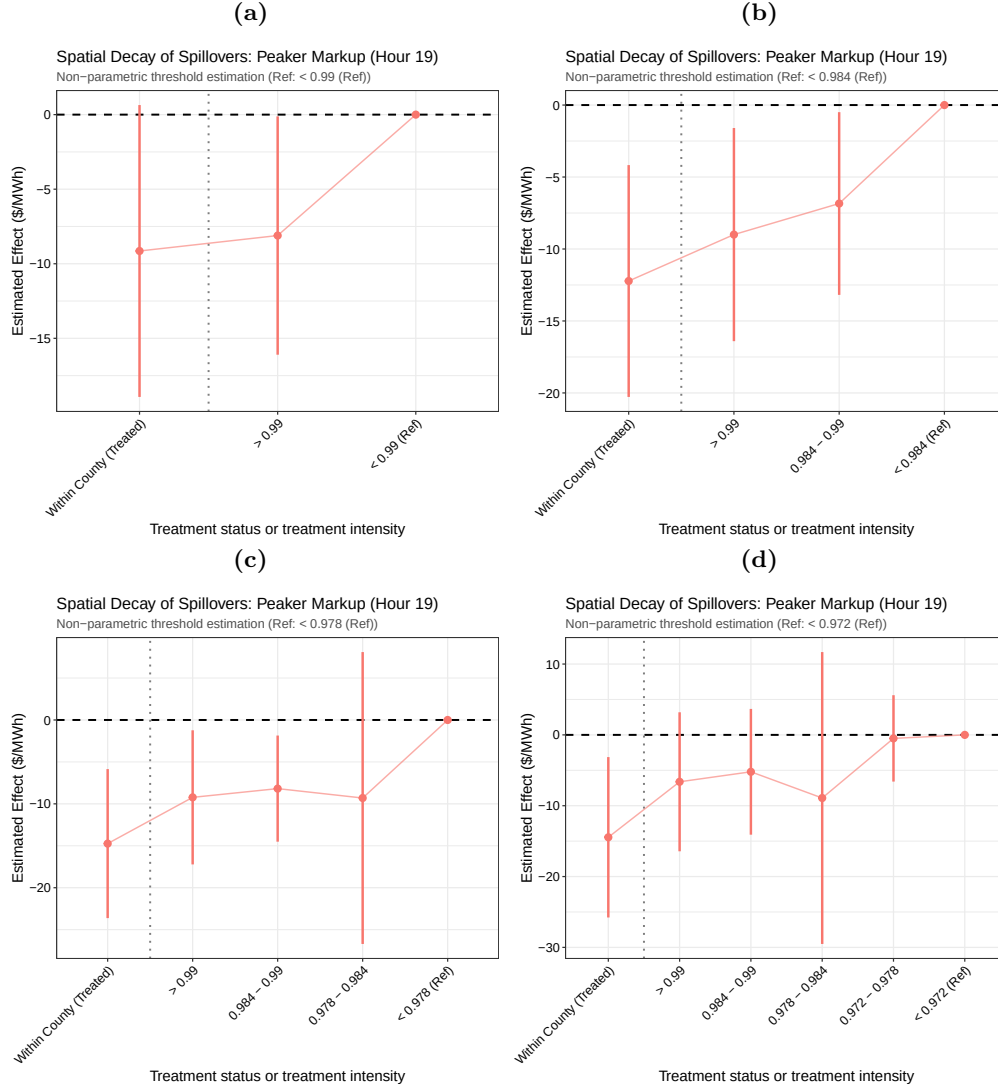
Figure C.9 presents the direct effect of storage in the same county from estimating equation (C.21). Overall, these results suggest that while there might be some spillover effect of storage entry on nearby generators, the main competitive effect of storage entry is concentrated on generators in the same county, and thus my main results are robust to potential spillover effects.

Figure C.9: Effect of storage entry on Peaker generators controlling for spillover effect



Note. Panel A show the coefficient estimates of β^{gh} for $g = \text{Peaker}$ from estimating equation (C.21). Panel B to D show the coefficient estimates of γ_j^{gh} for where $\mathcal{F}_j \in \{[0.99, 1], [0.984, 0.99], [0.978, 0.984], [0.972, 0.978]\}$. The line shows the 95% confidence interval clustered at the county level.

Figure C.10: Spatial spillover robustness check for Peaker generators

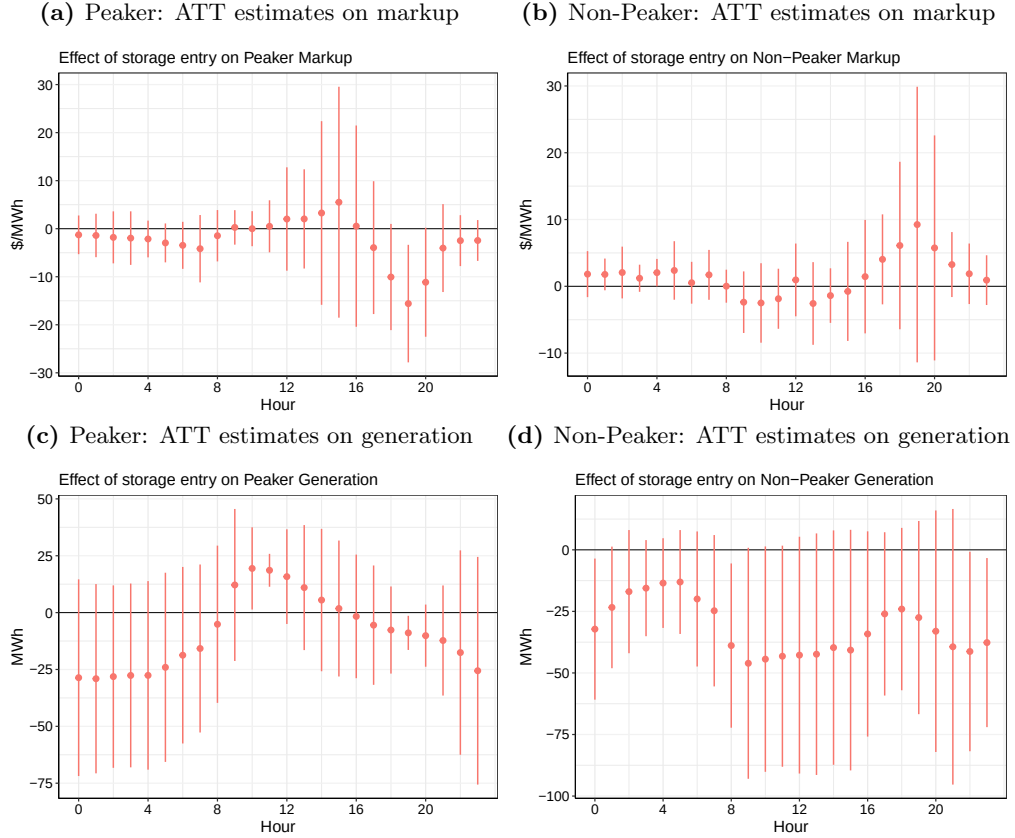


Note. These figures show the coefficient estimates of β^{gh} and γ_j^{gh} for where $\mathcal{F}_j \subset \{[0.99, 1], [0.984, 0.99], [0.978, 0.984], [0.972, 0.978]\}$ for $g = \text{Peaker}$ for $h = 19$ from estimating equation (C.21). The line shows the 95% confidence interval clustered at the county level.

C.3 Main Results using Other Staggered DD Estimator

Here I present the main results using the [Callaway and Sant'Anna \(2021\)](#) estimator. To simplify the procedure, I residualize the outcome variable by estimating a regression of the outcome on the control variables and all the fixed effects except the generator and week-of-sample fixed effects. I then use the residuals in the subsequent analysis. The results shown in Figure C.11 are consistent with the main results. The complete set of event study figures are shown in Figure C.12 and C.13. The results are consistent with the main results, and the magnitude of the estimates are similar.

Figure C.11: Main results for Peaker and Non-Peaker generators



Note. Panels A and B present the coefficient estimates of β^{gh} from estimating Equation (11) for peakers ($g = \text{Peaker}$) and non-peakers ($g = \text{Non-Peaker}$), with markup as the outcome. Panels C and D show the analogous estimates with generation as the outcome. The line shows the 95% confidence interval clustered at the county level. The peaker sample includes 177 treated generators and 34 never-treated generators; the non-peaker sample includes 250 treated generators and 97 never-treated generators. Balance tables are reported in Appendix Tables E.2 and E.3.

Figure C.12: Complete event study figure for Peaker markup using [Callaway and Sant'Anna \(2021\)](#) estimator

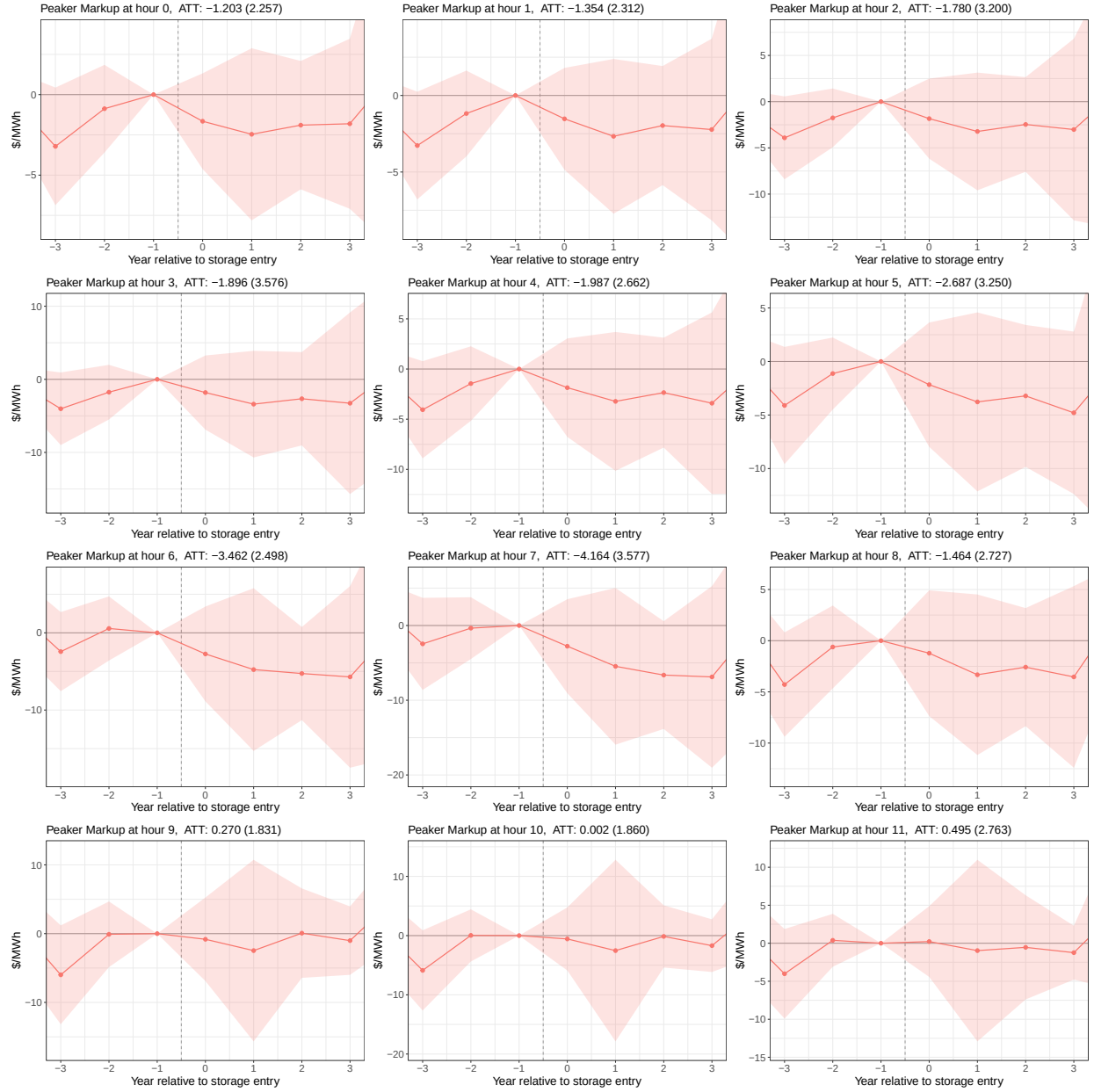


Figure C.12: (Continued) Complete event study figure for Peaker markup using [Callaway and Sant'Anna \(2021\)](#) estimator

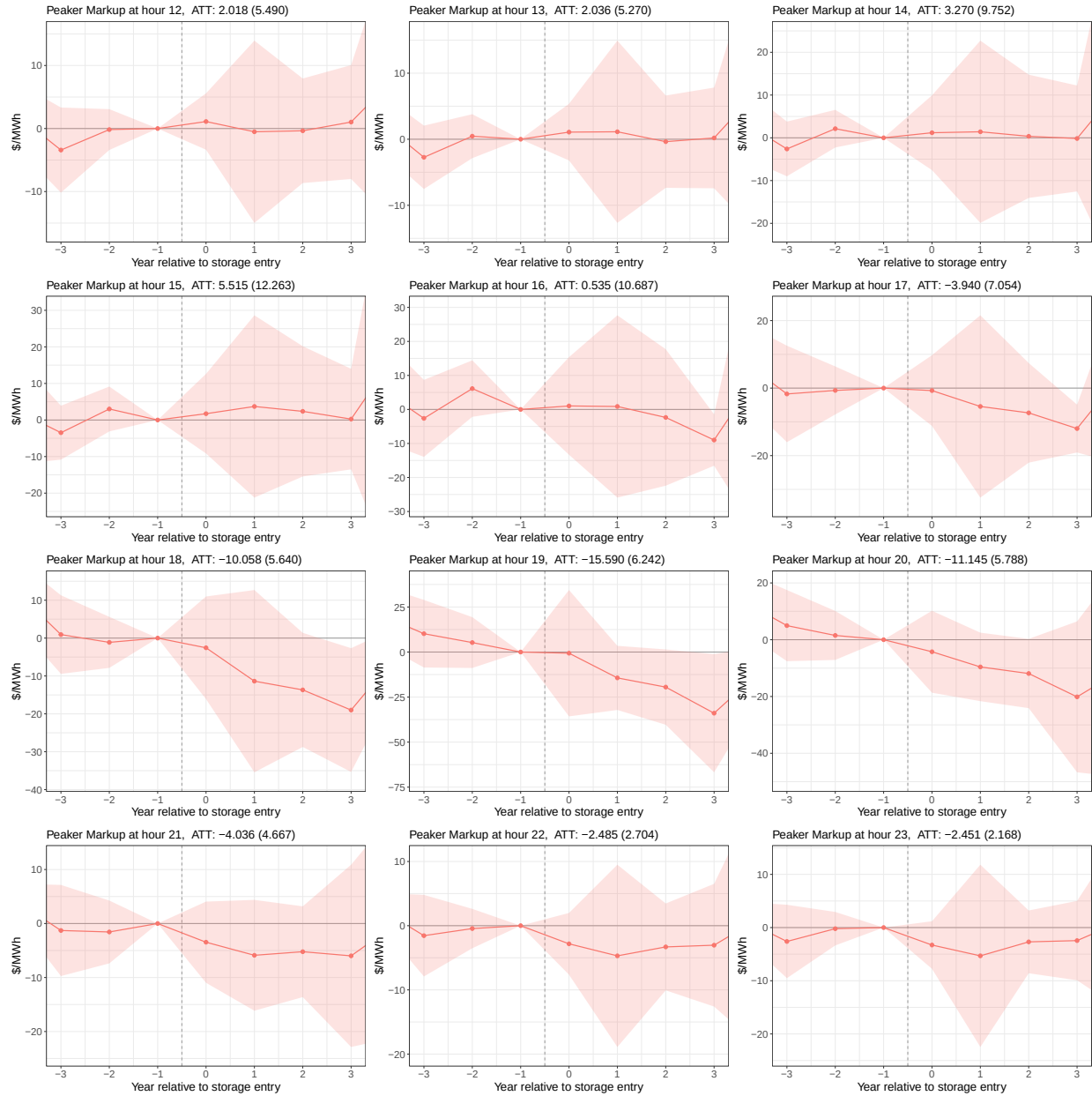


Figure C.13: Complete event study figure for Peaker generation using [Callaway and Sant'Anna \(2021\)](#) estimator

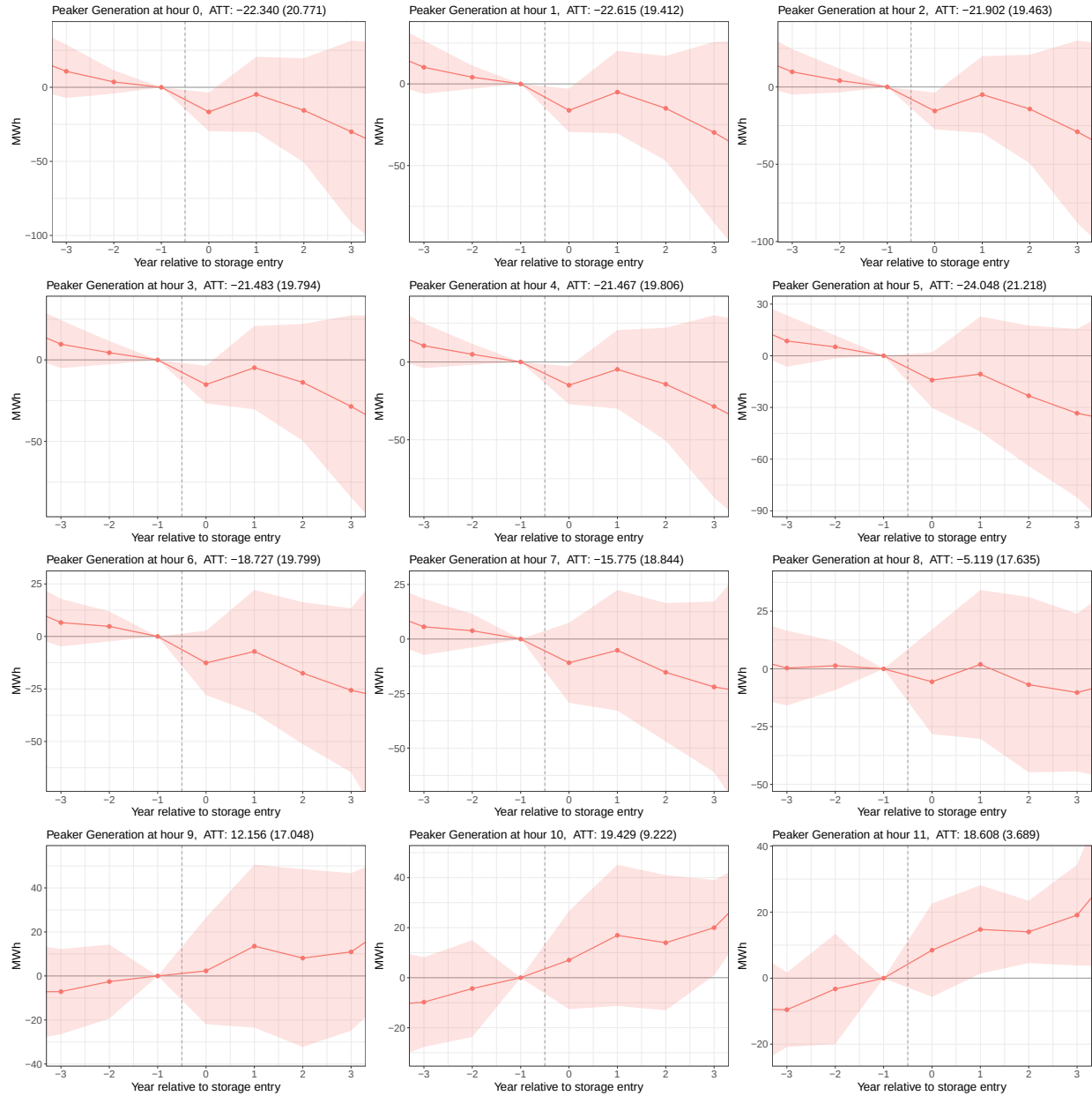
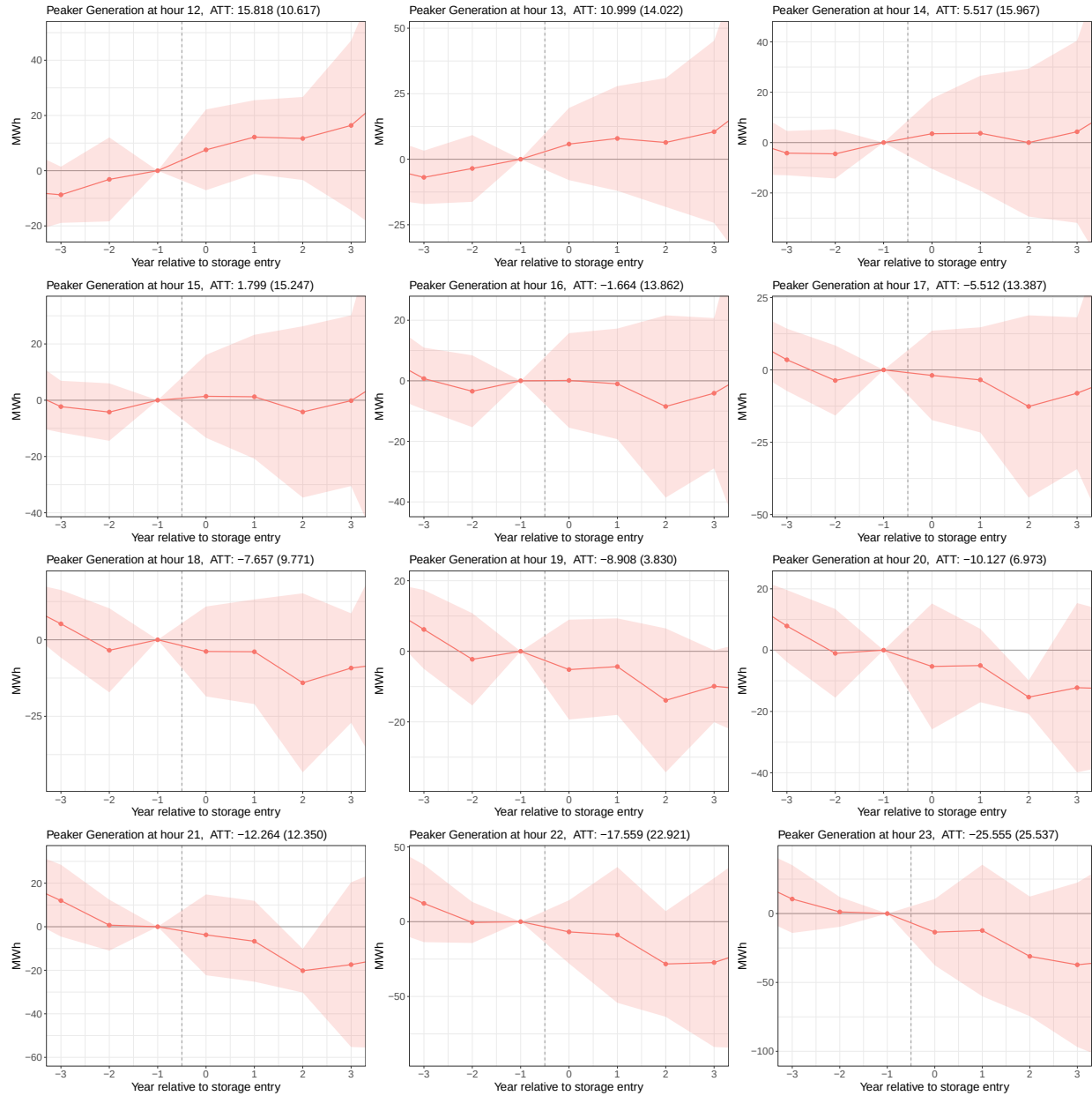


Figure C.13: (Continued) Complete event study figure for Peaker generation using [Callaway and Sant'Anna \(2021\)](#) estimator

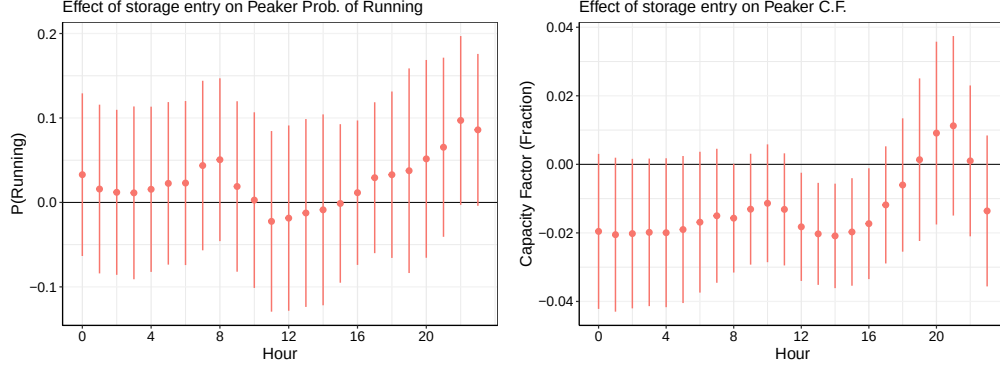


D Additional Results

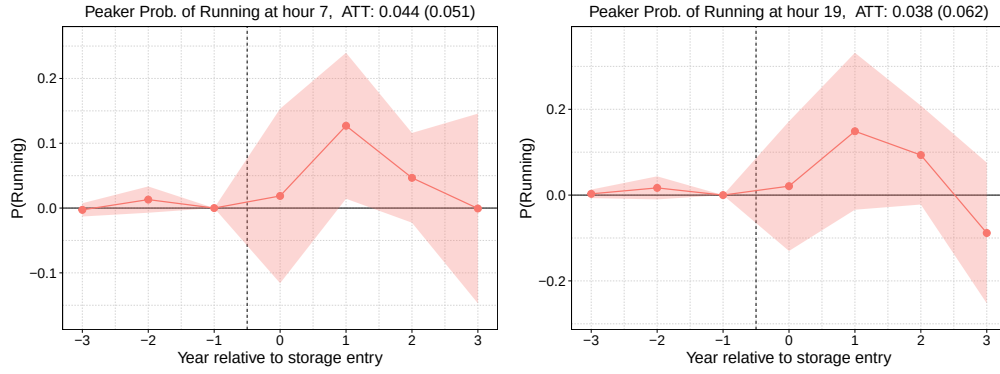
D.1 Results for Other Outcome Variables

Figure D.14: Results for Peaker generators using other outcome variables

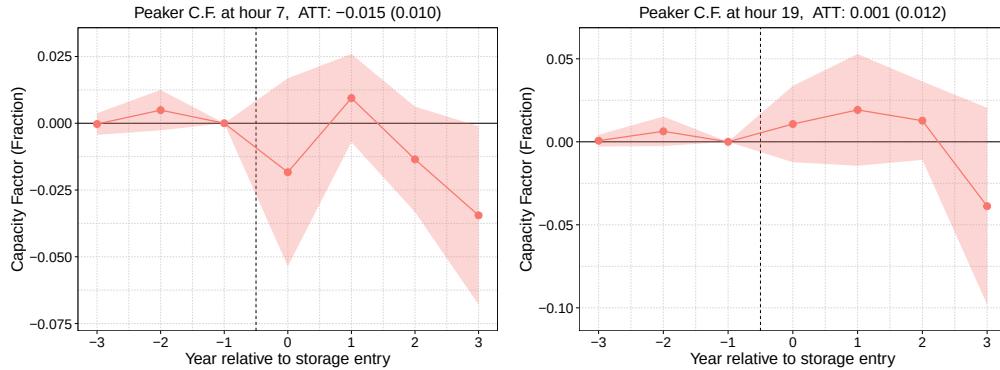
(a) Hourly ATT estimates on P(Running) (b) Hourly ATT estimates on Capacity Factor



(c) Event study estimates on markup for 7AM and 7PM



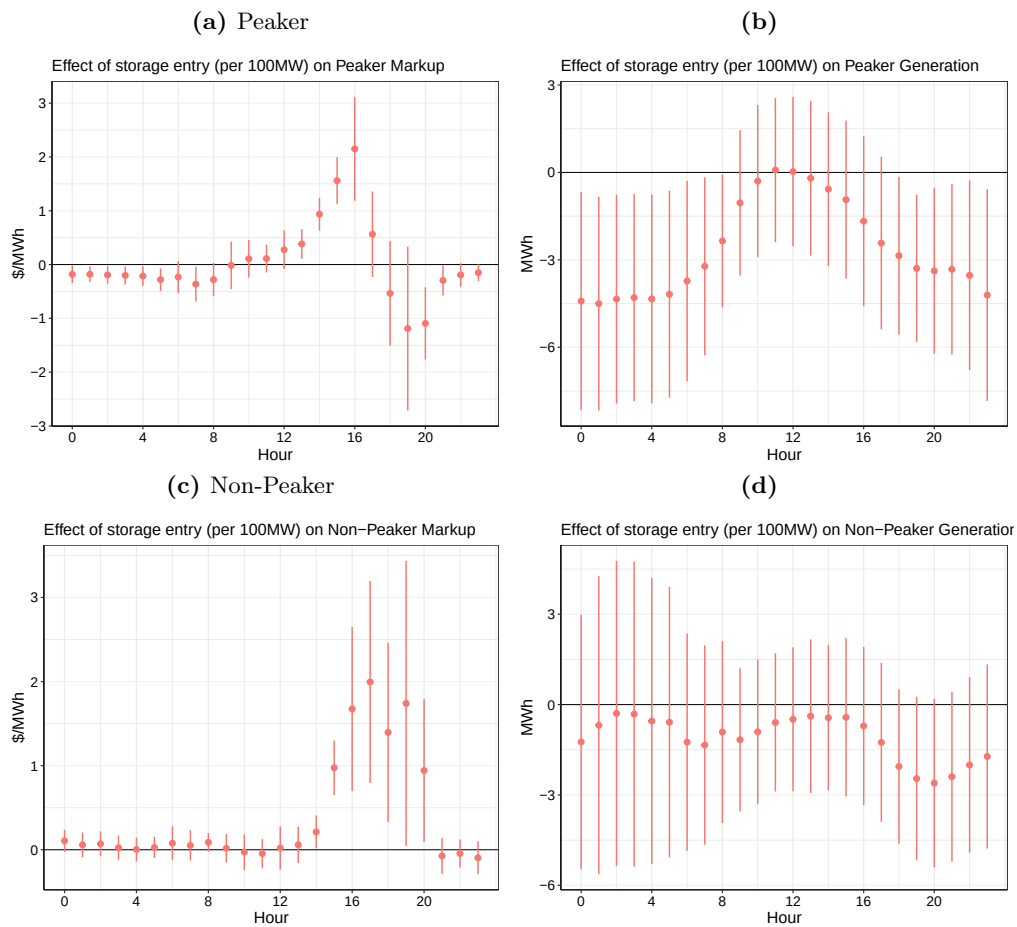
(d) Event study estimates on generation for 7AM and 7PM



Note. Panel A and B presents the coefficient estimates of β^{gh} for $g = \text{Peaker}$ from estimating equation (11). Panel C and D presents figures show the coefficient estimates of δ_s^{gh} for $g = \text{Peaker}$ for $h = 7, 19$ from estimating an event study in equation (13). The line shows the 95% confidence interval clustered at the daily level. The peaker sample include 177 treated generator and 34 never treated generator. Table E.2 presents the balance table for peaker generators.

D.2 Results for Continuous Treatment

Figure D.15: Effect of storage capacity on Markup and Generation

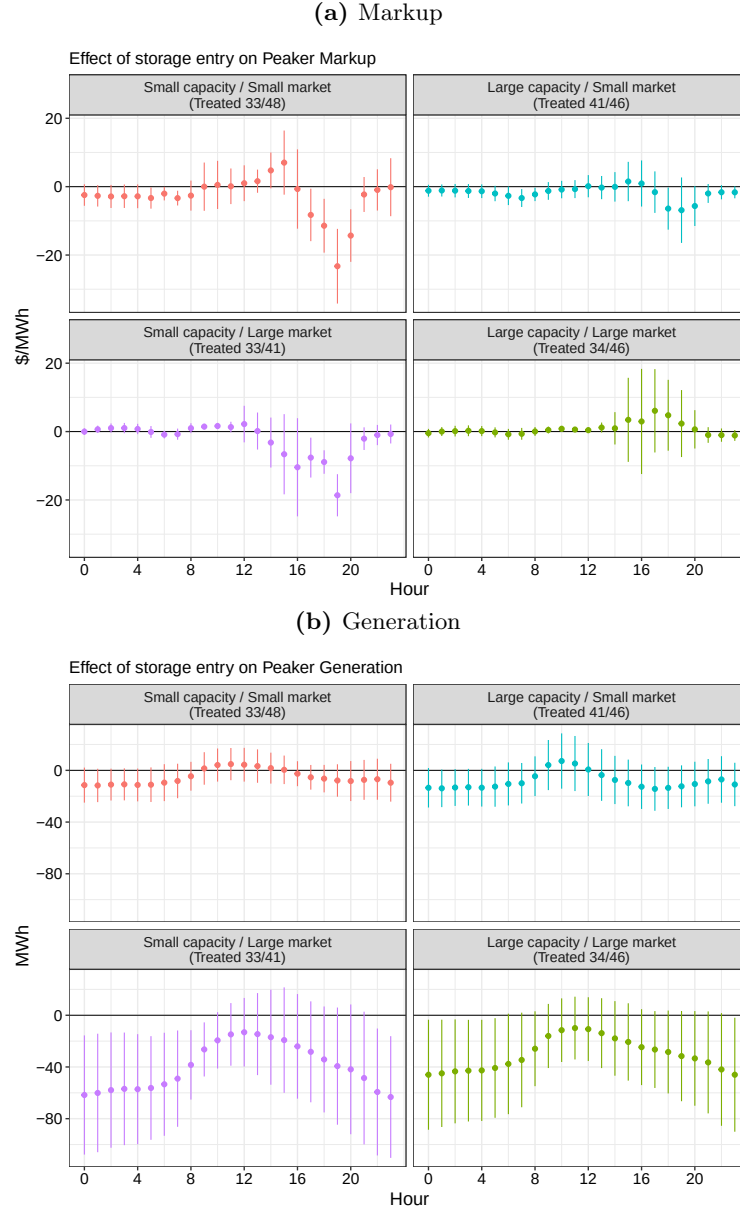


Note. These figures show the coefficient estimates of β^{gh} for $g = \text{Peaker}$ from estimating equation (15). The line shows the 95% confidence interval clustered at the county level.

D.3 Results for Heterogeneity Analysis

This section presents the treatment effect for all hours from the heterogeneity analysis in Section 5.2 and 5.3.1. The results are presented separately for Peaker and Non-Peaker generators, and for the discrete and continuous treatment specifications.

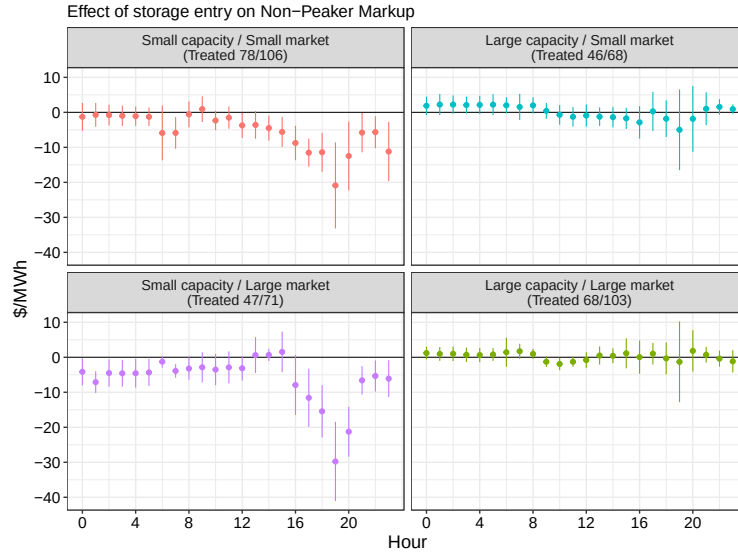
Figure D.16: Heterogeneity results for Peaker generators



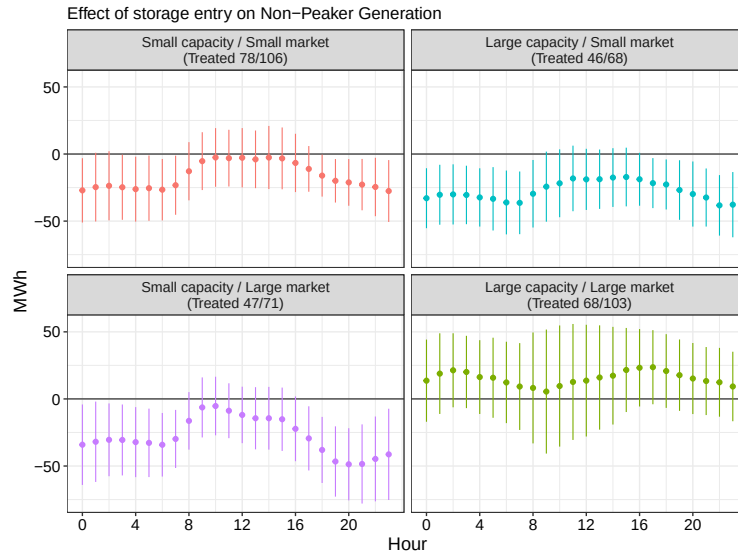
Note. These figures show the four heterogeneous effects constructed from the linear combinations: small capacity, small market: β^{gh} ; large capacity, small market: $\beta^{gh} + \gamma_C^{gh}$; small capacity, large market: $\beta^{gh} + \gamma_M^{gh}$; large capacity, large market: $\beta^{gh} + \gamma_C^{gh} + \gamma_M^{gh} + \gamma_{CM}^{gh}$, from estimating Equation (14) for $g = \text{Peaker}$. Standard errors are obtained by the delta method and clustered at the county level. The line shows the 95% confidence interval. The number of treated generators in each group is shown on each label.

Figure D.17: Heterogeneity results for Non-Peaker generators

(a) Markup

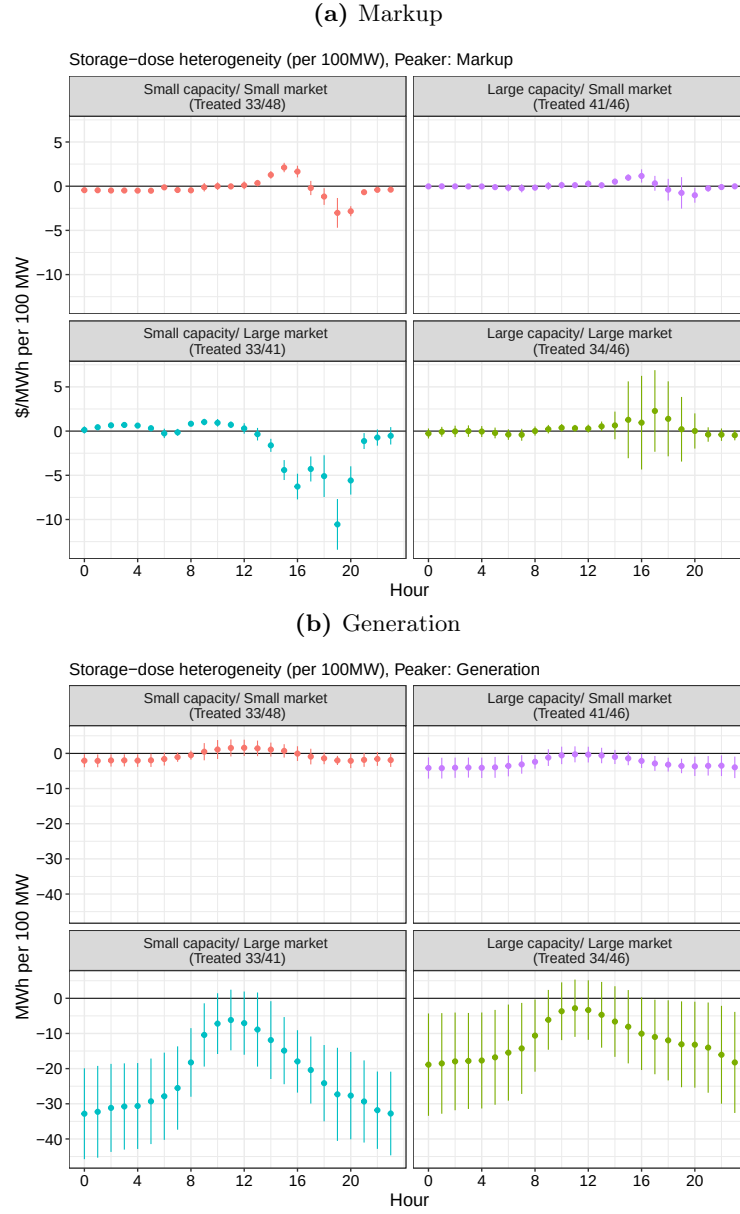


(b) Generation



Note. These figures show the four heterogeneous effects constructed from the linear combinations from estimating Equation (14) for $g = \text{Non-Peaker}$. Standard errors are obtained by the delta method and clustered at the county level. The line shows the 95% confidence interval. The number of treated generators in each group is shown on each label.

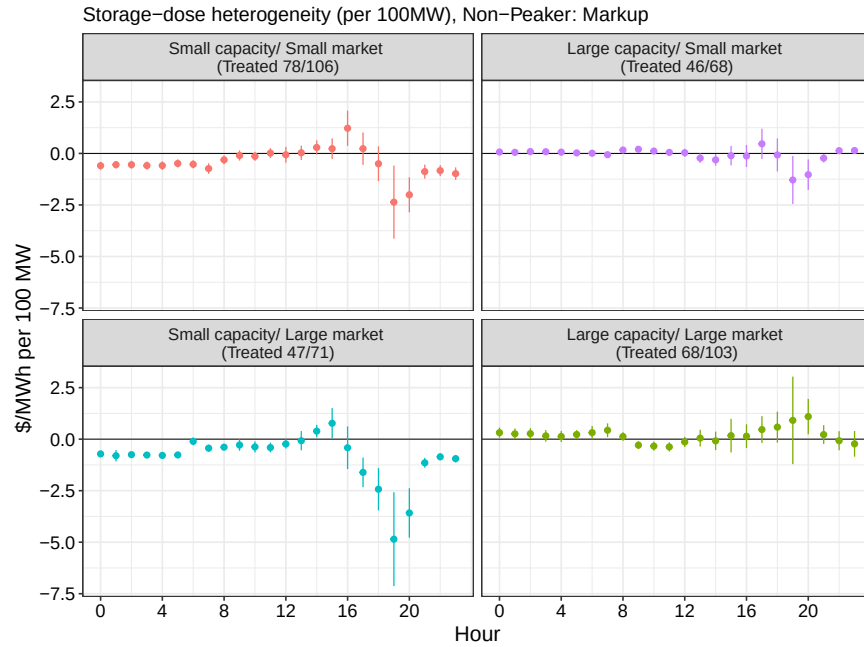
Figure D.18: Heterogeneity results using continuous treatment for Peaker generators



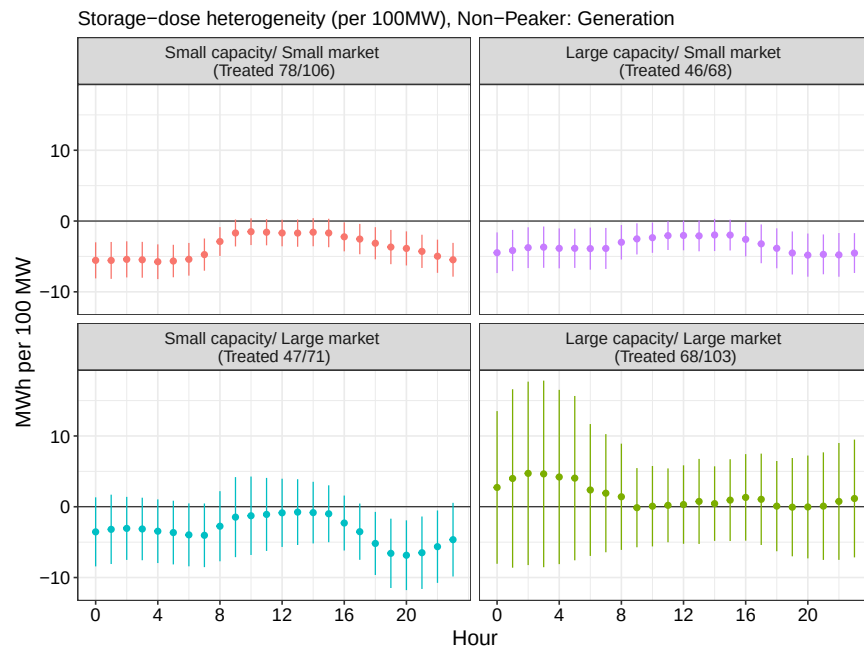
Note. These figures show the four heterogeneous effects from estimating an analog of Equation (14) using continuous treatment for $g = \text{Peaker}$. Standard errors are obtained by the delta method and clustered at the county level. The line shows the 95% confidence interval. The number of treated generators in each group is shown on each label.

Figure D.19: Heterogeneity results using continuous treatment for Non-Peaker generators

(a) Markup



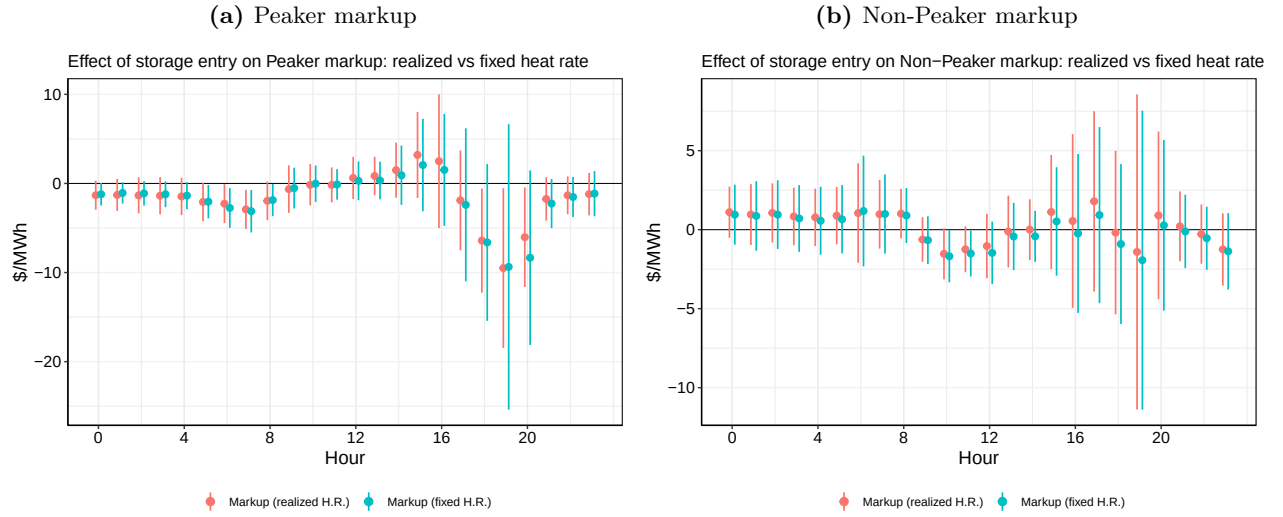
(b) Generation



Note. These figures show the four heterogeneous effects from estimating an analog of Equation (14) using continuous treatment for $g = \text{Non-Peaker}$. Standard errors are obtained by the delta method and clustered at the county level. The line shows the 95% confidence interval. The number of treated generators in each group is shown on each label.

D.4 Results for Mechanism Decomposition

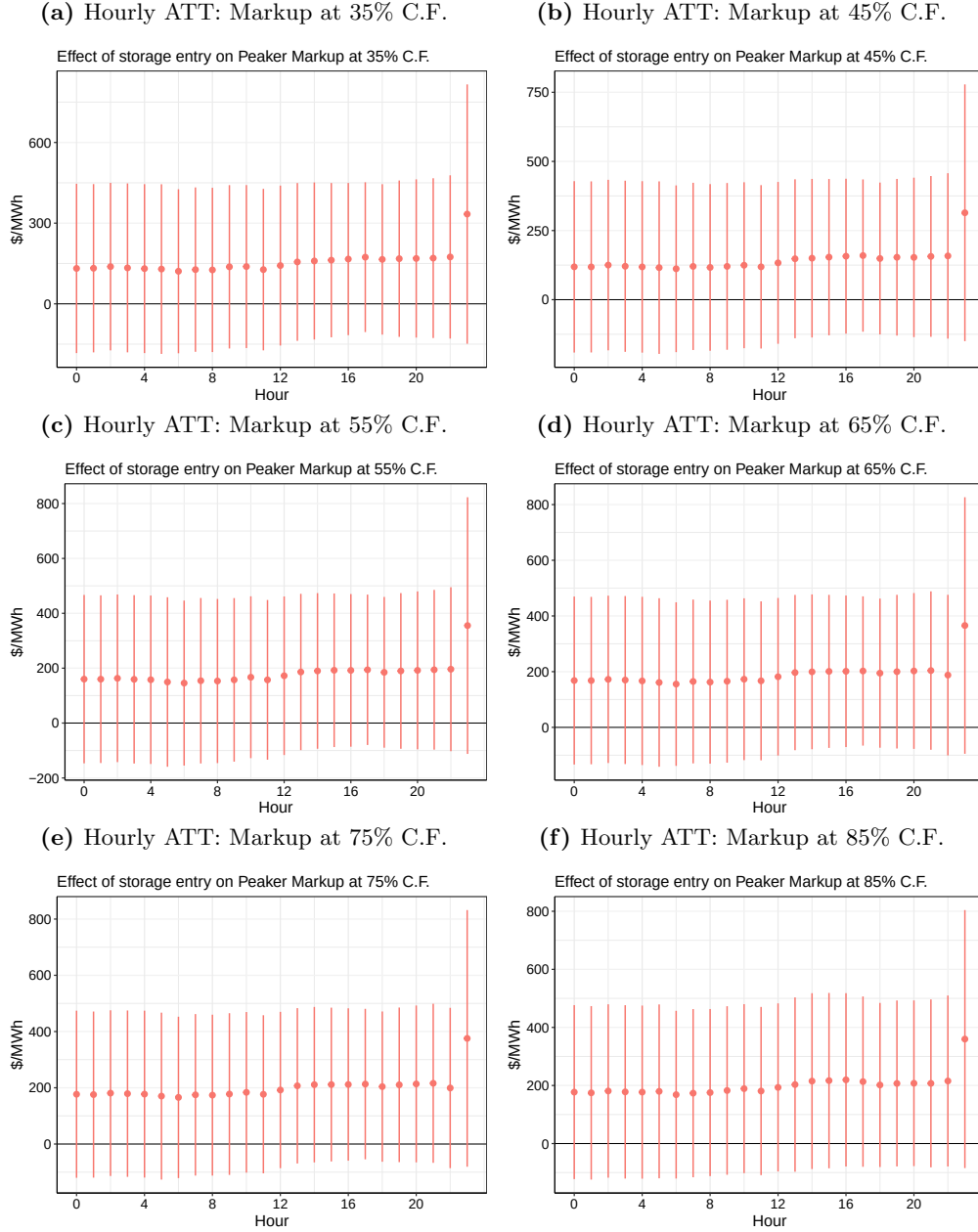
Figure D.20: Decomposing the competitive channel from the efficiency channel



Note. These figures present the coefficient estimates of β^{gh} from estimating Equation (11) using markup computed using the realized heat rate and pre-storage heat rate. The line shows the 95% confidence interval clustered at the county level.

D.5 Results for Bidding Behavior

Figure D.21: Bid Results for Peaker generators



Note. Panel A and B presents the coefficient estimates of β^{gh} for $g = \text{Peaker}$ from estimating equation (11). The line shows the 95% confidence interval clustered at the daily level. The peaker sample include 177 treated generator and 34 never treated generator. Table E.2 presents the balance table for peaker generators.

Figure D.21: (Continued) Bid Results for Peaker generators

(g) Hourly ATT: Offer Slope at 35-60% C.F. (h) Hourly ATT: Offer Slope at 60-85% C.F.

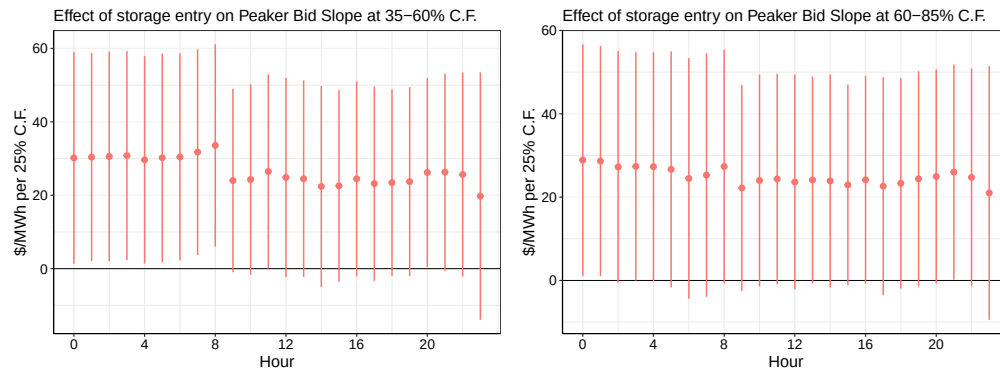
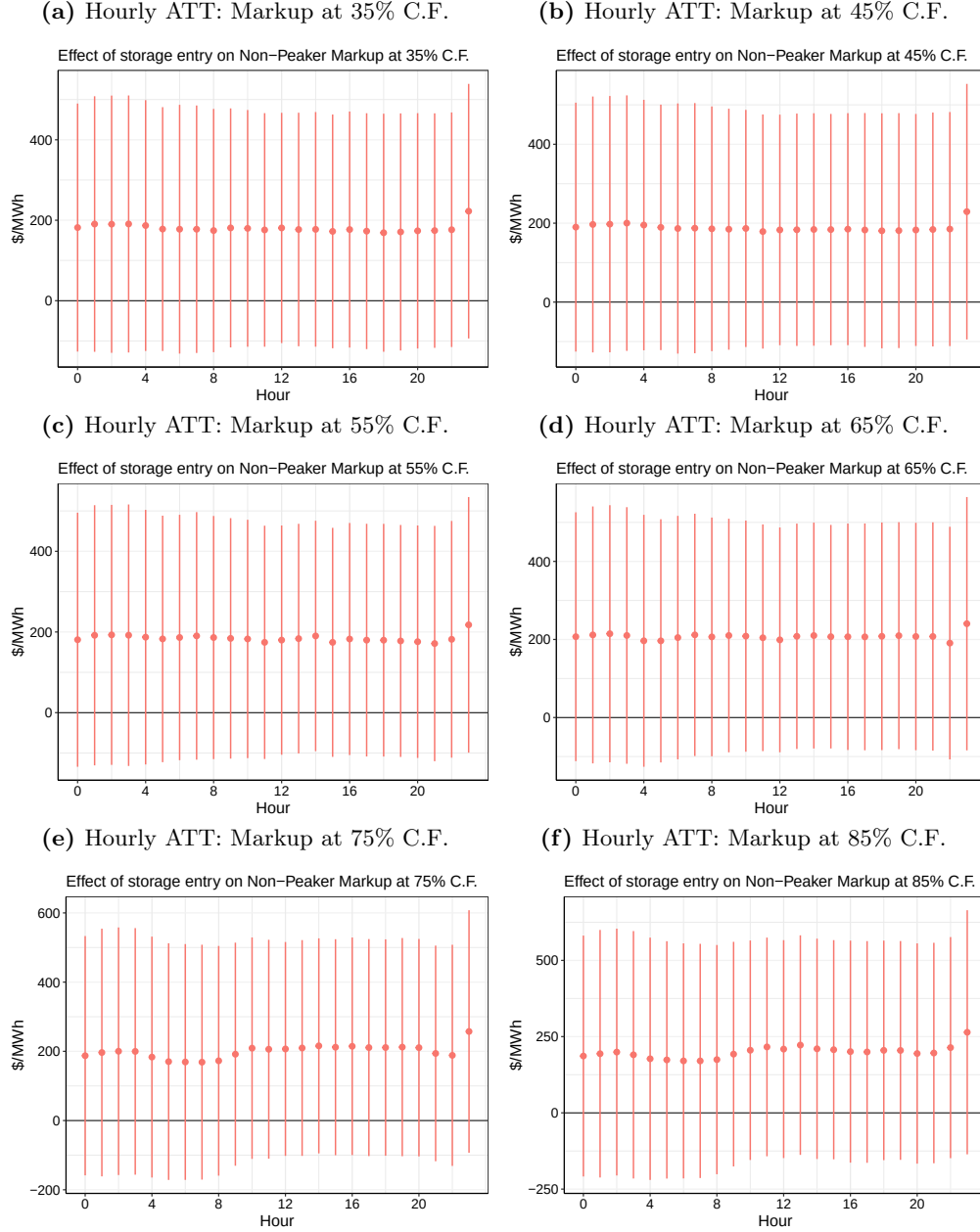


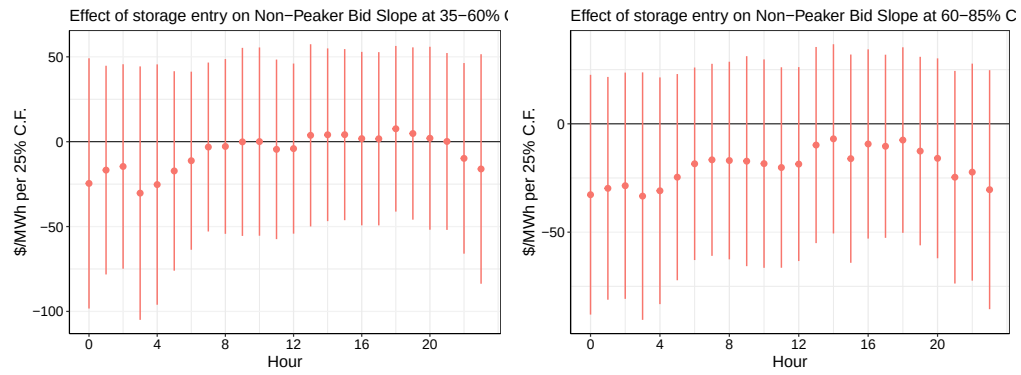
Figure D.22: Bid Results for Non-Peaker generators



Note. Panel A and B presents the coefficient estimates of β^{gh} for $g = \text{Peaker}$ from estimating equation (11). The line shows the 95% confidence interval clustered at the daily level. The peaker sample include 177 treated generator and 34 never treated generator. Table E.2 presents the balance table for peaker generators.

Figure D.22: (Continued) Bid Results for Non-Peaker generators

(g) Hourly ATT: Offer Slope at 35-60% C.F. (h) Hourly ATT: Offer Slope at 60-85% C.F.



E Additional Tables

Table E.1: Generation Mix Summary

	Capacity			Mean Generation		Units	
	GW	Duration (hr)	%	GWh	%	Count	%
<i>Panel A: Fuel Type</i>							
Natural Gas	65.23		38.9	20.482	44.5	159	24.2
Coal	20.19		12.1	8.252	17.9	17	2.6
Wind	41.57		24.8	10.453	22.7	192	29.2
Solar PV	26.63		15.9	2.062	4.5	133	20.2
Batteries	8.04	1.33	4.8	0.043	0.1	142	21.6
Nuclear	5.23		3.1	4.666	10.1	2	0.3
Other	0.64		0.4	0.079	0.2	12	1.8
Total	167.54		100.0	46.036	100.0	657	100.0
<i>Panel B: Top 5 Natural Gas Firms</i>							
Luminant	12.91		19.8	4.767	21.7	18	11.3
Calpine Corporation	10.04		15.4	6.877	31.3	17	10.7
Constellation Energy	6.09		9.3	2.007	9.1	10	6.3
NRG Energy	5.64		8.6	0.943	4.3	12	7.6
City of San Antonio	5.53		8.5	1.449	6.6	10	6.3
Others	25.03		38.4	5.943	27.0	92	57.9
Total	65.23		100.0	21.985	100.0	159	100.0
<i>Panel C: Top 5 Storage Firms</i>							
Engie S.A.	1.83	1.02	22.8	0.017	27.8	24	16.9
Enel S.p.A.	1.19	1.69	14.8	0.014	22.6	15	10.6
Plus Power	0.79	1.92	9.8	0.009	14.8	4	2.8
Key Capture Energy	0.63	1.83	7.8	0.002	3.5	11	7.8
Luminant	0.44	2.05	5.5	0.003	5.4	4	2.8
Others	3.16	1.22	39.3	0.016	25.8	84	59.1
Total	8.04		100.0	0.061	100.0	142	100.0

Notes. Panel A show capacity, generation, and number of units by fuel type. Panel B show capacity, generation, and number of units by firms for natural gas unit, while Panel C show specifically for storage firms. Discrepancies in generation between each panel arise from differences in operating intervals when aggregating generation over multiple units. I define capacity using maximum observed generation from my data, this creates discrepancies in generation capacity compared to (ERCOT, 2025). The largest firm in ERCOT owns 15 percent of total capacity and contributes, on average, about 20 percent of electricity generation in each interval. The second to fifth largest firms own between 4 to 6 percent of total capacity, contributing on average between 5 to 14.6 percent of electricity generation. For storage firms, the largest owns 21.9 percent of storage capacity, while the second to fifth largest own between 3.6 to 10 percent of storage capacity. Based on just the top five storage firms, there is significant heterogeneity in storage characteristics in terms of unit capacity and duration.

Table E.2: Balance Table for Peaker Generator

	All Generators			Treated Generators			Never-Treated Generators		
	All	12AM -12PM	12PM -12AM	All	12AM -12PM	12PM -12AM	All	12AM -12PM	12PM -12AM
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Outcome Variables (2018–2021)									
LMP (\$/MWh)	29.1 (93.9)	22.0 (50.5)	36.2 (122.4)	29.4 (95.1)	22.2 (51.7)	36.6 (123.8)	27.2 (86.6)	20.6 (43.4)	33.8 (114.1)
Capacity Factor	0.06 (0.14)	0.04 (0.12)	0.08 (0.16)	0.06 (0.14)	0.04 (0.12)	0.08 (0.16)	0.05 (0.14)	0.03 (0.10)	0.07 (0.17)
Probability Marginal	0.07 (0.25)	0.07 (0.26)	0.06 (0.24)	0.08 (0.27)	0.09 (0.28)	0.07 (0.26)	0.02 (0.13)	0.01 (0.11)	0.02 (0.15)
Markup (\$/MWh)	1.36 (74.15)	-1.95 (32.39)	4.67 (99.63)	1.02 (73.93)	-2.35 (32.46)	4.39 (99.27)	3.33 (75.37)	0.34 (31.90)	6.32 (101.62)
Panel B: Market Structure (2018–2021)									
HHI	1027 (1334)	970 (1210)	1084 (1445)	1020 (1386)	960 (1261)	1080 (1498)	1062 (1047)	1019 (923)	1105 (1157)
Market Size (GW)	85.7 (32.1)	87.2 (30.9)	84.2 (33.2)	87.7 (30.0)	89.3 (28.6)	86.1 (31.3)	75.9 (39.3)	77.2 (38.8)	74.6 (39.7)
# Firms per Market	79.4 (28.7)	81.0 (27.8)	77.8 (29.5)	81.0 (27.5)	82.6 (26.4)	79.3 (28.4)	71.7 (33.2)	73.0 (32.9)	70.5 (33.4)
Panel C: Generator Characteristics									
Heat Rate (MMBtu/MWh)		11.8 (2.8)			11.8 (2.9)			11.7 (2.3)	
CO ₂ Rate (tons/MWh)		0.5 (0.2)			0.5 (0.2)			0.5 (0.2)	
NO _x Rate (lbs/MWh)		2.202 (2.804)			1.978 (2.348)			3.372 (4.353)	
SO ₂ Rate (lbs/MWh)		0.014 (0.070)			0.015 (0.077)			0.006 (0.004)	
Marginal Cost (\$/MWh)		35.03 (9.89)			35.10 (10.34)			34.63 (7.17)	
Avg. Capacity (MW)		108 (106)			107 (96)			112 (151)	
Panel D: Sample Composition									
# of Unique Generators		211			177			34	
# of Counties		25			19			9	
# in Houston Zone		121			111			10	
# in North Zone		16			13			3	
# in South Zone		54			43			11	
# in West Zone		20			10			10	

Notes.

Table E.3: Balance Table for Non-Peaker Generator

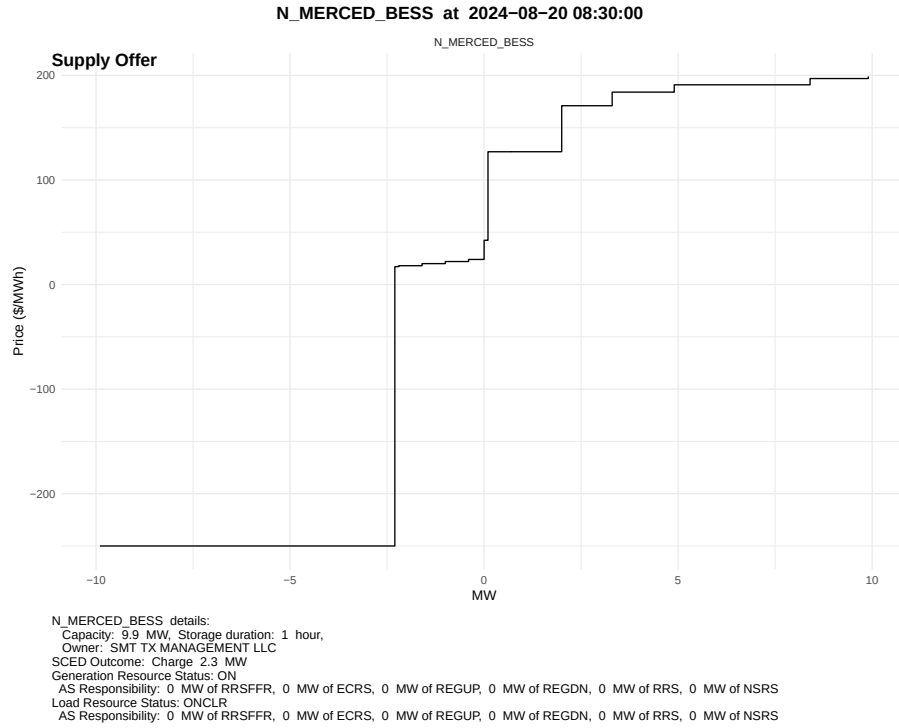
	All Generators			Treated Generators			Never-Treated Generators		
	All	12AM –12PM	12PM –12AM	All	12AM –12PM	12PM –12AM	All	12AM –12PM	12PM –12AM
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Outcome Variables (2018–2021)									
LMP (\$/MWh)	28.2 (92.4)	21.3 (49.0)	35.1 (120.8)	28.4 (94.1)	21.4 (50.3)	35.4 (122.8)	27.8 (87.4)	21.1 (45.2)	34.5 (114.7)
Capacity Factor	0.16 (0.20)	0.14 (0.18)	0.19 (0.22)	0.15 (0.20)	0.13 (0.18)	0.17 (0.21)	0.20 (0.21)	0.17 (0.19)	0.23 (0.22)
Probability Marginal	0.08 (0.27)	0.09 (0.28)	0.07 (0.25)	0.08 (0.27)	0.09 (0.28)	0.07 (0.25)	0.07 (0.26)	0.09 (0.28)	0.06 (0.24)
Markup (\$/MWh)	3.09 (75.83)	-1.16 (32.21)	7.34 (102.11)	2.52 (74.92)	-1.54 (31.19)	6.58 (101.10)	4.76 (78.38)	-0.05 (34.96)	9.57 (104.97)
Panel B: Market Structure (2018–2021)									
HHI	992 (1221)	937 (1084)	1046 (1342)	977 (1236)	917 (1082)	1038 (1370)	1030 (1179)	990 (1087)	1070 (1263)
Market Size (GW)	87.7 (29.8)	89.1 (28.5)	86.2 (30.9)	88.8 (28.5)	90.3 (27.0)	87.2 (29.8)	84.7 (32.8)	85.9 (32.0)	83.5 (33.5)
# Firms per Market	80.7 (27.0)	82.2 (26.1)	79.2 (27.9)	81.6 (26.1)	83.2 (25.0)	80.0 (27.1)	78.3 (29.3)	79.5 (28.7)	77.0 (29.7)
Panel C: Generator Characteristics									
Heat Rate (MMBtu/MWh)		10.1 (2.9)			10.2 (2.9)			9.7 (3.2)	
CO ₂ Rate (tons/MWh)		0.6 (0.2)			0.6 (0.2)			0.6 (0.2)	
NO _x Rate (lbs/MWh)		0.512 (0.611)			0.516 (0.655)			0.502 (0.479)	
SO ₂ Rate (lbs/MWh)		0.008 (0.009)			0.008 (0.010)			0.008 (0.006)	
Marginal Cost (\$/MWh)		29.75 (11.73)			30.41 (10.07)			28.04 (15.13)	
Avg. Capacity (MW)		523 (498)			521 (523)			527 (431)	
Panel D: Sample Composition									
# of Unique Generators		347			250			97	
# of Counties		44			29			18	
# in Houston Zone		99			85			14	
# in North Zone		120			55			65	
# in South Zone		98			92			6	
# in West Zone		30			18			12	

Notes.

F Additional Figures

Figure F.23: Example Storage Offer

(a) Example of unconstrained storage offer



(b) Example of energy constrained storage offer

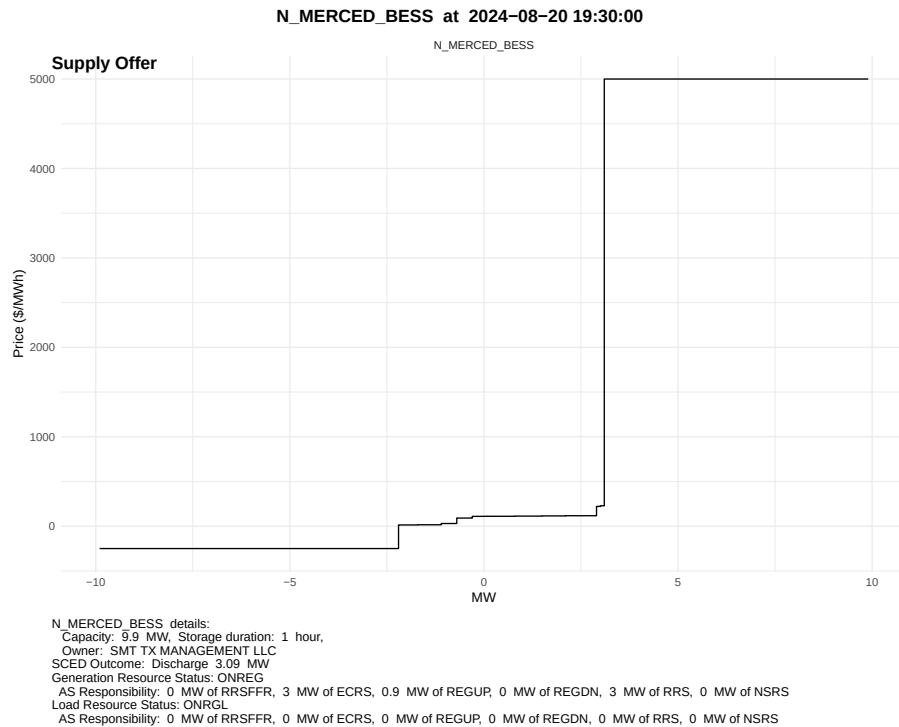
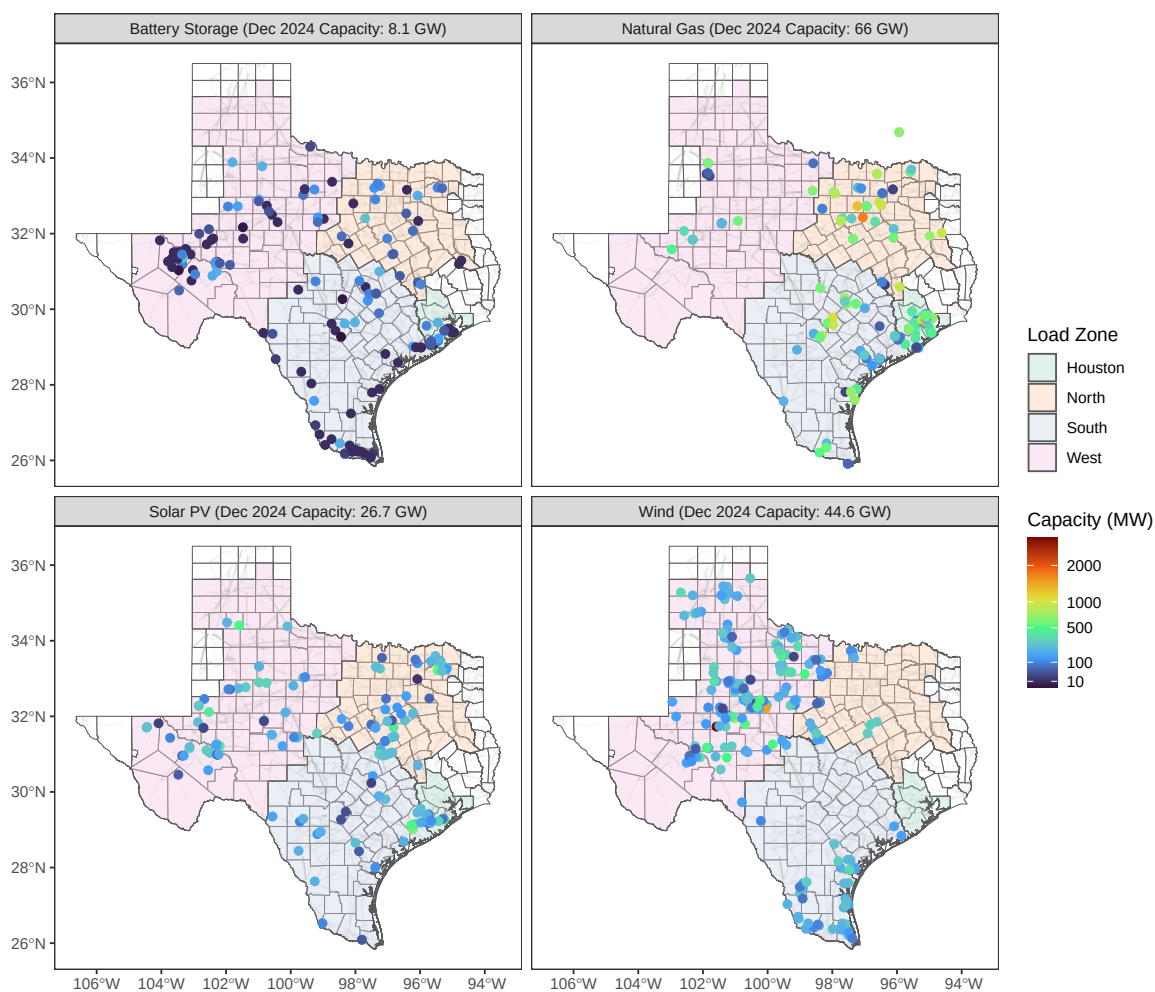


Figure F.24: Location of Generator in ERCOT



Notes: This figure shows the spatial distribution of each generator unit based on their fuel type in ERCOT. Natural gas and coal units are concentrated in the populated area of Texas, where the demand center is, such as the cities of Austin, Dallas, Houston, and San Antonio. Solar PV and storage are relatively spread evenly throughout the state, while wind is concentrated in the West and South load zone, the area with the highest wind potential.

Figure F.25: Installed capacity by Fuel Type from 2018-2024

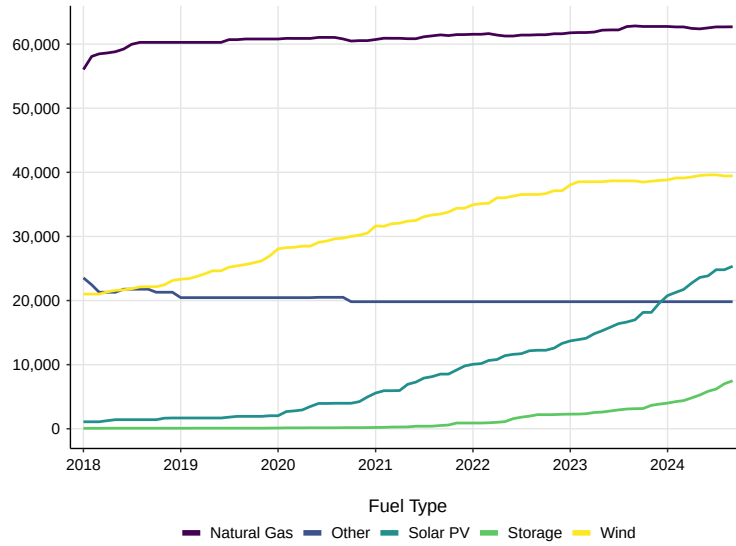
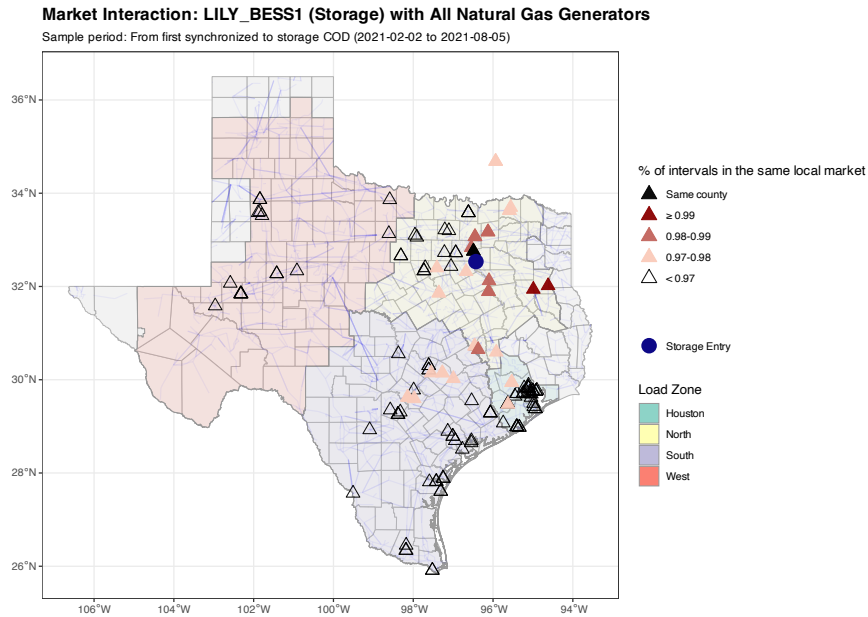
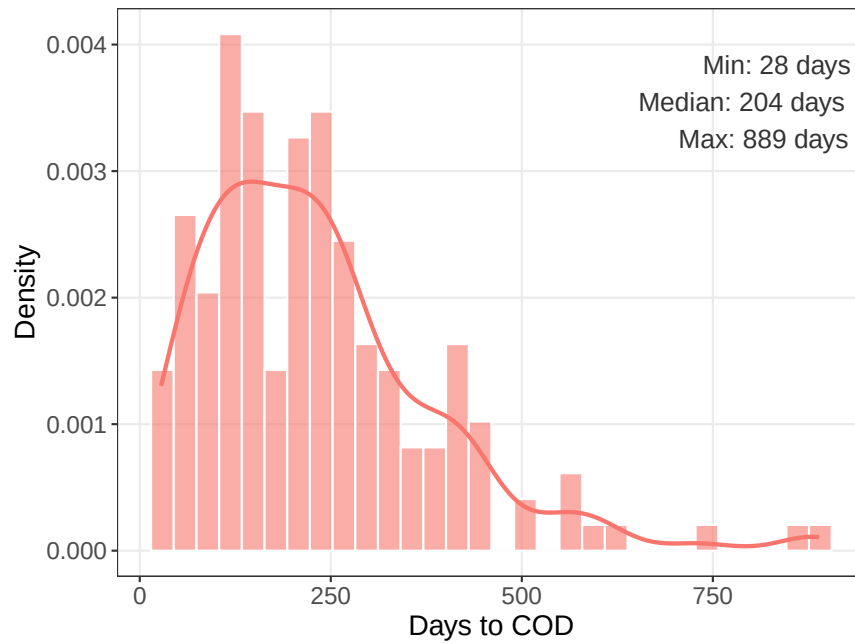


Figure F.26: Treatment assignment and market interaction frequency



Notes: This figure shows which natural gas generators are treated by the LILY_BESS1 storage unit. The blue dot shows the location of the storage unit. The solid black triangle represents generators located in the same county as the storage unit. The red triangle represents generators sharing the same local market with the storage unit more than 97% of the intervals prior to storage COD. The void triangle represents generators that do not share the same local market with the storage unit more than 97% of the intervals prior to storage COD.

Figure F.27: Distribution of storage first observation to COD

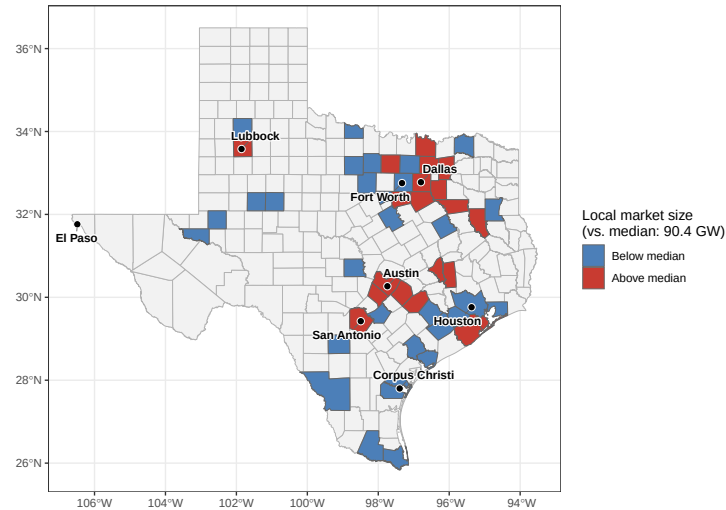


Notes: This figure shows the distribution of the time gap between the first date a storage unit shows up in the SCED data (synchronized) and the COD date.

Figure F.28: Market size by county

Average Local Market Size by County Pre-Storage

Texas counties siting natural gas generators.



Notes: This figure show counties siting natural gas generators in Texas, the color of the county indicates whether the county's local market size is above or below the median local market size.

Figure F.29: Complete event study figure for Peaker Markup

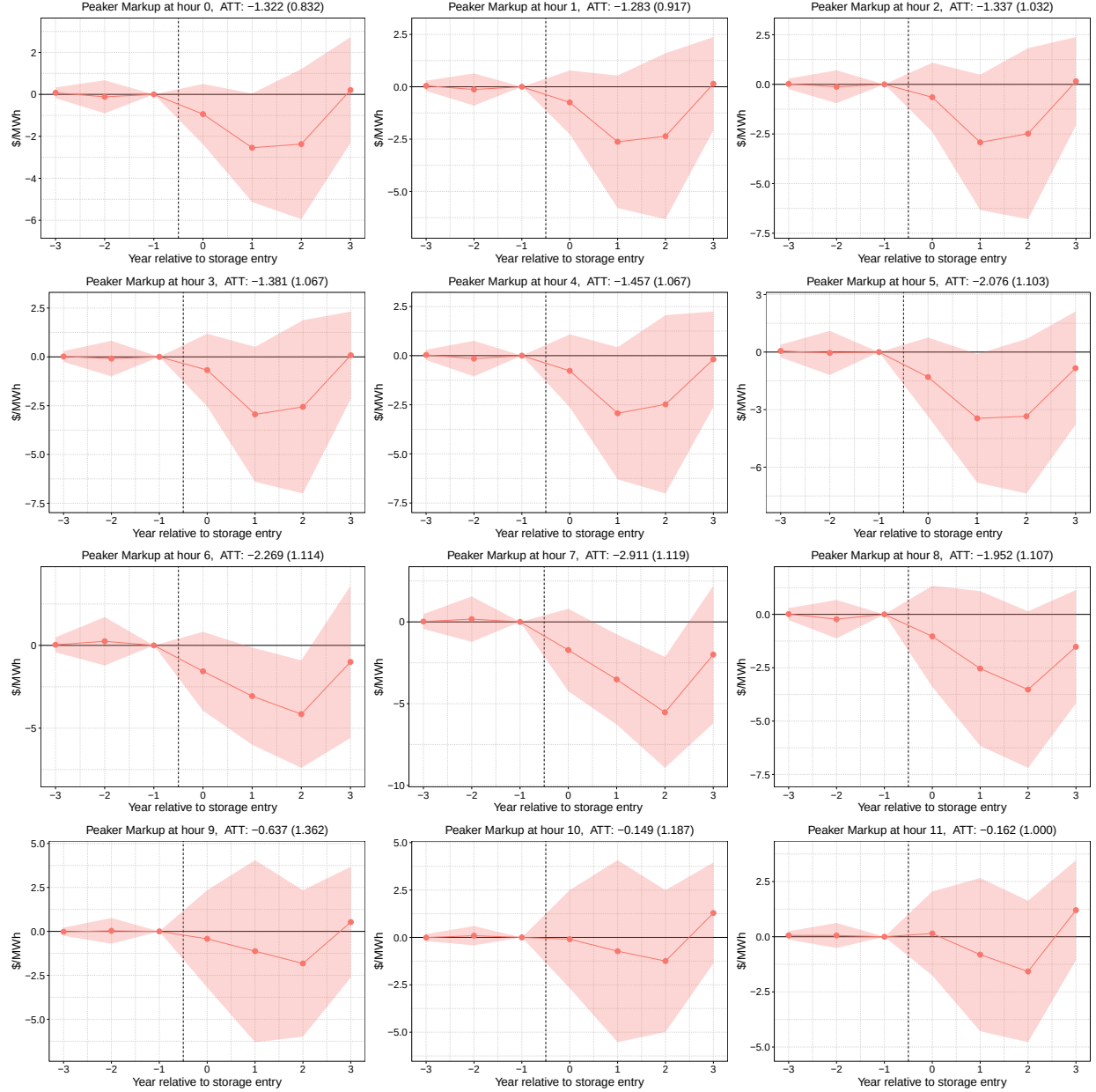


Figure F.29: (Continued) Complete event study figure for Peaker Markup

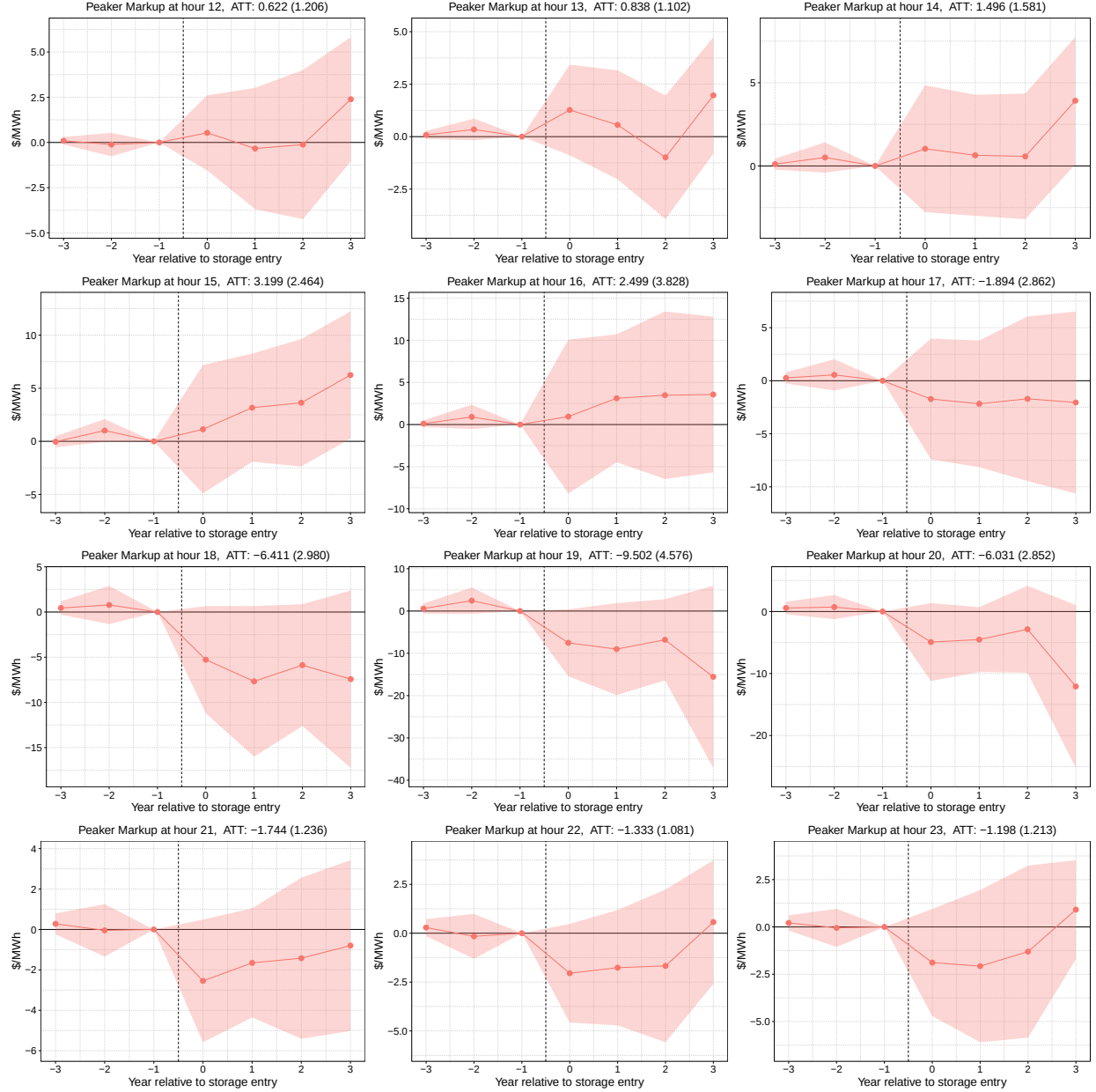


Figure F.30: Complete event study figure for Peaker Markup, monthly resolution

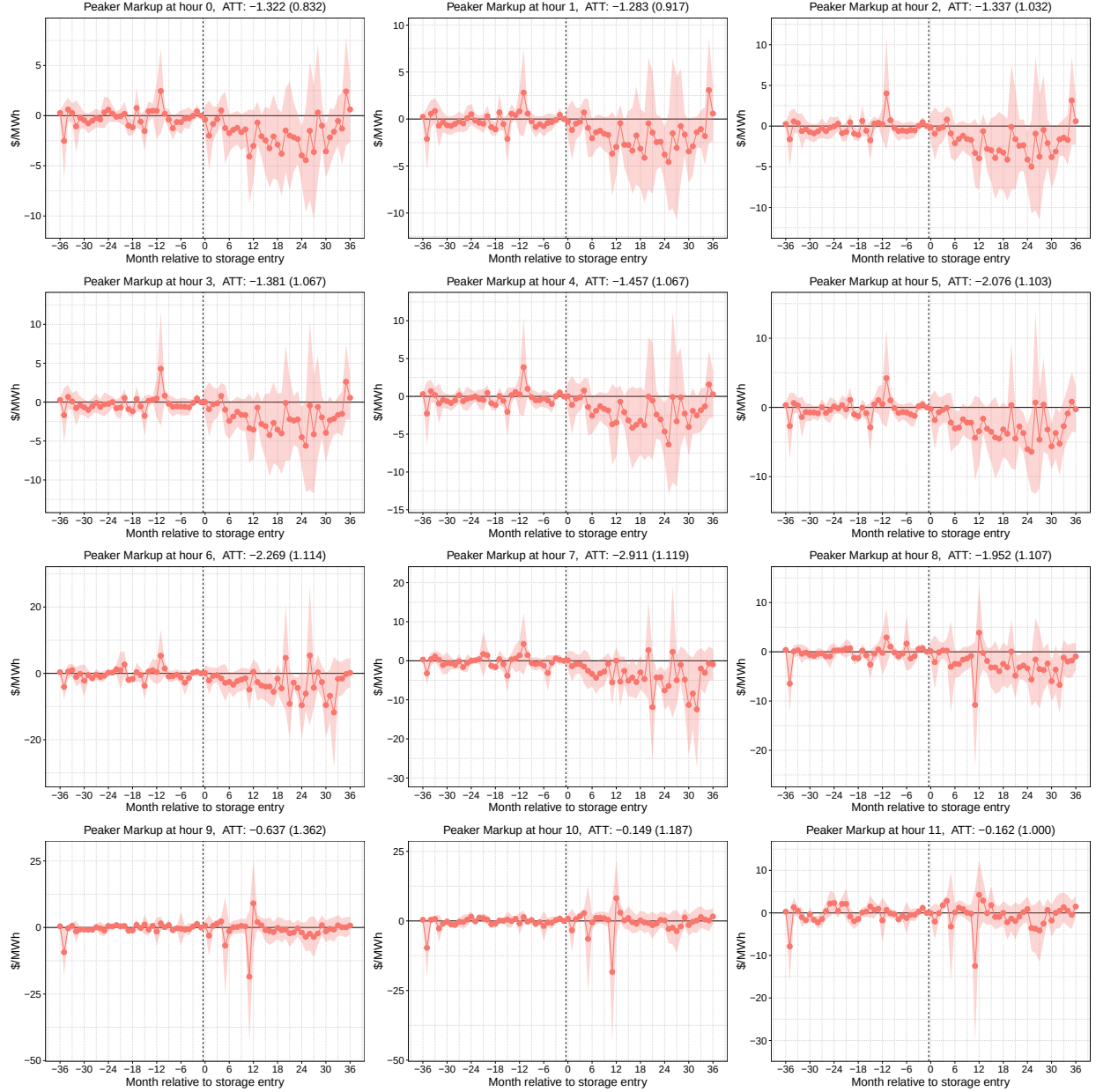


Figure F.30: (Continued) Complete event study figure for Peaker Markup, monthly resolution

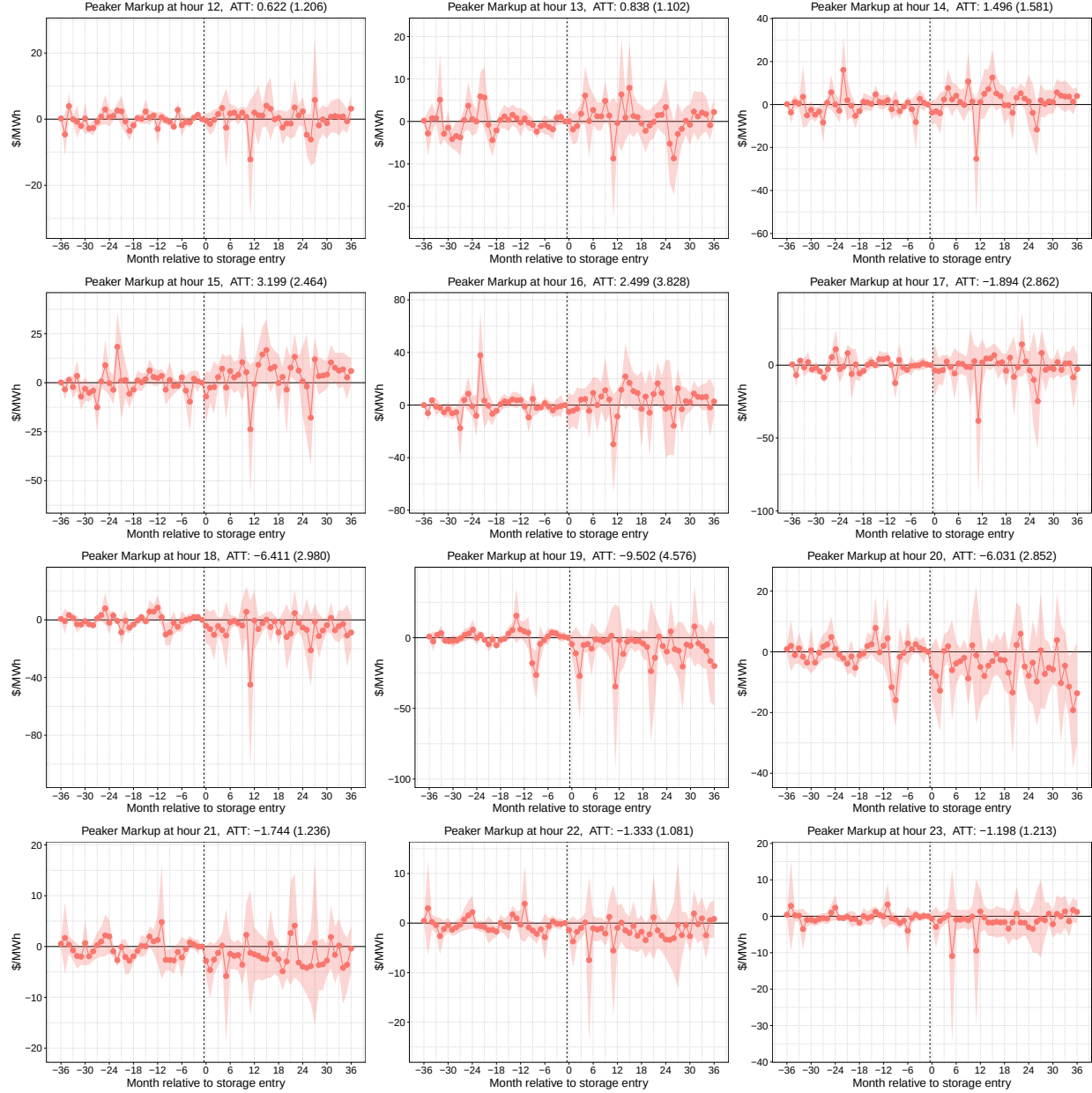


Figure F.31: Complete event study figure for Peaker Generation

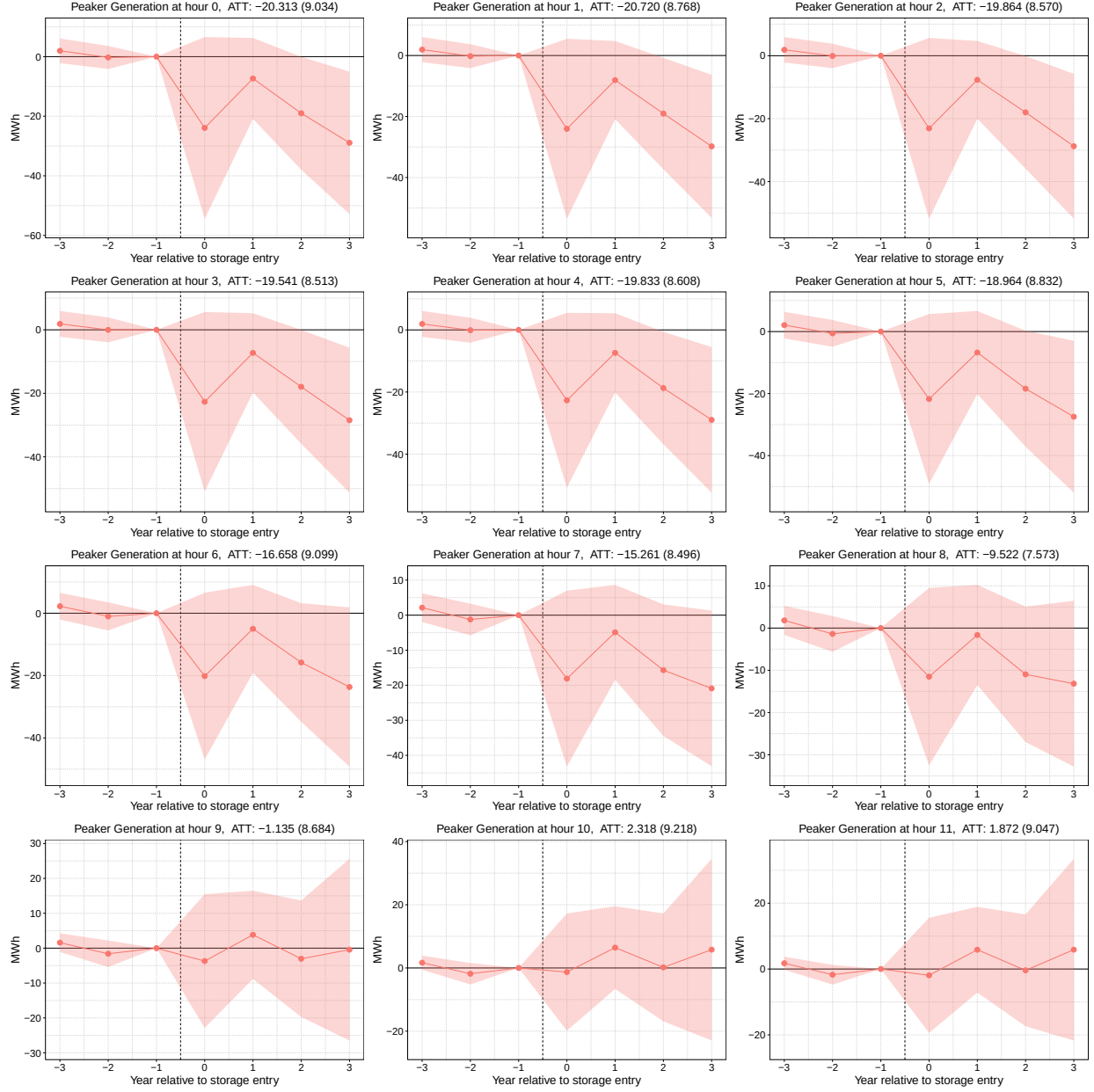


Figure F.31: (Continued) Complete event study figure for Peaker generation

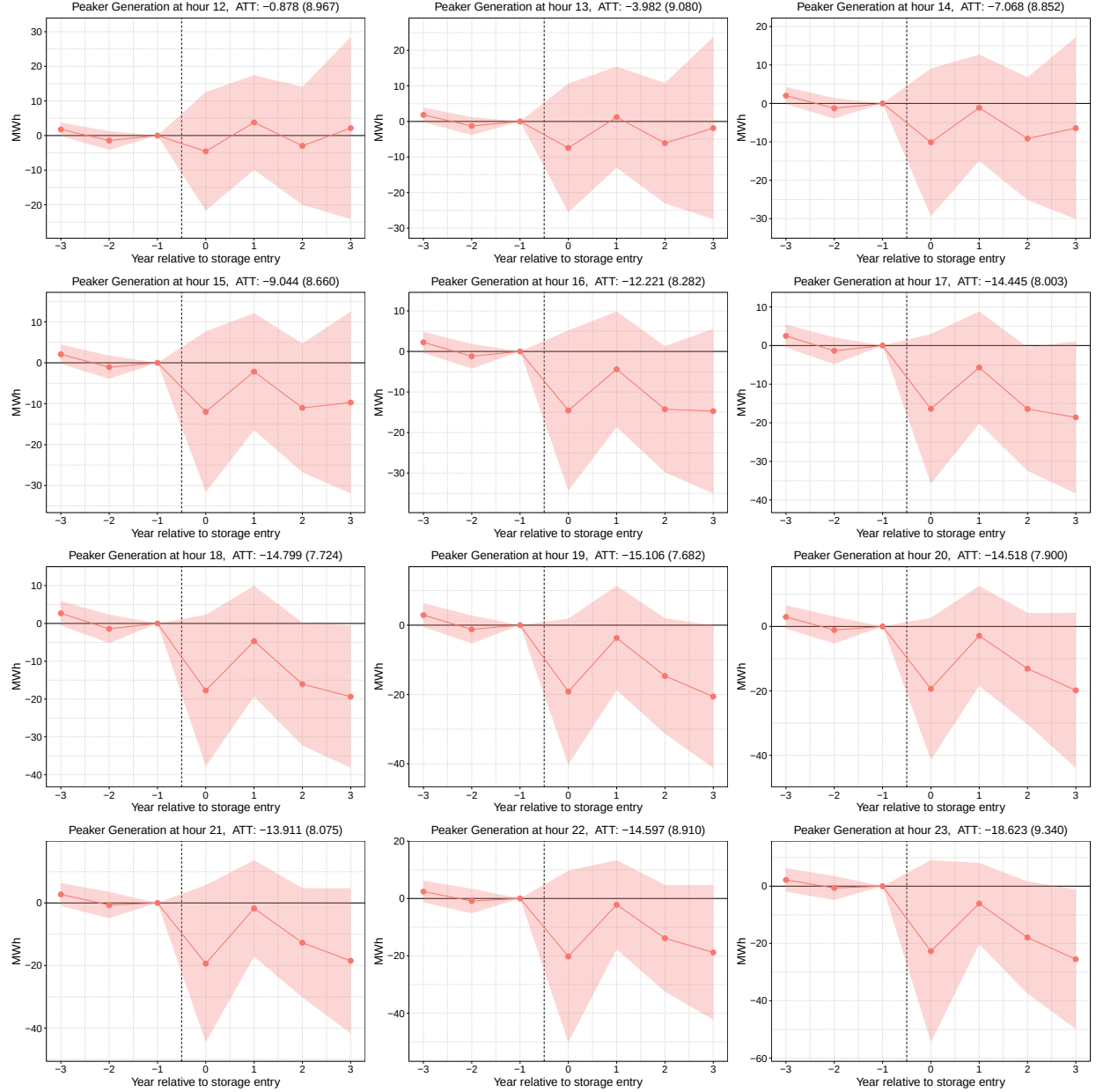


Figure F.32: Complete event study figure for Peaker Generation, monthly resolution

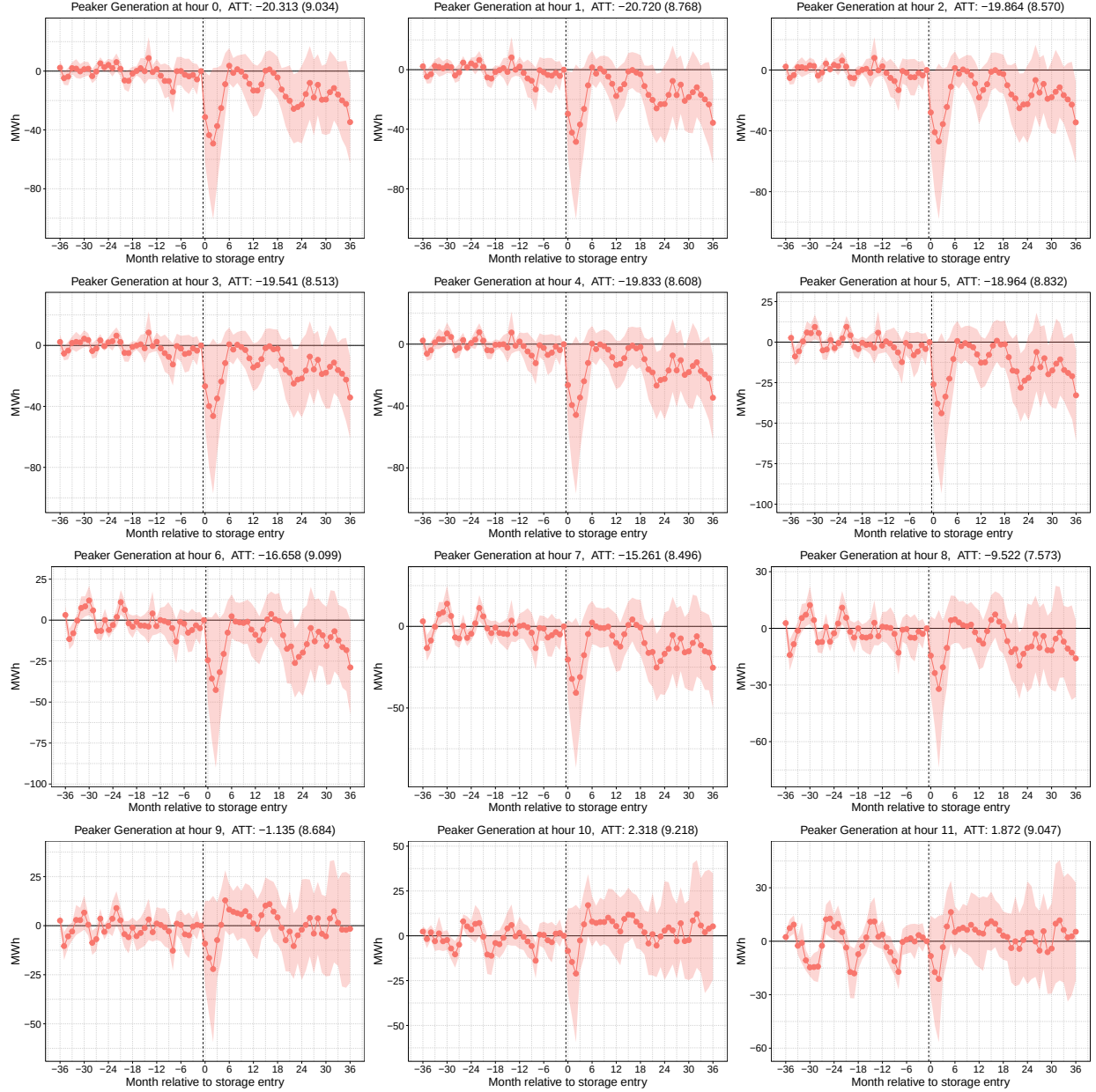


Figure F.32: (Continued) Complete event study figure for Peaker generation, monthly resolution

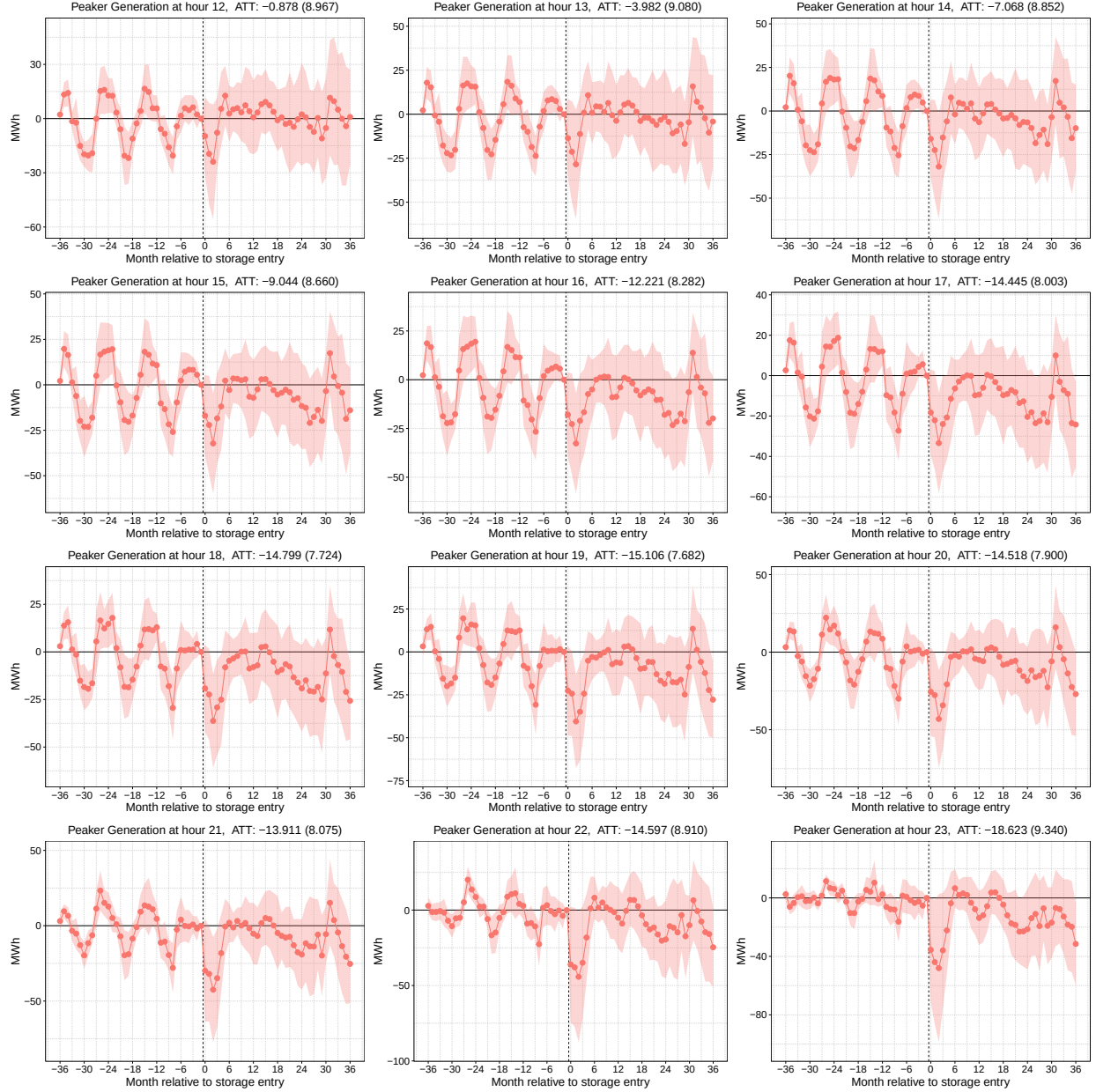


Figure F.33: Complete event study figure for Non-Peaker Markup

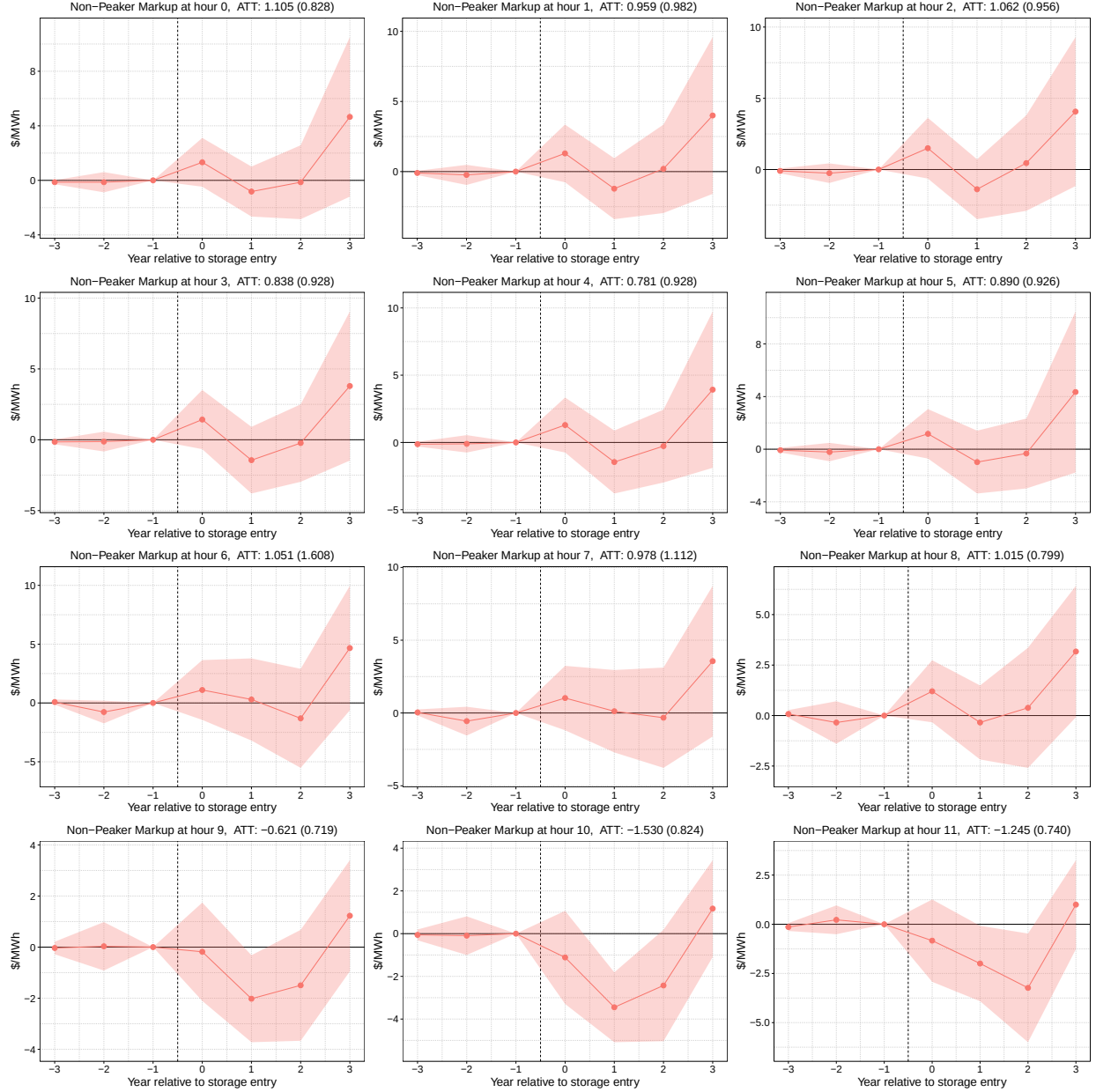


Figure F.33: (Continued) Complete event study figure for Non-Peaker Markup

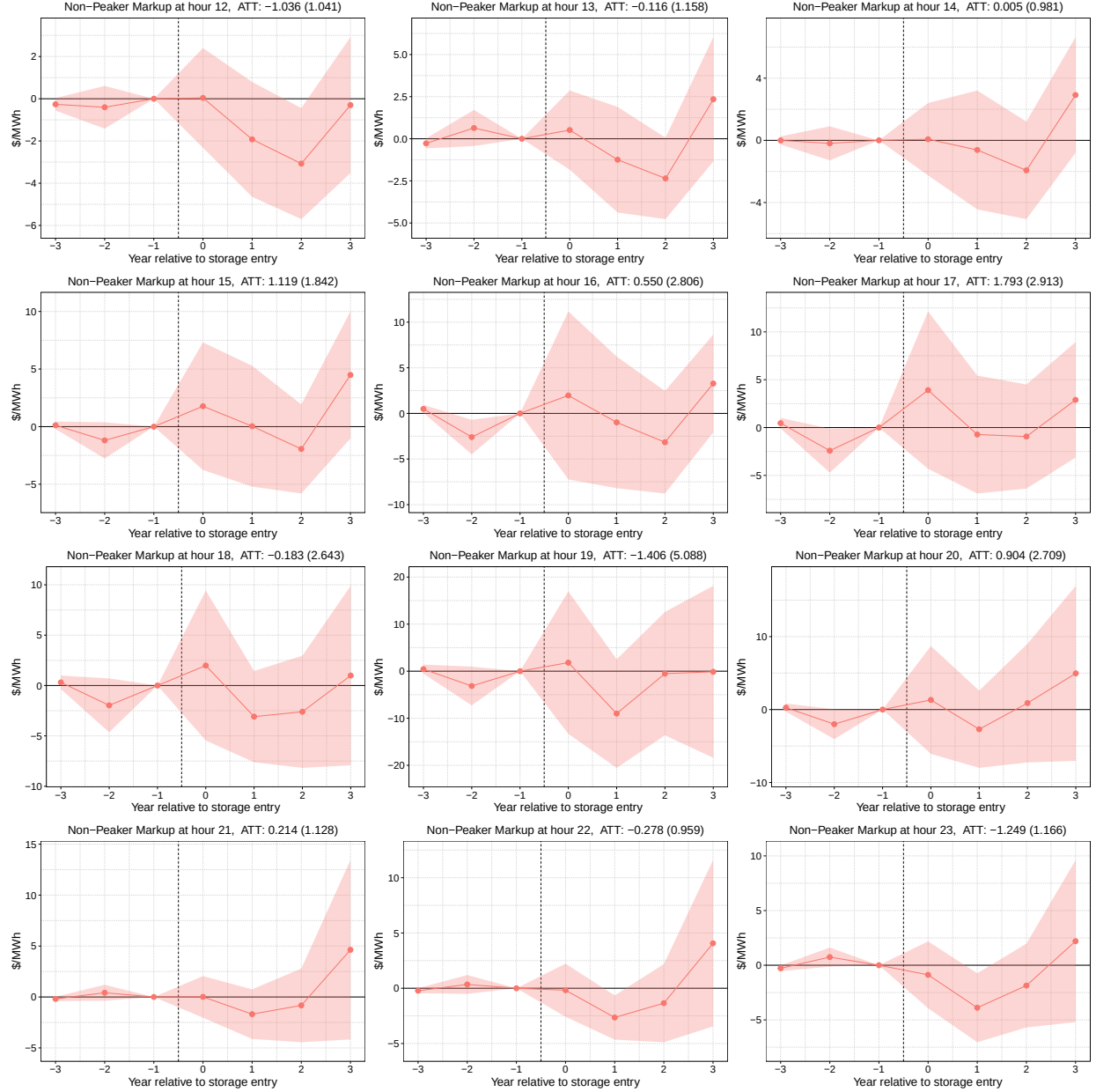


Figure F.34: Complete event study figure for Non-Peaker Markup, monthly resolution

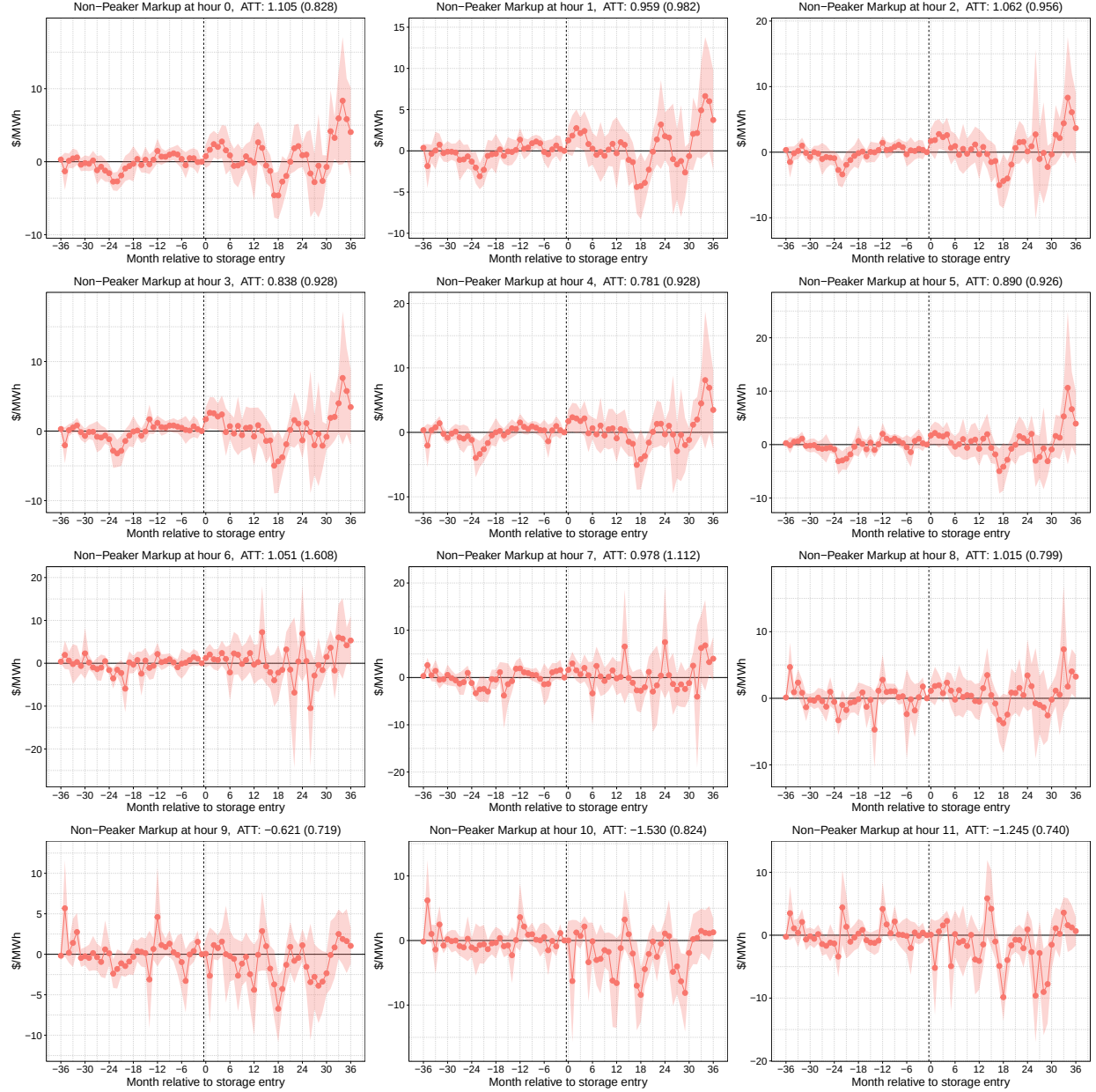


Figure F.34: (Continued) Complete event study figure for Non-Peaker Markup, monthly resolution

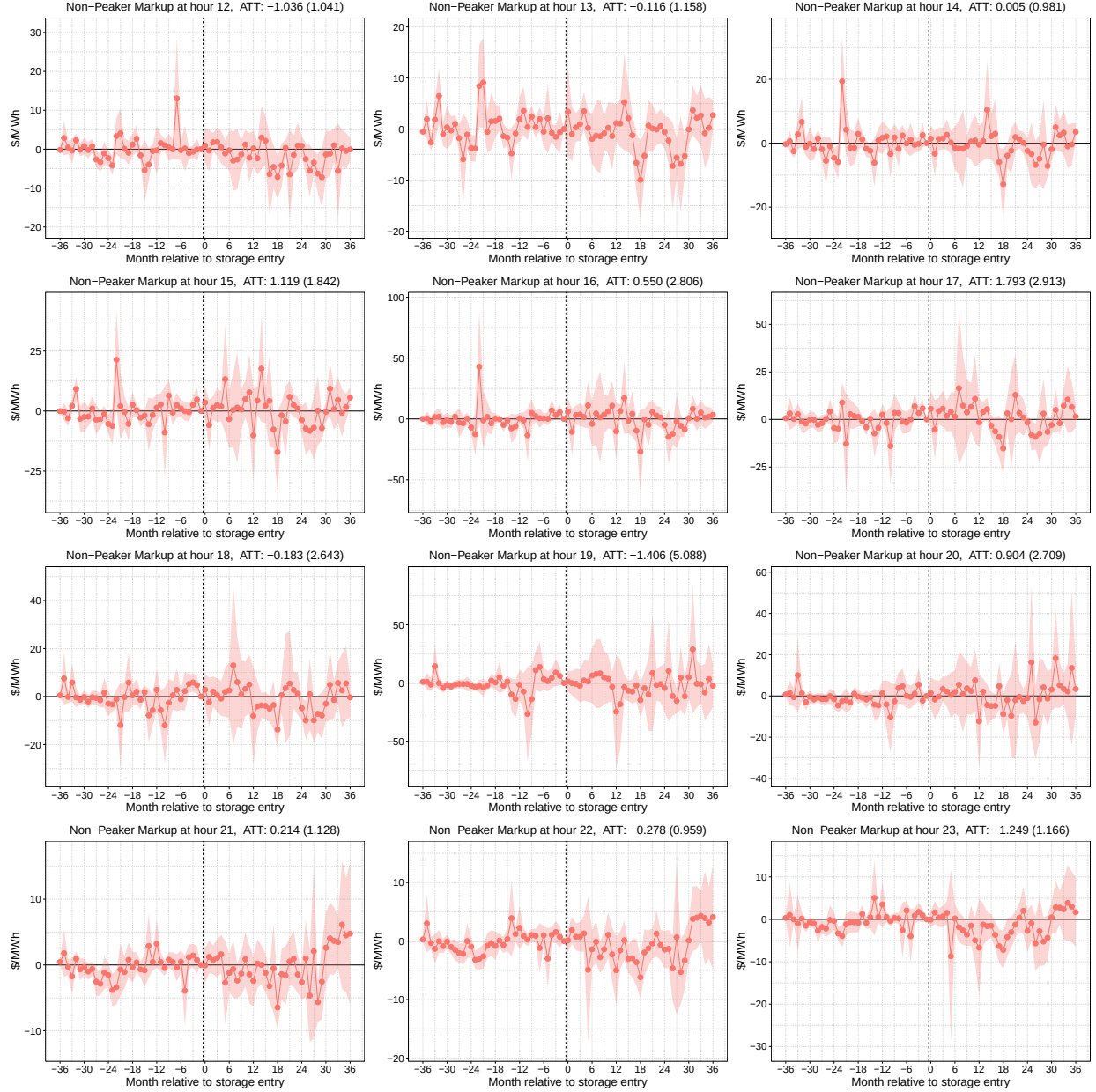


Figure F.35: Complete event study figure for Non-Peaker Generation

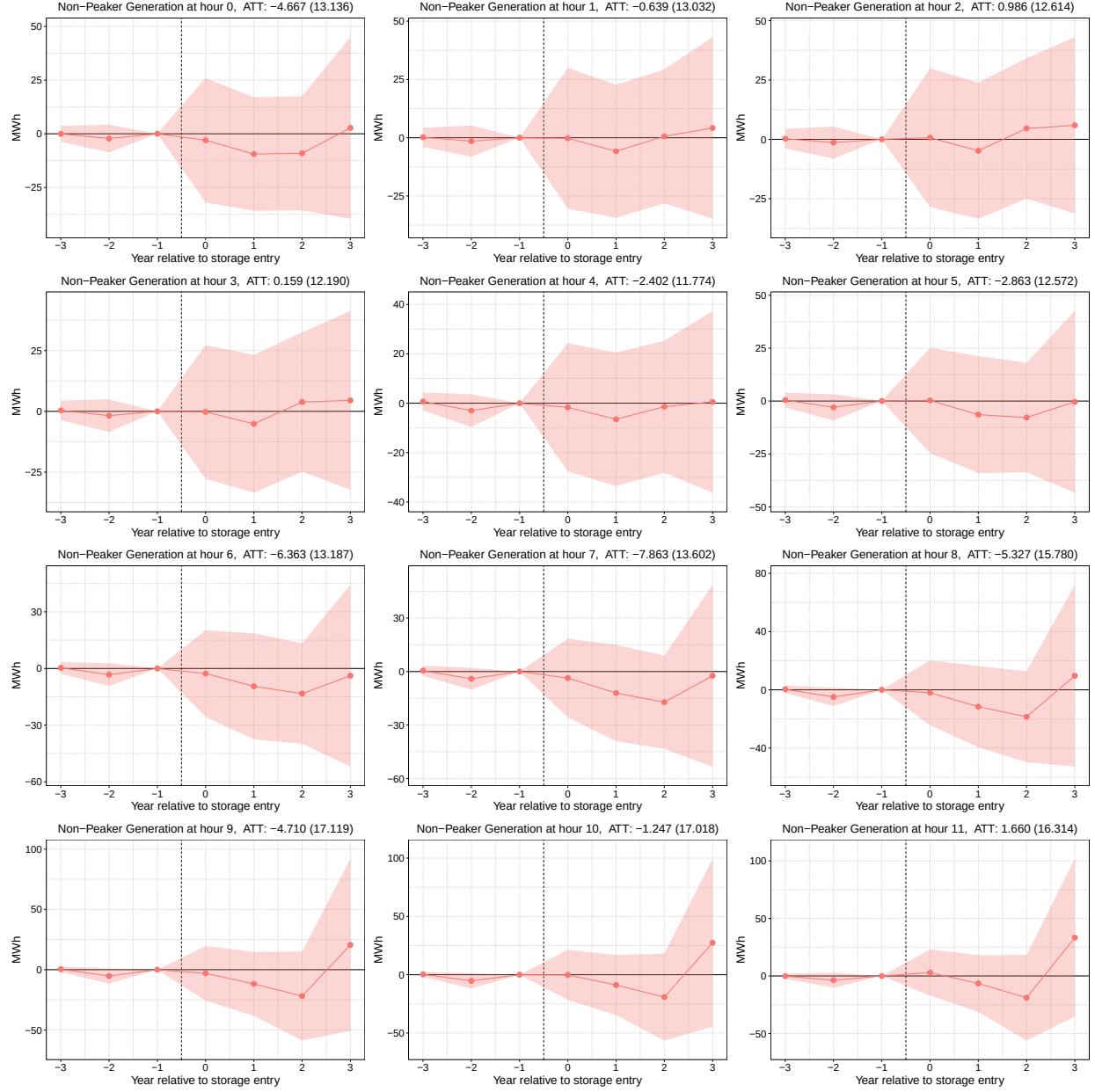


Figure F.35: (Continued) Complete event study figure for Non-Peaker Generation

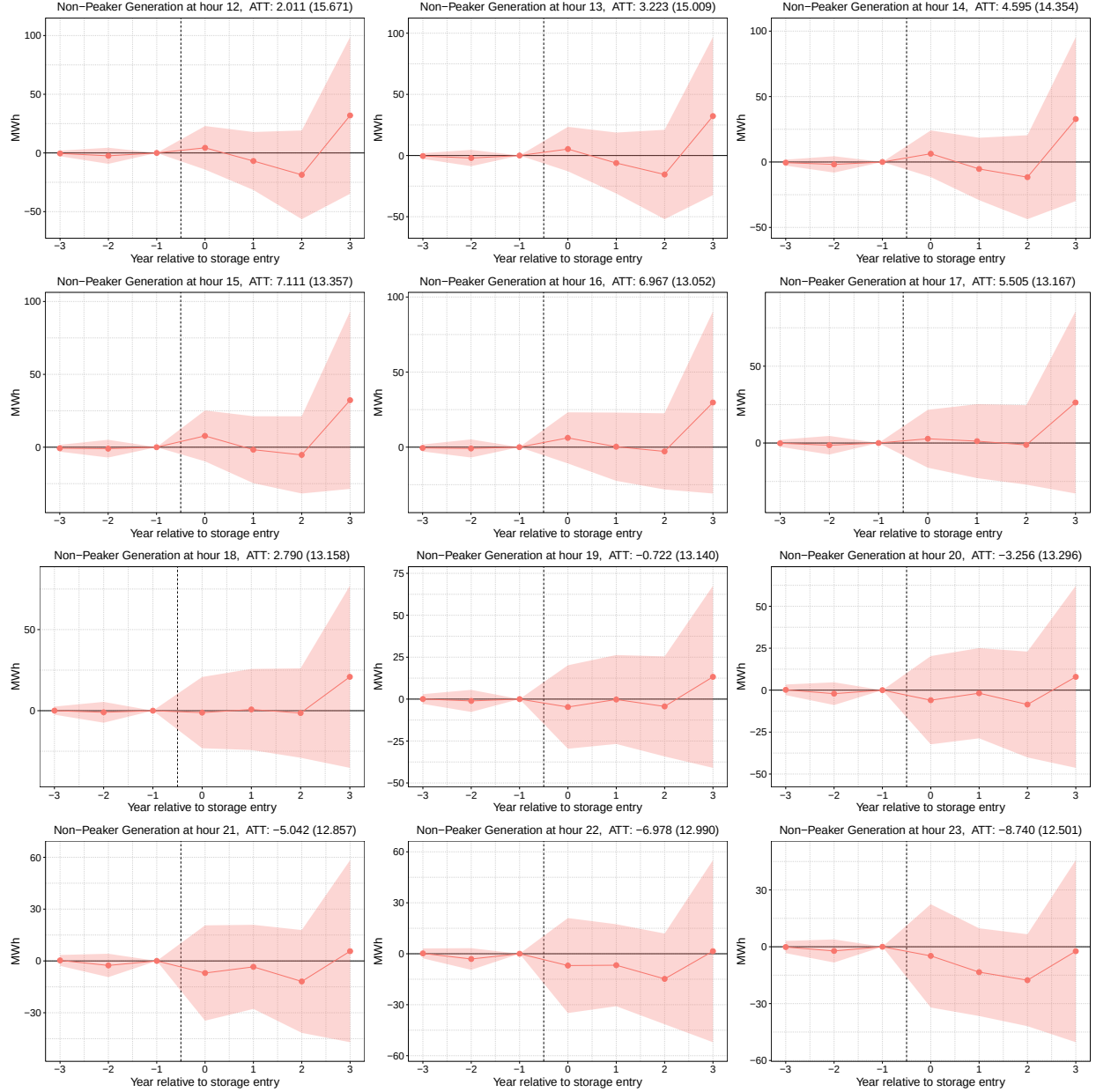


Figure F.36: Complete event study figure for Non-Peaker Generation, monthly resolution

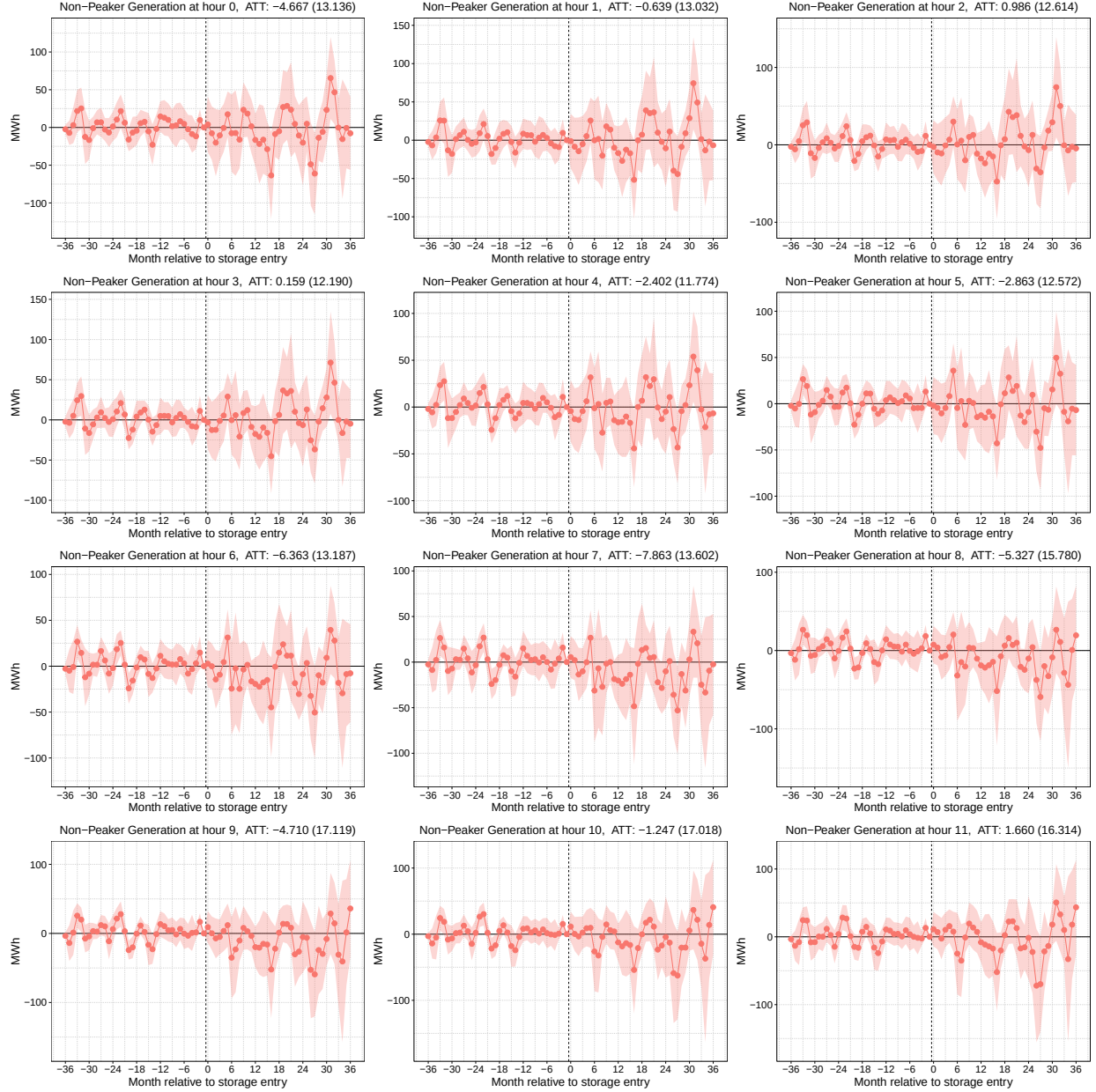


Figure F.36: (Continued) Complete event study figure for Non-Peaker Generation, monthly resolution

